United States Coast Guard Storm Water Management Job Guide

February 2023

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United States Department of Homeland Security
United States Coast Guard

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PART A – STORM WATER MANAGEMENT JOB GUIDE

1. STORM WATER MANAGEMENT JOB GUIDE

The United States Coast Guard (USCG) is subject to storm water management, pollution prevention, and sustainability requirements promulgated under federal laws and regulations, executive orders (EOs), and Department of Homeland Security (DHS) and USCG policy. This Storm Water Management Job Guide (henceforth referred to as the "Guide") provides policies and guidance on pollution prevention practices for USCG activities that have potential to discharge pollutants to the municipal storm water system or local waterbodies. The purpose of the Guide is to reduce or eliminate the discharge of pollutants resulting from these activities by identifying and providing guidance and education on storm water pollution prevention practices. The Guide provides practical advice using simple and non-technical language, wherever possible, to ensure that it is useful to unit personnel at all levels, especially those who are assigned storm water management responsibilities as a collateral duty.

The Guide is organized as follows:

Part A is an overview of storm water management requirements and USCG programs. It provides guidance for managing common USCG storm water needs, and discusses related requirements for training, inspections, recordkeeping, and reporting.

Within Part A:

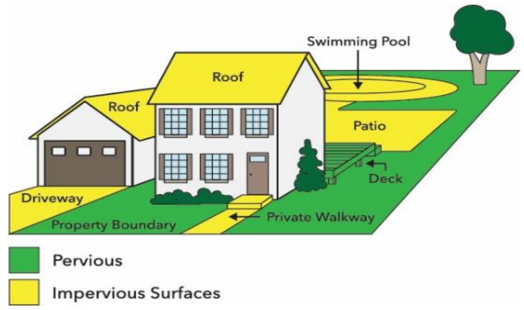
- Chapter 1 provides an overview of storm water management requirements and permitting, describes what storm water is, and summarizes the USCG storm water management approach.
- Chapter 2 provides an overview of the three National Pollutant Discharge Elimination System (NPDES) storm water permits (Industrial, Construction and Municipal Separate Storm Sewer Systems (MS4)).
- Chapter 3 provides guidance for managing storm water, regulatory inspection information, and special considerations to note.

Part B includes Environmental Procedure Cards (EPCs) providing USCG installations with guidance for managing storm water.

List of Attachments:

- **Attachment A – List of Acronyms**
- **Attachment B – Permit Matrix:** A generalization of unit operations nationwide that can be helpful to identify specific NPDES permitting applicable to unit operations
- **Attachment C – Multi-Sector General Permit (MSGP) Details:** Detailed information on facility permit classifications and MSGP requirements as well as enterprise-wide determinations
- **Attachment D – Special Considerations by State:** List of state-specific differences in permitting expectations
- **Attachment E – Advanced Structural BMPs:** Information on proprietary engineered systems that may be used to remove pollutants from storm water runoff or prevent pollutants from entering storm water.
- **Attachment F – USCG-Owned Wastewater Treatment Plant:** Table providing units that currently operate a WWTP

1.1 STORM WATER BASICS



IMPERVIOUS SURFACES *are areas which prevent or impede the infiltration of storm water into the soil as would have in natural conditions prior to development. Common impervious areas include, but are not limited to, rooftops, sidewalks, walkways, patio areas, driveways, parking lots, storage areas, compacted gravel and soil surfaces, awnings and other fabric or plastic coverings.*

Storm water has a major impact on the water quality of streams, lakes, and other waterbodies. Storm water (sometimes called storm water runoff) is created when it rains, and the water does not soak into the soil. Storm water runoff is generated from rain and snowmelt events that flow over land or *impervious surfaces*, such as paved streets, parking lots, and building rooftops, and does not soak into the ground**.** The storm water picks up pollutants (like sediment, oil and grease, animal waste, fertilizers, and other chemicals) and carries them into storm drains or directly into waterbodies. Preventing contamination of storm water is critically important. Otherwise, polluted runoff will be discharged untreated into the waterbodies we use for swimming, fishing, and drinking water. Common storm water pollutants include the following.

- **Sediments** – A significant water quality problem in the United States is sediment dislodged from exposed soil, stream channel banks, and channel beds. Excessive sediment can be detrimental to aquatic life by interfering with photosynthesis, respiration, growth, and reproduction.

- **Nutrients** – Runoff from urban watersheds contain increased nutrients such as nitrogen and/or phosphorus compounds. In an urban setting, nutrients in storm water primarily come from fertilizers, lawn and landscape wastes, and pet wastes. Increased nutrient levels promote weed and algae growth, which can block sunlight from reaching underwater grasses. This depletes dissolved oxygen when the organic matter decomposes.

- **Pathogens** – Pathogens harmful to human health consist of bacteria, protozoa, viruses, and other microscopic organisms. The sources of pathogens in urban storm water include leaking private or public sewer lines, combined sewer overflows, malfunctioning septic tanks, and wastes from animals.

- **Hydrocarbons** – Oils, greases, and gasoline contain a wide array of hydrocarbon compounds, some of which have been shown to be carcinogenic, tumorigenic, and mutagenic in various species of fish and other lifeforms. In large quantities, oil can impact drinking water supplies and affect recreational use of waters. Oils and other hydrocarbons wash off roads and parking lots, primarily due to vehicle leaks. Other sources include improper disposal of motor oil in storm drains and streams, spills at fueling stations, and restaurant grease traps.

- **Reduced Oxygen in Streams** – Storm water laden with lawn and landscape waste, pet waste, and other sources delivers organic matter to local streams. While organic material is necessary for aquatic life, an overabundance of organic matter can contribute stream impacts. The decomposition of this matter uses up dissolved oxygen (DO) in the water, which is vital for fish and other aquatic life. When combined with increased algae and weed growth due to nutrient loading, DO levels rapidly deplete causing fish kills and other stream health impacts.

- **Toxic Materials** – Besides oils and greases, urban storm water runoff can contain a wide variety of other toxicants and compounds, including heavy metals like lead, zinc, copper, and cadmium, as well as organic pollutants that include pesticides, polychlorinated biphenyls (PCBs), and phenols. These contaminants are of concern because they are toxic to aquatic organisms and can bio-accumulate in the food chain. Sources of these contaminants include industrial and commercial areas, urban surfaces like rooftops and roadways, vehicles, and other machinery, improperly disposed household chemicals, landfills, hazardous waste sites, and atmospheric deposition.

❖ **Thermal Pollution** – As storm water flows over impervious surfaces, such as asphalt and concrete, it increases in temperature before reaching a stream or waterbody. Since warm water holds less DO than cold water, this "thermal pollution" further reduces oxygen levels in urban streams. Temperature changes can severely impact certain sensitive aquatic species, such as trout and stoneflies which can survive only within a narrow temperature range.

STONEFLY *nymphs develop in cool, well-oxygenated water and do not tolerate pollution; therefore, their existence within a waterbody is an indicator of good water quality.*

❖ **Trash and Debris** – Trash and other debris that are washed through storm drain systems and into streams and lakes can harm terrestrial and aquatic animals that consume or become entangled/engulfed in the solid waste.

Here are a few terms often mentioned in this guide and other storm water-related materials:

What is a pollutant?

"*Pollutant* means dredged spoil, solid waste, incinerator residue, filter backwash, sewage, garbage, sewage sludge, munitions, chemical wastes, biological materials, radioactive materials ..., heat, wrecked or discarded equipment, rock, sand, cellar dirt and industrial, municipal, and agricultural waste discharged into water." [40 CFR. § 122.2]

What is a point source?

"*Point source* means any discernible, confined, and discrete conveyance, including but not limited to, any pipe, ditch, channel, tunnel, conduit, well, discrete fissure, container, rolling stock, concentrated animal feeding operation, landfill leachate collection system, vessel or other floating craft from which pollutants are or may be discharged. This term does not include return flows from irrigated agriculture or agricultural storm water runoff." [40 CFR § 122.2]

What is a nonpoint source?

According to the EPA website (https://www.epa.gov/nps/basic-information-about-nonpoint-source-nps-pollution) , the term "nonpoint source" is defined to mean any source of water pollution that does not meet the legal definition of "point source" in section 502(14) of the Clean Water Act.

Nonpoint source pollution can include:

- Excess fertilizers, herbicides and insecticides from agricultural lands and residential areas
- Oil, grease and toxic chemicals from urban runoff and energy production
- Sediment from improperly managed construction sites, crop and forest lands, and eroding streambanks
- Salt from irrigation practices and acid drainage from abandoned mines
- Bacteria and nutrients from livestock, pet wastes and faulty septic systems
- Atmospheric deposition and hydromodification

What is a storm water discharge?

A storm water discharge means the release of storm water that contains pollutants that may affect water quality from any conveyance that is used for collecting and conveying storm water. It can be a discharge of storm water laden with soil from a construction site, or a discharge related to manufacturing or processing at an industrial facility.

1.2 STORM WATER REGULATORY REQUIREMENTS

This section provides an overview of key federal requirements governing the management of storm water at USCG shore installations.

1.2.1 Federal Laws and Regulations

The **Clean Water Act (CWA)**, as amended in 1972, provided the statutory basis for the NPDES permit program, which is intended to control water pollution by regulating point sources that may discharge pollutants into waters of the United States. Section 402 of the CWA mandated that the United States Environmental Protection Agency (EPA) develop and implement the NPDES permit program.

The Code of Federal Regulations (CFR) contains all rules published in the Federal Register by agencies of the federal government. The CFR is divided into 50 titles, with environmental regulations contained mostly within Title 40. The primary regulations developed by the EPA to administer the NPDES permit program are available in Title 40 of the CFR at Part 122 – EPA Administered Permit Programs: The National Pollutant Discharge Elimination System.

The CWA gives EPA the authority to set effluent limits on an industry basis. Facilities that discharge pollutants in effluent must first obtain an NPDES permit, or the discharge is considered illegal. Most storm water discharges are considered point sources and require coverage under an NPDES permit.

The NPDES storm water program regulates some storm water discharges from three potential sources: municipal separate storm sewer systems (MS4s), construction activities, and industrial activities. Operators of these sources might be required to obtain an NPDES permit before they can discharge storm water. For more information on these sources, refer to Section 2.

In many cases, permitting, administrative, and enforcement aspects of the NPDES program are performed by states granted authorization by the EPA to do so. The EPA remains the permitting authority in a few states, territories, and on most tribal lands. Figure 1-1 from **www.epa.gov** depicts each state's NPDES program management authorization status as of July, 2019 (verified in February, 2023).

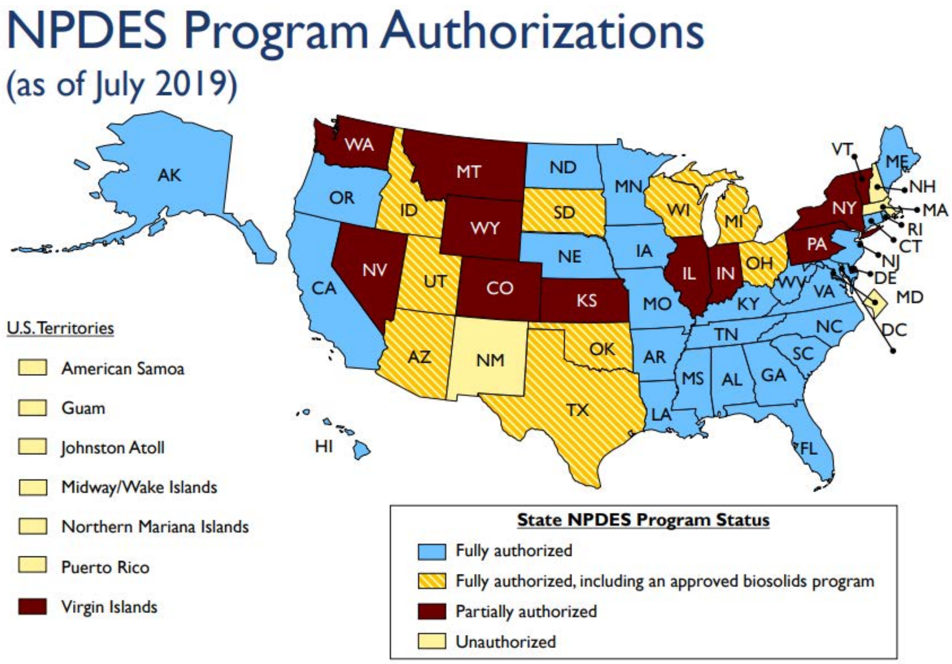


Figure 1-1. State NPDES Program Authority

1.2.2 Executive Orders

EO 13990, Protecting Public Health and the Environment and Restoring Science to Tackle the Climate Crisis, was signed into effect on 20 January 2021 and directs federal agencies to manage their buildings, vehicles, and overall operations to optimize energy and environmental performance, reduce waste, and cut costs. The EO directs agencies to meet all statutory requirements related to energy and environmental performance, and identifies the following priority areas:

- Building energy use and efficiency;
- Renewable energy and electricity;
- Potable and non-potable water consumption and storm water management;
- Use of performance contracting;
- Sustainable buildings;
- Waste prevention, recycling, and toxics reduction; and
- Sustainable procurement

1.2.3 United States Department of Homeland Security and USCG Policy and Guidance

DHS and USCG headquarters have established policy and guidance for environmental issues. The directives, plans, and instructions below are not specifically related to storm water but can be applied in certain instances.

DHS Environmental Management Directive, Environmental Management Program, #023-02, Revision 1 (8 May 2013) and DHS Environmental Management Manual 023-02-002-01 (15 April 2015) establish DHS direction for environmental compliance management, requiring all DHS components to comply with EOs, federal, state, interstate, and local environmental laws as well as DHS policies and procedures. Furthermore, all DHS officials and employees are required to comply with environmental laws applicable to their jobs and responsibilities.

DHS Strategic Sustainability Performance Plan, last updated in 2017, establishes a department-wide goal to "reduce waste generation through elimination, source reduction, and recycling" as one of the Department's top five strategic priorities for sustainability. The Plan also establishes sustainable acquisition (i.e., green procurement) goals and strategies, requiring that 95 percent of all applicable new contracts and modifications contain requirements for procuring products and services that are environmentally preferable, recycled content, non-toxic and/or less toxic.

DHS Acquisition Manual, Appendix Q, requires contracting personnel to maximize the use of sustainable (green) products, including giving purchasing preference to recovered materials, energy- and water-efficient products, alternative fuels and fuel efficiency, sustainable buildings, biobased products, non-ozone-depleting substances, and environmentally preferred products.

COMDTINST 5100.47 (series), Safety and Environmental Health Manual sets forth key elements of the USCG's Safety and Environmental Health Program. This manual defines responsibilities for program implementation and specifies Safety and Environmental Health standards, policies, and responsibilities applicable across all missions and mission support activities. This manual also outlines requirements and procedures for the safe handling and use of hazardous materials.

COMDTINST 16477.5, Coast Guard Qualified Recycling Program (QRP) Policy provides policy guidance for establishing USCG QRPs, and reporting requirements for recycling activities to "prevent pollution, reduce contribution to overcrowded landfills, conserve our finite supply of natural and fiscal resources, and minimize routine waste disposal costs by reducing waste, reusing resources, and procuring products made of recycled materials, as well as materials that are readily recyclable and therefore environmentally preferable."

COMDTINST 16478.1 (series), Hazardous Waste Management Policy, establishes USCG policy and responsibilities for the handling, storage, transportation, and disposal of hazardous waste generated by USCG units during operational activities or mission support.

COMDTINST 16478.5, Environmental Compliance Evaluation Program, requires the USCG to implement an ECE Program at shore facilities.

COMDTPUB P5090.1C, Commanding Officer's Environmental Guide, is an easy-to-use, desktop guide to environmental stewardship for Commanding Officers and Officers-in-Charge of USCG units and includes inspection checklists to aid in compliance.

EO 13990, Protecting Public Health and the Environment and Restoring Science to Tackle the Climate Crisis, which directs Federal agencies to review and take action to address the promulgation of Federal regulations and other actions during the previous four years that conflict with national objectives to improve public health and the environment, ensure access to clean air and water, and prioritize environmental justice and employment.

1.3 WHO MANAGES STORM WATER?

All USCG employees are responsible for executing their work responsibilities and assignments in a manner that is compliant with regulations and environmentally sound. The responsibilities for day-to-day management of storm water are primarily assigned to personnel involved with outdoor facilities and those unit personnel responsible for executing and overseeing the unit's compliance management program. Installation-level environmental personnel include Environmental Protection Specialists (EPSs) and supporting environmental staff. EPSs often receive technical and other logistical support from their chain of command (e.g., base, sector, Civil Engineering Unit [CEU], or Shore Infrastructure Logistics Center-Environmental Management Division [SILC-EMD] environmental personnel). Smaller stations typically do not have an EPS; storm water management responsibilities are assigned as a collateral duty to operational personnel. In all cases, installation-level environmental personnel should coordinate closely with other installation functions, including facilities management, supply and contracting, and operational commands. Senior executive officers (CO/OIC) have ultimate responsibility for the overall unit environmental compliance program, including storm water management.

1.3.1 Operations-Level

All USCG shore-based facilities whose operations have the potential to produce storm water discharges are encouraged to use this Guide to achieve compliance with storm water regulations.

1.3.2 Depot-Level

SILC-EMD provides program-level mission support for environmental compliance. SILC-EMD is responsible for maintaining and updating this Guide. The Civil Engineering Unit Environmental Management Branches (CEU-EMBs) are the primary points of contact providing depot-level mission support to USCG operations within their area of responsibility (AOR). CEU-EMBs provide regional subject matter expertise (SME) guidance to units regarding environmental compliance and environmental management. CEU-EMBs will provide depot-level mission support regarding the interpretation and implementation of this Guide.

1.3.3 Applicable Assets (and/or Services)

This Guide applies enterprise-wide for all unit operations with the potential to produce a storm water discharge. This Guide contains storm water-specific Environmental Procedure Cards (EPCs) to be implemented enterprise-wide for those units conducting activities that have the potential to impact storm water. These cards are not meant to regulate the activities of the operations, rather to guide them to best management practices as they relate to minimizing the potential for pollution to storm water during daily activities at the unit. Each of these EPCs were developed using a combination of state regulations, USCG policy, BMPs accumulated for other federal agencies and local municipalities, as well as site visits to each type of USCG facility across the nation to determine applicability. Additional details regarding EPCs can be found in Section 3.2.

1.3.4 When to Perform

Each unit is required to implement storm water management measures to minimize the discharge of pollutants in runoff from the site, even if not covered under an NPDES storm water permit. The procedures for managing storm water discharges shall be performed continuously and in perpetuity, even if the unit is not required to obtain coverage under an NPDES permit.

1.3.5 Do You Need a Storm Water Permit?

If you do not have a storm water permit at your unit but are concerned you may need one or have been informed by a regulator that you need one, proceed as follows:

Step 1 – Consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD to determine the best way forward.

Step 2 - If it is decided that a permit is needed, then procure the funds necessary to secure permit.

Step 3 - Work with your storm water SME to submit an NOI and SWPPP.

Step 4 - Comply with all aspects of the permit once you are covered under it.

Step 5 - Don't forget to track the permit termination date and reapply in time to maintain coverage!

There are some activities that produce allowable non-storm water discharges (i.e., fire hydrant/water line flushings, irrigation drainage, and exterior building wash water that does not contain detergents) but it is best to follow the steps above prior to allowing any discharge to the ground, storm sewer system or local waterbodies.

A Permit Matrix is provided in **Attachment B – Permit Matrix**. The matrix is a generalization of unit operations nationwide that can be helpful to identify current NPDES permitting applicable to unit operations.

2. NPDES STORM WATER PERMITS

2.1 NPDES PERMITTING OVERVIEW

As discussed in Section 1.2, the CWA gives EPA the authority to control water pollution by regulating all point sources that discharge pollutants into waters of the United States via permitting. Entities discharging pollutants must obtain an NPDES permit prior to discharge, otherwise the discharge is considered illegal.

Many storm water discharges are considered point sources and require coverage under an NPDES permit. As mentioned in Section 1.2.1, most states have been delegated authorization by the EPA to conduct permitting, administrative, and enforcement aspects of the NPDES program. The EPA remains the permitting authority in a few states, territories, and on most tribal lands.

NPDES permitting is typically approached using one of two different permit types, general and individual. A general permit is assigned to facilities or activities that generate discharges containing common and similar potential pollutants. General permits lessen the burden on permittees and regulators by providing standardized permit requirements tailored to the common pollutants and eliminating the need for each facility or operation to apply for an individual permit. General permits typically cover most facilities within the USCG.

Individual permits are triggered when an activity or pollutants generated from an activity are not included within the activities covered under a general permit (i.e., major industrial facilities with specialized discharges, POTWs, drinking water systems, etc.). Since almost all USCG operations fall under general permitting parameters, general permit types will be discussed in detail within the following text. For additional information on individual industrial permits, please consult SMEs at your servicing Sector, CEU, or SILC EMD.

The NPDES permit program regulates storm water discharges from three potential sources: **industrial activities, municipal separate storm sewer systems (MS4s), and construction activities.**

Industrial Activities – Industrial activities may require coverage under EPA's Multi-Sector General Permit (MSGP) for storm water discharges. 40 CFR § 122.26(b)(14)(i)-(xi), identifies 11 categories of industrial activity with storm water discharges that are required to be covered under an NPDES permit. Additional information on permitting requirements for industrial activities regulated under the MSGP can be found in Section 2.1 of this Guide and also in **Attachment C – Industrial Permit Requirements**.

MS4s – According to the EPA, an MS4 is defined as a conveyance or system of conveyances that is:

- owned by a state, city, town, village, or other public entity that discharges to waters of the U.S.,
- designed or used to collect or convey storm water (e.g., storm drains, pipes, ditches),
- not a combined sewer, and
- not part of a sewage treatment plant, or publicly owned treatment works (POTW).

Operators of large, medium, and regulated small MS4s may be required to obtain authorization to discharge storm water. Additional information on MS4s can be found in Section 2.2 of this Guide.

Construction Activities – Operators of construction sites that are 1 acre or larger (including smaller sites that are part of a larger common plan of development or sale) may be required to obtain authorization to discharge storm water under an NPDES construction storm water permit. Authorization to discharge may also be required for dewatering discharges and hydrostatic testing discharges of water/sewer utilities. Where EPA is the permitting authority, operators must meet the requirements of EPA's Construction General Permit (CGP). In some cases, the EPA or the State permitting authority may require coverage under an "individual permit" if it is determined that an activity does not meet the conditions for coverage under the CGP. An individual permit will include permit requirements and conditions tailored to the site-specific activity.

Additional information on permitting requirements for construction activities can be found in Section 2.3 of this Guide.

FOR ADDITIONAL INFORMATION OR CLARIFICATION OF REQUIREMENTS OR PROCEDURES DISCUSSED IN THIS GUIDE, UNIT PERSONNEL SHOULD SEEK ASSISTANCE THROUGH THEIR STORM WATER SMEs AT SERVICING SECTOR, CEU, OR SILC EMD.

2.2 INDUSTRIAL ACTIVITIES

2.2.1 Industrial Multi-Sector General Permit (MSGP)

The EPA NPDES program regulates storm water runoff from industrial facilities through the Multi-Sector General Permit (MSGP). EPA's MSGP covers 29 different industrial sectors (see Figure 2-1).

The permitting, administrative, and enforcement aspects of the NPDES program are performed by states granted authorization by the EPA to do so. The EPA remains the permitting authority in a few states, territories, and on most tribal lands. Refer to Section 1.2.1 for additional details on this.

A: Timber Products	P: Land Transportation
B: Paper Products	Q: Water Transportation
C: Chemical Products	R: Ship/Boat Building, Repair
D: Asphalt/Roofing	S: Air Transportation
E: Glass, Clay, Cement	T: Treatment Works (WWTPs)
F: Primary Metals	U: Food Products
G: Metal Mining	V: Textile Mills
H: Coal Mines	W: Furniture and Fixtures
I: Oil and Gas	X: Printing, Publishing
J: Mineral Mining	Y: Rubber, Misc. Plastics
K: Hazardous Waste	Z: Leather Tanning/Finishing
L: Landfills	AA: Fabricated Metal Products
M: Auto Salvage Yards	AB: Transportation Equip.
N: Scrap Recycling	AC: Electronic, photo goods
O: Steam Electric Facilities	AD: Non-classified facilities

Figure 2-1
Sectors of Industrial Activity
(Source: https://www.epa.gov/)

Permit eligibility is limited to discharges from facilities in the sectors of industrial activity provided to the left. USCG facilities primarily fall under Sector P (Land Transportation), Sector Q (Water Transportation), and Sector S (Air Transportation). It should be noted that a single facility may fall under multiple activities. For detailed information on facility permit classifications and MSGP requirements as well as enterprise-wide determinations, refer to **Attachment C – Industrial Permit Requirements.**

As mentioned in Section 1.2.1, many states have been granted authorization by the EPA to conduct permitting, administrative, and enforcement aspects of the NPDES program. The EPA remains the permitting authority in a few states, territories, and on most tribal lands. Figure 1-1 from **www.epa.gov** depicts each state's NPDES program management authorization status.

2.2.1.1 No Exposure Certification

NPDES MSGP permit coverage may be waived if the potential discharger can certify that a condition of "no exposure" exists at the facility. A condition of no exposure exists when all industrial materials and activities (that would typically require MSGP coverage) are protected by a storm-resistant shelter to prevent exposure to rain, snow, snowmelt, and/or runoff (40 CFR §122.26 (g)). A No Exposure Certification Form must be completed for each facility and submitted to the NPDES permitting authority as well as the MS4 operator, where applicable. A condition of no exposure must be maintained at the facility for the exclusion to remain applicable.

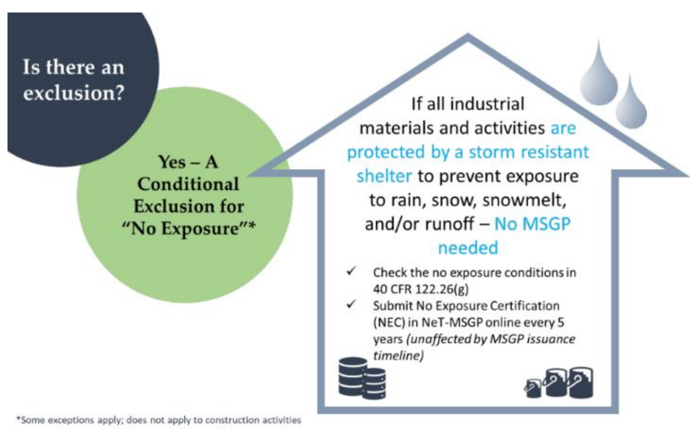


Figure 2-1. No Exposure Certification

2.2.2 Individual Industrial Storm Water Permit Examples

Some activities that fall outside of the 29 industrial permitting sectors may require an Individual NPDES Permit. Several examples of when individual coverage may apply are discussed in the following sections. Additionally, there are some allowable non-storm water discharges such as fire hydrant/water line flushings, irrigation drainage, and exterior building wash water that does not contain detergents. This list is not exhaustive, and it may be necessary to ***consult the appropriate storm water SME for any questions related to permitting requirements at your unit.***

2.2.2.1 Air Compressor Blowdown

Air compressors are an integral part of the USCG mission support. In the process of operation of air compressors, excessive water is a by-product when humid air is compressed. Air compressor blowdown and discharge water may contain lubricating oil or other potential pollutants such as metals. Hydrocarbons can potentially contaminate surface and groundwater when improperly managed. Refer to the EPC-SW-05 – COMPRESSOR BLOWDOWN DISCHARGE ACTIVITIES in Part B of this Guide for storm water management measures for compressor blowdown discharges.

2.2.2.2 Potable Drinking Water Systems

The source of potable water for the majority of USCG units is from municipal water supplies. Most municipalities make potable water available to both residential, commercial, and government users. In some cases, the USCG is not located within a municipal boundary and must rely upon either wells or a water treatment facility. Both are described below:

Potable Water Treatment Facility

Although only a few exist within the USCG, these systems may contain multi-stage treatment systems that rely on both physical and chemical treatment of water before being pumped to potable water users. These systems may include a type of coagulation/flocculation that utilizes chemicals such as alum that promote settling as well as filtration to remove additional solids. This process concludes with a form of disinfection such as chlorination or ultraviolet light (UV) to remove impurities/pathogens.

Well Water

Well water is much more prominent in the USCG at non-urban locations. These wells are typically housed within a small structure large enough to hold a pump and other small appurtenances. In some cases, a small amount of chemical treatment occurs, mainly in the form of chlorine or other disinfectant.

2.2.2.3 Swimming Pools

Swimming pools and hot tubs exist at various units of all types where swimming training is a key component to the mission. These swimming pools are located in both indoor and outdoor environments, depending on the local climate.

Water from pools often contains high levels of chlorine. Discharging chlorinated pool/spa water into waterbodies is harmful to fish, microorganisms, vegetation, and other aquatic life. The discharge of any sewage or industrial waste, including swimming pool water, to waters of the United States without a permit is a violation of the CWA. The complete contents of swimming pools are seldom discharged. Outdoor swimming pools are usually discharged only at the end of summer.

Two separate types of discharge should be regulated from swimming pools: backwash water disposal and pool water disposal. Refer to EPC-SW-15 – SWIMMING POOLS in Part B of this Guide for storm water management measures related to swimming pool discharges.

2.2.2.4 Snow Storage Management

Snow storage management is a concern for most cold climate states. As snow melts, road salt, sand, litter, and other pollutants are transported into local surface water or through the soil where they may eventually reach the groundwater. Road salt and other pollutants can contaminate water supplies and are toxic to aquatic life at certain levels. Sand washed into surface waters can create sediment deposits, sand bars, or fill in wetlands and ponds, impacting aquatic life, causing flooding, and affecting our use of these resources.

There are several steps that the USCG can take to minimize the impacts of snow removal and storage on public health and the environment. These steps will help the USCG avoid the costs of a contaminated water supply, degraded waterbodies, and potential flooding. The purpose of these storm water control measures is to help the USCG select, prepare, and maintain appropriate snow storage sites before the snow begins to accumulate through the winter. Refer to EPC-SW-14 – SNOW STORAGE MANAGEMENT in Part B of this Guide for storm water management measures regarding snow storage management.

2.2.2.5 Wastewater Treatment Plants (WWTPs)

WWTPs employ the process of removing contaminants from wastewater and household sewage, including domestic, commercial, and industrial. It includes physical, chemical, and biological processes to remove physical, chemical, and biological contaminants. Its objective is to produce an environmentally safe fluid waste stream (or treated effluent) and a solid waste (or treated sludge) suitable for disposal or reuse (usually as farm fertilizer). Sewage treatment generally involves three stages, called primary, secondary and tertiary treatment.

Most wastewater facilities or activities are required to obtain an individual NPDES permit for a point source discharge which includes permit requirements and conditions tailored to the specific wastewater treatment and disposal systems regulated in the permit. This includes decentralized wastewater treatment systems, commonly referred to as a "packaged plant". Wastewater permits generally contain requirements for, depending on the type of facility and disposal means, the treatment of the wastewater, disposal to surface water (NPDES), discharge to ground water, the land-application of reclaimed water, the beneficial use of reclaimed water (e.g., landscape irrigation), influent and effluent monitoring and reporting, and, in the case of a domestic wastewater facility, industrial pretreatment, and domestic residuals management.

There are currently more than 10 WWTPs operated by the USCG. These WWTPs have specific permitting requirements required by the individual states in which they exist. In addition to the individual point source permit, facilities containing a WWTP must also have an NPDES storm water permit if the treatment capacity exceeds 1.0 million gallons per day (mgd). The units that currently operate a WWTP are listed in **Attachment F – Wastewater Treatment Plant Operations**.

Refer to EPC-SW-17 – WASTEWATER TREATMENT PLANT OPERATIONS for storm water management measures regarding WWTPs.

2.2.2.6 Cooling Tower Discharges

Cooling towers are an integral component of many refrigeration systems, providing comfort or process cooling across a broad range of applications. They are the point in the system where heat is dissipated to the atmosphere through the evaporative process. Applications most common to USCG units are heating, ventilation, and air circulation (HVAC) cooling towers. Figure 2-2 shows an example of an HVAC cooling tower located at a USCG Station.

Figure 2-2. HVAC Cooling Tower

When water evaporates from the tower, dissolved solids (such as calcium, magnesium, chloride, and silica) are left behind. As more water evaporates, the concentration of dissolved solids increases. If the concentration gets too high, the solids can cause scale to form within the system or the dissolved solids can lead to corrosion problems. The concentration of dissolved solids is controlled by blowdown. Anti-fouling/anti-scaling agents are sometimes added to cooling tower make-up water to reduce the frequency of blowdowns. These agents may be acidic or alkaline, which affects the pH of the water and can also be harmful if discharged to the environment.

Similar to compressors, the blowdown discharge from these cooling towers should not be directly plumbed to the storm sewer. These discharges should be either 1) plumbed to the sanitary sewer via permit or 2) allowed to discharge into a grassy area where the water can evaporate or infiltrate into the soil *if* the discharge has been characterized and found to be free of petroleum, oils and lubricants (POLs) and any other pollutants.

2.2.2.7 Underground Injection Wells

The installation and use of Underground Injection Control (UIC) Class V Injection Wells to infiltrate storm water will require individual permit coverage. Storm Water Drainage Wells are regulated through the UIC program Class V injection well programs as requirements to protect underground sources of drinking water (USDWs). Class V storm water drainage wells are typically shallow wells designed to place rainwater and melted snow below the land surface. UIC Class V injection wells may provide a pathway for contaminates to reach subsurface drinking water sources and should not be used unless BMPs have been developed to mitigate potential for such harm.

Units with septic systems used in an industrial setting are typically required to meet UIC Program requirements and are considered a Class V well if either one of the following conditions are met:

- The septic system, regardless of size, receives any amount of industrial or commercial wastewater (also known as industrial waste disposal wells or motor vehicle waste disposal wells); or
- The septic system receives solely sanitary waste from multiple family residences or a non-residential establishment and has the capacity to serve 20 or more persons per day (also known as large-capacity septic systems).

2.2.2.8 Wash Racks, Refueling Stations, and Oil-Water Separators

Water quality structural control systems such as wash racks, refueling stations, and oil-water separators (OWS) may require individual permits under multiple compliance programs. The intended use of OWSs determines whether the OWS is subject to the SPCC, wastewater, or storm water regulations.

If the intended use of the OWS is for oil recovery then the OWS is subject to SPCC regulations, while the water discharges maybe subject to either storm water (i.e., via retention/detention ponds to surface water) or wastewater regulations (i.e., connected to POTWs).

OWSs associated with wash rack activities may be subject to storm water regulations; while OWSs associated with refueling operations would be subject to SPCC provisions. Combined wash rack and refueling systems would be subject to both sets of regulations; or if connected to a POTW then subject to wastewater treatment standards and SPCC provisions.

Additional details on oil-water separators and wash racks can be found in **Attachment E – Advanced Structural BMPs.**

Additional details on fueling, defueling, and fuel storage can be found in EPC-SW-09 – FUELING, DEFUELING & FUEL STORAGE, EPC-SW-09A – MOBILE REFUELING, and EPC-SW-09B – BULK FUEL FARMS.

2.2.2.9 Hydrostatic Testing and Excavation Dewatering

Hydrostatic Test Water Discharge - Discharging of hydrostatic test water will likely require a general or individual permit from the authorized State in which the discharge is located within or the EPA. States have varying approaches to permitting this discharge so consult with storm water SMEs at your servicing Sector, CEU, or SILC EMD to determine the best way forward. Additional details on hydrostatic testing discharge can be found in EPC-SW-06– CONSTRUCTION SITES FOR SMALL PROJECTS.

Construction Excavation Dewatering Discharge - EPA's Construction General Permit (CGP) defines dewatering as "the act of draining accumulated storm water from building foundations, vaults, and trenches, or other similar points of accumulation" (CGP, Appendix A). The CGP includes uncontaminated construction dewatering water as an authorized (i.e., allowed) non-storm water discharge, provided the operator satisfies the conditions specified in Part 2.4 of the CGP.

Dewatering will likely require a general or individual permit from the authorized State or EPA. States have varying approaches to permitting this discharge so consult with storm water SMEs at your servicing Sector, CEU, or SILC EMD to determine the best way forward. Additional details on excavation dewatering discharges can be found in EPC-SW-06– CONSTRUCTION SITES FOR SMALL PROJECTS.

2.2.2.10 Firefighting Training Areas

Firefighting training burn pits are often associated with commingling of storm water and point source wastewater generated from training activities. Commingling of firefighting wastewater often happens in retention/detention pond systems designed to address velocity dissipation issues associated with runoff. Chemicals in firefighting foam may contain contaminants that are harmful to the environment. Please ***consult the appropriate storm water SME*** for any questions related to firefighting training areas.

2.3 SMALL MUNICIPAL SEPARATE STORM SEWER SYSTEMS (MS4S)

The NPDES program under the CWA was initially developed to reduce pollutants from industrial and municipal sewage discharges but storm water runoff, a major source of water pollution, was not covered under the program. In 1987, Congress drafted Section 402(p) of the CWA, which added storm water discharge into the NPDES program. In 1990, the EPA issued the Phase I Storm Water Rules which regulated storm water runoff from the following:

- "Medium" and "large" municipal separate storm sewer systems (MS4s) generally serving populations of 100,000 or greater,
- construction activity disturbing 5 acres of land or greater, and
- ten categories of industrial activity.

In 1999, EPA issued the Phase II Storm Water Rule to expand the requirements to small MS4s and construction sites between one and five acres in size. Small MS4s are defined as any MS4 that does not meet the definition of a Large or Medium MS4 including Federally owned military bases.

Only a select sub-set of small MS4s, referred to as *regulated* small MS4s, are covered by the Phase II Final Rule. Small MS4s are designated as a *regulated* small MS4 in one of three ways:

1. Automatic Nationwide Designation (An urbanized area (UA) is a densely settled core of census tracts and/or census blocks that have population of at least 50,000, along with adjacent territory containing non-residential urban land uses as well as territory with low population density included to link outlying densely settled territory with the densely settled core. It is a calculation used by the Bureau of the Census to determine the geographic boundaries of the most heavily developed and dense urban areas.)
2. Potential Designation by the NPDES Permitting Authority – Required Evaluation (Designation Criteria EPA recommended that the NPDES permitting authority use a balanced consideration of the following designation criteria on a watershed or other local basis: ⎱ Discharge to sensitive waters; ⎱ High population density; ⎱ High growth or growth potential; ⎱ Contiguity to a UA; ⎱ Significant contributor of pollutants to waters of the United States; and ⎱ Ineffective protection of water quality concerns by other programs.)
3. Potential Designation by the NPDES Permitting Authority – Physically Interconnected (Physically interconnected means that one MS4 is connected to a second MS4 in such a way that it allows for direct discharges into the second system)

What is an MS4?

According to 40 CFR 122.26(b)(8), *"municipal separate storm sewer means a conveyance or system of conveyances (including roads with drainage systems, municipal streets, catch basins, curbs, gutters, ditches, man-made channels, or storm drains):*

(i) Owned or operated by a State, city, town, borough, county, parish, district, association, or other public body (created by or pursuant to State law)...including special districts under State law such as a sewer district, flood control district or drainage district, or similar entity, or an Indian tribe or an authorized Indian tribal organization, or a designated and approved management agency under section 208 of the Clean Water Act that discharges into waters of the United States.

(ii) Designed or used for collecting or conveying storm water;

(iii) Which is not a combined sewer; and

(iv) Which is not part of a Publicly Owned Treatment Works (POTW) as defined at 40 CFR 122.2."

Storm water discharges from MS4s in urbanized areas are a concern because of the high concentration of pollutants found in these discharges. Concentrated development in urbanized areas substantially increases impervious surfaces, such as city streets, driveways, parking lots, and sidewalks, on which pollutants from concentrated human activities settle and remain until a storm event washes them into nearby storm drains. Common pollutants include pesticides, fertilizers, oil and grease, salt, cigarette butts, plastic bottles, other litter and debris, and sediment. Another concern is the possible illicit connections of sanitary sewers, which can result in fecal coliform bacteria entering the storm sewer system. Storm water runoff picks up and transports these and other harmful pollutants then discharges them – untreated – to waterways via storm sewer systems. When left uncontrolled, these discharges can result in fish kills, the destruction of spawning and wildlife habitats, a loss in aesthetic value, and contamination of drinking water supplies and recreational waterways that can threaten public health.

In accordance with EPA guidance, the term MS4 does not solely refer to municipally owned storm sewer systems, but also has a much broader application that can include facilities like USCG installations. An MS4 also is not always just a system of underground pipes – it can include roads with drainage systems, gutters, and ditches.

Certain larger USCG facilities that meet the definition of a small MS4 have been designated as a requiring a small MS4 permit. This requirement stems from the population density of the facility as well as the operations at the USCG facility that resemble the operations of a small city. At several USCG Training Centers (TRACENs), these facilities have residential housing, grocery stores, fuel stations/garages, restaurants, and other facilities. For several of the units that are already in a municipality that is covered by an MS4 (either as a Phase I or Phase II), they still require compliance with the MS4 regulations if they independently operate their storm water system.

It is important to note that units *may* be required to comply with a surrounding jurisdiction's MS4 permit if they are geographically located within a jurisdiction designated as such. For example, Base New Orleans is located within a complex owned by NASA and must follow NASA's permit requirements.

One of the first steps upon gaining permit coverage under an MS4 permit is the implementation of a Storm Water Management Plan outlining the facility's compliance with the six minimum control measures (40 CFR §122.34(b)) that are listed below. These six minimum controls differ from the Industrial permits in that they deal with non-point source pollution rather than point-source pollution issues.

The minimum controls are as follows:

1. ***Public Education and Outreach***

   Distributing educational materials and performing outreach to inform citizens about the impacts polluted storm water runoff discharges can have on water quality.

2. ***Public Participation/Involvement***

   Providing opportunities for citizens to participate in program development and implementation, including effectively publicizing public hearings and/or encouraging citizen representatives on a storm water management panel.

3. ***Illicit Discharge Detection and Elimination***

   Developing and implementing a plan to detect and eliminate illicit discharges to the storm sewer system (includes developing a system map and informing the community about hazards associated with illegal discharges and improper disposal of waste).

4. ***Construction Site Runoff Control***

   Developing, implementing, and enforcing an erosion and sediment control program for construction activities that disturb 1 acre or more of land (controls could include silt fences and temporary storm water detention ponds).

5. ***Post-Construction Runoff Control***

   Developing, implementing, and enforcing a program to address discharges of post-construction storm water runoff from new development and redevelopment areas. Applicable controls could include preventative actions such as protecting sensitive areas (e.g., wetlands) or the use of structural best management practices (BMPs), such as grassed swales or porous pavement.

6. ***Pollution Prevention/Good Housekeeping***

   Developing and implementing a program with the goal of preventing or reducing pollutant runoff from municipal operations. The program must include municipal staff training on pollution prevention measures and techniques (e.g., regular street sweeping, reduction in the use of pesticides or street salt, or frequent catch-basin cleaning).

2.4 CONSTRUCTION GENERAL PERMIT (CGP)

Storm water runoff from construction sites may carry sediment from exposed cleared and graded areas. Sedimentation can cause severe impacts on local waterbodies, especially small streams. During storms, sediment-laden runoff may cause habitat degradation for fish and other aquatic species, the loss of reservoir storage capacity, and negative impacts on the navigational capacity of waterways among other issues. In addition to sediment, activities that commonly occur on a construction site may yield pollutants such as pesticides, petroleum products, solvents, and chemicals that can contaminate storm water runoff.

The NPDES storm water program requires construction site operators engaged in clearing, grading, and excavating activities that disturb 1 acre or more, including smaller sites in a larger common plan of development or sale, to obtain coverage under an NPDES CGP for construction storm water discharges. "Disturbance" refers to exposed soil resulting from clearing, grading, and excavating. A "larger common plan of development or sale" means a contiguous area where multiple separate and distinct construction activities are occurring under one plan (e.g., the operator is building on multiple lots less than 1 acre individually that will cumulatively exceed 1 acre). A "plan" is broadly defined as any type of documentation or physical demarcation indicating that construction activities may occur in a specific or "common" area.

It should be noted that some construction activities may require an individual permit. CG unit CO/OICs/FE/EOs should communicate and/or collaborate with CEUs to ensure construction activities have necessary NPDES permits in place prior to commencement of ground disturbing activities.

NPDES permitting authorities have the option of providing a waiver to operators of small construction activity (land disturbance equal to or greater than 1 acre and less than 5 acres) who <u>certify</u> to either one of the two following conditions:

1. Low predicted rainfall potential, where the rainfall erosivity factor is less than 5 during the period for construction activity. The rainfall erosivity factor is one of six variables used by the Revised Universal Soil Loss Equation (RUSLE) – a tool used to predict soil loss from construction sites. *OR*

2. A determination that storm water controls are not necessary based on either:

    a. A total maximum daily load (TMDL) that addresses the pollutant(s) of concern, *OR*

    b. An equivalent analysis that determines allocations are not needed to protect water quality based on consideration of in-stream concentrations, expected growth in pollutant concentrations from all sources, and a margin of safety.

<u>USCG units are responsible for ensuring that their general contractors have obtained the proper CGP coverage and that the proper construction structural BMPs are placed and maintained throughout the life of the construction project.</u> Structural BMPs require regular maintenance to operate correctly. Accumulated sediments should be removed frequently, and materials should be checked periodically for wear. Regular inspections should be performed after all major rain events. Uncontrolled runoff from construction sites is a water quality concern because that sedimentation can have on local waterbodies, particularly small streams. Furthermore, construction activities can yield pollutants such as pesticides, petroleum products, chemicals, solvents, asphalts and acids that can contaminate storm water. If left uncontrolled, sediment-laden runoff from construction sites can result in loss of in-stream habitat, difficulty in filtering drinking water, the loss of reservoir storage capacity, and negative impacts on the navigational capacity of waterways.

The NPDES permitting authority will define the specific requirements for storm water controls on small construction activity on a state-by-state basis. In general, the development of a Storm Water Pollution Prevention Plan (SWPPP) with appropriate BMPs to minimize discharge of pollutants from the construction site will be required for coverage under the CGP. Refer to **EPC-SW-06 – CONSTRUCTION SITES FOR SMALL PROJECTS** for effective storm water management controls to implement on small construction sites.

3. STORM WATER MANAGEMENT PRACTICES

The USCG has developed ENVIRONMENTAL PROCEDURE CARDS (EPCS) to be utilized enterprise-wide for those units with operations or activities that have the potential to impact storm water. The EPCs, found in Part B of this Guide, have been organized by activity for ease of use and have been designed to assist units by providing best practices to control and prevent unallowable storm water discharges. Unit personnel responsible for each activity covered by an EPC in Part B of this Guide should have or obtain the appropriate EPC for that activity. **For quick and easy reference, a copy of the EPC should also be posted in the designated physical area where the activity occurs**. Attachment E – **Advanced Structural BMPs** includes a discussion of advanced structural BMPs, which consist of proprietary engineered systems that may be installed to remove pollutants from storm water runoff or prevent pollutants from entering storm water. These may be used when needed and resources allow. Structural BMPs typically require regular inspection and maintenance.

Regardless of required permit coverage, each unit is required to implement storm water control measures to minimize the discharge of pollutants in runoff from the site. The basic operational control measures listed below for Maintenance, Employee Training, and Illicit Discharges should be implemented at every facility.

3.1 OPERATIONAL CONTROL MEASURES

Maintenance

- ☑ Storm water drains should be inspected regularly and cleaned at least once annually (or more often if required by permit). Sediment in the bottom of a storm drain should NOT be allowed to accumulate a depth of more than 2 inches, as it may compromise performance and increase turbidity and/or total suspended solids (TSS) at the discharge point.
- ☑ Oil/water separators (OWS) should be inspected, maintained, and cleaned annually (or more often if required by permit) to avoid sediment buildup or potential operational failure.
- ☑ Grease traps should be inspected semi-annually and cleaned annually (or more often if required by permit) to avoid buildup or potential operational failure.
- ☑ Spill response supplies should be inspected monthly to ensure that material levels are maintained and ready for use. Spill response supplies should be replaced immediately after each use.

Employee Training

- ☑ Train all employees who work in areas where industrial materials or activities are exposed to storm water, or who are responsible for implementing activities necessary to meet the conditions of an NPDES permit. Training should be conducted at least annually or in accordance with the applicable permit.

Illicit Discharges

The following policies are to be implemented to eliminate the potential for illicit connections to the storm sewer system:

- ☑ Plug all floor drains known to be connected to the storm sewer system or if the connection is unknown. Consult the appropriate CEU for further engineering options, which may be subject to permitting requirements.

☑ Perform smoke or dye testing to determine connections to the sanitary sewer system or the storm sewer system when/if a drainage problem is perceived (e.g., restriction in flow, backup of drains).

☑ Maintain facility schematics to reflect all connections accurately. Identify and annotate on existing site plan any known inaccuracies and physical changes at the facility until an updated site map can be obtained.

3.2 ENVIRONMENTAL PROCEDURE CARDS

Part B of this guide presents the environmental procedure cards (EPC) for storm water pollution prevention associated with typical USCG unit operations in more detail, organized by activity (See Table 3-1). Per USCG policy, these control measures shall be implemented at each facility that performs the relevant activity.

Table 3-1. Environmental Procedure Card Selection Guide by Activity

Activity	Number
Aircraft De-Icing Activities	01
Aircraft Maintenance and Repair Activities	02
Boat and Vessel Maintenance and Repair	03
Buoy Operations	04
Compressor Blowdown Discharge Activities	05
Construction Sites for Small Projects	06
Dumpsters and Solid Waste Management	07
Engine Flushing	08
Fueling, Defueling, and Fuel Storage Activities	09
Mobile Refueling	09A
Bulk Fuel Farms	09B
Grounds Maintenance	10
Hazardous Materials and Waste Storage Activities	11
Outdoor Storage of Bulk Materials	12
Small Arms Firing Ranges	13
Snow Storage Management	14
Swimming Pools	15
Boat and Vessel Washing Operations	16A
Aircraft Washing Operations	16B
Vehicle and Equipment Washing Operations	16C
Pressure Washing Activities	16D
Wastewater Treatment Plant Operations	17

3.3 SPECIAL CONSIDERATIONS

This section discusses several water quality considerations each unit should be aware of and factor in if applicable. Additionally, some states differ slightly in how they expect some activities to be conducted. **Attachment D – Special Considerations by State** provides a list of these differences.

3.3.1 Invasive Species Management

An invasive species is defined in EO 13112 as an alien (or non-native) species whose introduction does or is likely to cause economic or environmental harm or harm to human health. Invasive species occur when non-native species are introduced to an ecosystem in which they did not evolve. In some cases, the populations of these non-native species grow in a way that is not sustainable for the ecosystem, causing the native species to struggle to survive. The invasive species may, in some cases, rival the native species and cause a disruption in the native communities. If the native species are eliminated, the ecosystem becomes less diverse and can collapse. Two management procedures are ballast water exchange (BWE) and the development of boat, trailer, and asset inspection protocol. BWE is mentioned here for informational purposes only. This Guide will focus on boat, trailer, and asset washing operations since these procedures have the potential to affect storm water.

The USCG prohibits ships to discharge untreated ballast water in waters of the United States. BWE requires that ships on their way to the next port release the lower-salinity coastal water they brought aboard in their last port and replace it with higher-salinity open-ocean water. Information regarding BWE requirements can be found at https://www.dco.uscg.mil/Our-Organization/Assistant-Commandant-for-Prevention-Policy-CG-5P/Commercial-Regulations-standards-CG-5PS/Marine-Safety-Center-MSC/Ballast-Water/. Ballast water should never be incorporated into a storm water discharge.

All watercraft, deployable assets (including buoys and chain), and response gear (e.g., booms, skimmers) used in water are potential carriers of aquatic nuisance species, and inspections are essential to the prevention of aquatic nuisance species transport. To help reduce and prevent the spread of invasive species, all vessels should go through a decontamination process before being reintroduced into another body of water. The inspection and decontamination procedures associated with the management of invasive species can be found under the **EPC-SW-08 – ENGINE FLUSHING** and **EPC-SW-16 – WASHING OPERATIONS** in Part B of this Guide.

3.3.2 Total Maximum Daily Loads (TMDLs)

Under Section 303(d) of the CWA, states, territories, and authorized tribes are required to develop lists of impaired waters. These are waters for which technology-based controls required by NPDES permits are not achieving state water quality standards. The law requires that states establish priority rankings for waters on the lists and develop TMDLs. The TMDL process establishes the maximum quantity of pollutants a waterbody can assimilate before water quality is impaired, then requires that this maximum level not be exceeded. A TMDL is established for each pollutant that is found to be contributing to the impairment of a waterbody or a segment of a waterbody. Pollutants that originate from a point source are given allowable levels of contaminants to be discharged; this is the Waste Load Allocation (WLA).

The CWA includes two basic approaches for protecting and restoring the nation's waters. One is a technology-based approach whereby EPA promulgates effluent guidelines that rely on available technologies to remove pollutants from waste streams. These effluent guidelines are used to derive NPDES permit limits. The other approach is water-quality based and may ultimately result in more stringent NPDES permit limits. The Section 303(d) program is at the core of the water quality-based approach and serves to link water quality goals and NPDES permit limits.

Section 303(d) of the CWA does not specifically require implementation plans for TMDLs; however, it requires that WLAs be implemented through the NPDES permit program. After a TMDL has been developed, water quality-based discharge limits in NPDES permits must be consistent with the assumptions and requirements of the WLA.

Industrial activities can generate a number of pollutants in storm water that can lead to exceedances of water quality standards, and construction activities typically generate sediment, which can affect related parameters such as turbidity and TSS. Depending on the TMDL, facilities may need to install additional storm water control measures that are specifically designed to achieve reduction in the pollutant(s) causing the impairment to the receiving water. Any additional requirements for TMDL watersheds will be included in the NPDES permit. Facilities should consult permit requirements pertaining to discharges to impaired waters in the MSGP and/or CGP, whichever applies to the facility.

Significant non-compliance (SNC) violations associated with exceedances in TMDLs can result in Order of Compliance action and/or fines. Typically, a SNC violation can be resolved with the change in operation or implementation of a BMP. The magnitude of the violation will determine the enforcement action. A “one-size-fits-all” policy for response to violations is difficult to implement since the violations can vary vastly from one area to another. An example of an Order of Compliance is provided in Section 3.3.3 below.

3.3.3 Special Requirements for Chesapeake Bay

The EPA has established a Chesapeake Bay TMDL with rigorous accountability measures to restore clean water in the Bay and the region’s streams, creeks, and rivers. The TMDL encompasses a 64,000 square mile watershed that includes the District of Columbia and large sections of Delaware, Maryland, New York, Pennsylvania, Virginia, and West Virginia. Specifically, the TMDL sets Bay watershed limits for nitrogen, phosphorus, and sediment per year. The pollution limits are further divided by jurisdiction and major river basins.

USCG facilities located within the Chesapeake Bay watershed and operating wastewater treatment plants may be subject to effluent loading limits and monitoring requirements for nitrogen and phosphorus. Table 3-2 includes a listing of the USCG units that are geographically located within the Chesapeake Bay watershed.

Table 3-2. Units Located within Chesapeake Bay Watershed

Type	Unit	Latitude/Longitude
Station	Annapolis	38.92114, -76.681317
Station	Cape Charles	37.263914, -76.014313
Station	Crisfield	37.974998, -75.858822
Sector Field Office	Eastern Shore	37.930776, -75.381510
Station	Inigoes	38.153936, -76.427937
Station	Little Creek	36.910291, -76.176925
Station	Milford Haven	37.487348, -76.308739
Station	Oxford	38.678617, -76.170322
Base	Portsmouth	36.52829, -76.216460
Port Security Unit	PSU 305	37.15063, -76.58721
CAMSLANT	Pungo	36.910291, -76.176925
Station	Stillpond	39.336582, -76.130785
Yard	USCG Yard	39.20503, -76.56918
Station	Wachapreague	37.605662, -75.687473
TRACEN	Yorktown	37.210348, -76.484990

3.4 REGULATORY INSPECTIONS

3.4.1 USEPA Inspections

Regulatory inspections are an integral part of USEPA's enforcement operations. Inspections include a physical site inspection and are usually focused at one or two specific media areas, including storm water. Regulators may provide ample notification of the site visit in advance, but they may also arrive onsite without notice.

Like ECEs, USEPA regulatory inspections usually include pre-inspection activities, such as obtaining important facility documents and information. The site visit may include personnel interviews, record reviews, taking photographs of violations, collection of samples, and/or observing base operations. Violations discovered through these inspections are subject to civil or criminal penalties and enforcement, depending on severity.

EPA's primary enforcement authorities are set forth in CWA § 309. Section 313 requires federal agencies to comply with the CWA and therefore, EPA can enforce against federal agencies for violations, even if penalties are not an option. USCG legal assistance will not be provided to members charged with violating environmental regulations. The regulations are not subject to interpretation nor is ignorance of the regulations a valid defense.

3.4.2 State Inspections

State agencies may also conduct announced or unannounced inspections. These inspections can likewise include multiple environmental media areas. States that are authorized to enforce the CWA may conduct storm water inspections. Other potential state-authorized inspection programs include ASTs and USTs, wastewater, hazardous waste, and air emissions.

Authorized States can issue administrative compliance orders or take civil judicial action against violators of CWA provisions, including Federal facilities. However, states cannot recover penalties for past violations against Federal facilities. States can also file a criminal action against a Federal employee that may result in significant fines and/or prison sentences.

3.4.3 Internal Reporting of Environmental Enforcement Actions (EEAs)

Environmental Enforcement Actions (EEAs) include any notification by the U.S. Environmental Protection Agency or other authorized federal, state, local, or regional environmental regulatory authority of an alleged violation of an applicable environmental requirement. If a unit is inspected and receives any storm water violations (i.e. NOVs) and/or corrective actions, ALCOASTNOTE (ACN 005/22) outlines procedures for EEA notification to CG-47. Operational unit CO/OICs are responsible for responding to potential enforcement actions and following up with closure of enforcement actions with the regulatory agency. Units should seek assistance from LSC, SILC, and CEU for SME support where needed.

The unit CO/OIC is responsible for submitting the following email notifications of the violation and corrective actions to CG-47, as follows:

(1) EEA Notification – Notify CG-47 at EnvironmentalCompliance@uscg.mil immediately upon receipt of the EEA and include the following:

- Subject Line: "EEA Initial Notification for [insert unit name]"

- Electronic copy of the EEA
- Issuance date of the EEA
- Summary of the alleged violation(s)
- Description of any corrective action already taken or planned
- Amount of any issued or expected monetary fines or "TBD" if unknown
- Expected completion date of the corrective action

(2) Resolution Notification – Within five business days of completing corrective action to resolve the EEA, email CG-47 the following:

- Subject Line: "EEA Resolution Notification for [insert unit name]"
- Date resolution achieved
- Description of corrective action undertaken
- Updated monetary fine/penalty dollar amount
- Description of the expected, initiated or finalized regulatory agency response

(3) Closure Notification – Within five business days of receipt of the official EEA closure documentation, email CG-47 the following:

- Subject Line: "EEA Closure Notification for [insert unit name]"
- Electronic copy of the EEA written notification of closure
- Final dollar amount of any civil monetary penalty assessed as a result of the EEA

4. REFERENCES

40 CFR § 112.8. Spill Prevention, Control, and Countermeasure Plan requirements for onshore facilities (excluding production facilities)

40 CFR § 122.2. Definitions

40 CFR § 122.26. Storm water discharges (applicable to State NPDES programs, see § 123.25)

Occupational Safety and Health Administration. *Standard Industrial Classification (SIC) Manual*. August 6, 2013. https://www.osha.gov/data/sic-manual

U.S. EPA. Authorization Status for EPA's Construction and Industrial Storm Water Programs. Accessed online May 2022. https://www.epa.gov/npdes/authorization-status-epas-construction-and-industrial-stormwater-programs

U.S. EPA. NPDES MSGP for Storm Water Discharges Associated with Industrial Activity Fact Sheet. Accessed online May 2022. *https://www.epa.gov/sites/default/files/2021-01/documents/2021_msgp_-_fact_sheet.pdf*

U.S. EPA, Office of Wastewater Management. *NPDES Storm Water Multi-Sector General Permit for Industrial Activities (MSGP)*. March 2021. https://www.epa.gov/npdes/stormwater-discharges-industrial-activities-epas-2021-msgp

U.S. EPA. Chesapeake Bay TMDL. Accessed online May 2022. https://www.epa.gov/chesapeake-bay-tmdl

U.S. EPA. National Menu of Best Management Practices (BMPs) for Storm Water. October 2000. Accessed online May 2022. https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater

PART B – ENVIRONMENTAL PROCEDURE CARDS

Environmental Procedure Cards:

EPC-SW-01 – Aircraft De-Icing Activities

EPC-SW-02 – Aircraft Maintenance and Repair

EPC-SW-03 – Boat and Vessel Maintenance and Repair

EPC-SW-04 – Buoy Operations

EPC-SW-05 – Compressor Blowdown Discharge Activities

EPC-SW-06 – Construction Sites for Small Projects

EPC-SW-07 – Dumpsters and Solid Waste Management

EPC-SW-08 – Engine Flushing

EPC-SW-09 – Fueling, Defueling, and Fuel Storage

EPC-SW-09A – Mobile Refueling

EPC-SW-09B – Bulk Fuel Farms

EPC-SW-10 – Grounds Maintenance

EPC-SW-11 – Hazardous Materials and Waste Storage Activities

EPC-SW-12 – Outdoor Storage of Bulk Materials

EPC-SW-13 – Small Arms Firing Ranges

EPC-SW-14 – Snow Storage Management

EPC-SW-15 – Swimming Pools

EPC- SW-16A – Boat and Vessel Washing Operations

EPC- SW-16B – Aircraft Washing Operations

EPC- SW-16C – Vehicle and Equipment Washing Operations

EPC- SW-16D – Pressure Washing Activities

EPC-SW-17 – Wastewater Treatment Plant Operations

AIRCRAFT DE-ICING ACTIVITIES

Aircraft and airfield de-icing can generate storm water pollution if not conducted properly. De-icing fluid that is left on the ground gets picked up by runoff from rain or melting snow/ice and enters the storm water drainage system. The system carries the polluted water to local streams, rivers, and lakes, where the toxic fluid harms or kills aquatic life. The pollution prevention practices outlined in this sheet are critical to preventing storm water pollution from de-icing activities.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from airfield and aircraft de-icing activities from reaching the storm drainage system and local streams, rivers, and lakes	· Air Stations · Bases (with flying operations)	· De-icing chemicals (ethylene, propylene glycol, urea, etc.)

POLICIES AND GUIDANCE

Air stations are generally required to have permit coverage under the National Pollutant Discharge Elimination System (NPDES) program.

If permitted under Sector S - Air Transportation, operators must **track their usage of de-icing chemicals** on an annual basis. It is recommended that permit operators document usage during quarterly inspections to ensure meeting annual reporting goals.

A storm water pollution prevention plan (SWPPP) must be prepared for the air station. Thoughtful development and adherence to the SWPPP is the best safeguard against pollutant discharges from activities performed at the airport or airfield. The SWPPP should include the following best practices:

<u>AIRCRAFT</u>

- Perform de-icing activities in area(s) designated specifically for that purpose, called de-icing pads. The pads are designed to allow runoff from storm water/snow and ice melt to be collected and then treated before discharge, recycled, or properly disposed.
- If possible, store aircraft in hangars prior to flights to avoid de-icing.

STORAGE/EQUIPMENT

- Locate de-icing chemical transfer/storage areas and vehicle/equipment garages indoors. If they must be outdoors, employ the pollution prevention practices for storage/handling.
- Regularly inspect de-icing equipment, vehicles, storage areas, and collection/containment systems for damage that could result in leaks or spills of de-icing chemicals. Damaged equipment should not be used until repaired or replaced.
- Calibrate de-icing equipment to meter the amount being applied.
- Recover de-icing fluids for reuse where possible.
- Store contaminated de-icing fluids in tanks and only release controlled amounts to the wastewater treatment plant (WWTP), as allowed by the local sewer authority/municipality.

RUNWAYS/TAXIWAYS/PARKING AREAS

- De-icing fluid that does not reach fluid collection inlets/systems should be removed using cleaning methods appropriate for the pollutants in the fluid. For example, use granular activated carbon that can help remove pollutants like polyfluoroalkyl substances (PFAS) from combining with storm water runoff. Properly dispose of the waste from fluid cleanup activities.
- Use mechanical vacuum systems or other devices to collect de-icing fluid from the apron for proper disposal.
- Ensure employees are trained on pollution prevention and spill response and cleanup protocols for de-icing activities.

EMERGENCIES
EPA
Report oil or chemical spills at:
800-424-8802

AIRCRAFT MAINTENANCE AND REPAIR ACTIVITIES

Facilities that conduct aircraft maintenance and repair activities shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from aircraft repair and maintenance activities from entering the storm drainage system or streams, rivers, and lakes	· Air Stations · Bases (with flying operations)	· Heavy metals (lead, copper, zinc) · Oil, gasoline, & grease · Toxic chemicals · Trash & debris · Other fluids (antifreeze, solvents, etc.)

POLICIES AND GUIDANCE

Air stations and bases where aircraft maintenance usually occurs are generally required to have permit coverage under the National Pollutant Discharge Elimination System (NPDES) program. This requires development of a storm water pollution prevention plan (SWPPP) for the site. Thoughtful development and adherence to the SWPPP is the best safeguard against pollutant discharges from activities performed at the airport or airfield. The SWPPP should include the following best practices:

GENERAL

- Follow storm water control measures in **EPC-SW-16 - WASHING OPERATIONS** or **EPC-SW-08 - ENGINE FLUSHING** for aircraft washing or engine flushing.
- For de-icing, follow storm water control measures in **EPC-SW-01 – AIRCRAFT DE-ICING ACTIVITIES**.
- When possible, eliminate floor drains that are connected to the storm or sanitary sewer. If your floor drain is connected to a storm sewer, ditch, or discharges to a stream or other waterbody or directly to the ground, you must have a National Pollutant Elimination System (NPDES) permit issued from your state environmental agency or EPA. If necessary, install a sump that is pumped/cleaned regularly. If your floor drain is sealed or plugged and does not discharge at all, then collected wastes should be properly treated or disposed of by a licensed waste disposal company. Consult with your stormwater SME if you have questions regarding floor drain connections.
- DO NOT pour liquid waste into floor drains *(pictured at right)*, sinks, outdoor storm drain inlets, or other storm drains or sewer connections.

AIRCRAFT MAINTENANCE AREAS

- Prevent and contain spills and drips to the extent possible.
- Minimize exposure to aircraft maintenance areas by performing maintenance activities indoors. If work is performed outdoors, use berms, curbs, etc. to contain storm water runon and runoff. Collect storm water runoff from the maintenance area and/or provide treatment through an oil/water separator (OWS).
- Do not conduct activities that generate dust or debris outside. If done outside capture all for material proper disposal.
- Provide a centralized station for cleaning so the solvents stay in one area.
- Use drip pans, drain boards, and drying racks to direct drips back into a fluid holding tank for reuse.
- Drain all parts of fluids prior to disposal. Hot drain (drain oil close to or at engine operating temperature) all oil filters; crush and recycle.
- Transfer used fluids to the proper container promptly. Do not leave full drip pans or other open containers around the maintenance area. Empty and clean drip pans and containers.
- Clean up leaks, drips, and other spills without using water. Use absorbents for dry cleanup whenever possible.
- Label and track the recycling of waste material (e.g., used oil, spent solvents, batteries).
- Store batteries and other significant materials inside.
- DO NOT pour liquid waste into floor drains, sinks, outdoor storm drain inlets, or other storm drains or sewer connections.
- Label all materials and waste storage containers with the type of material contained therein (e.g., used oil, hydraulic fluids). Store materials and waste indoors or under a roof within secondary containment to minimize exposure.
- Prohibit the practice of washing down an area where the practice would result in the discharge of pollutants to a storm water system.
- Ensure that employees are properly trained in maintenance, storage, de-icing, fueling, and washing techniques.

AIRCRAFT STORAGE - INDOOR

- Store aircraft and ground support equipment indoors when possible or cover the storage area with a roof.
- Store aircraft and ground support equipment awaiting maintenance in designated areas only.
- Use absorbents for dry cleanup of spills and leaks.
- Use drip pans under all vehicles and equipment awaiting maintenance for the collection of fluid leaks.
- Regularly sweep area to minimize debris on the ground.

AIRCRAFT STORAGE - OUTDOOR

- Store aircraft and ground support equipment awaiting maintenance in designated areas only.
- Install perimeter drains, berms, and dikes around storage areas to limit storm water runon in these areas. Do not put in areas where they become a prop hazard.
- Use absorbents for dry cleanup of spills and leaks.
- Use drip pans under all vehicles and equipment awaiting maintenance for the collection of fluid leaks.
- Inspect the storage yard for full or near full drip pans regularly to ensure best management practices (BMPs) are implemented and that no spills occur.
- Regularly sweep area to minimize debris on the ground.

WASTE AND MATERIAL STORAGE

- Maintain good integrity of all storage containers (e.g., used oils, hydraulic fluids, spent solvents, waste/off spec aircraft fuel).
- Create a centralized storage area for waste and material storage.
- Locate storage areas away from high traffic areas and surface waters, when possible.
- Inspect storage tanks and piping systems (pipes, pumps, flanges, couplings, hoses, and valves) for failures or leaks and perform preventive maintenance.
- Maintain an inventory of fluids to be able to quickly identify and characterize leakage.
- Install fluid level indicators where applicable, when possible.
- Provide drip pads/pans where chemicals are transferred from one container to another to allow for recycling of spills and leaks.
- Properly dispose of chemicals with expired shelf-life dates.
- Store and handle reactive, ignitable, or flammable liquids in compliance with applicable local fire codes, local zoning codes, the National Electric Code, and NFPA 30 (Flammable and Combustible Liquids Code, published by the National Fire Protection Association).
- Dispose of greasy rags, oil filters, air filters, batteries, spent coolant, and degreasers in compliance with Resource Conservation and Recovery Act (RCRA) regulations.

BOAT AND VESSEL MAINTENANCE AND REPAIR

Facilities that conduct boat and vessel maintenance and repair shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. When conducting ship repair availabilities for USCG vessels, follow the general requirements described in SFLC Standard Specification SFLC 0000.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from boat and vessel maintenance and repair activities from entering the storm drainage system or streams, rivers, and lakes	· Stations · Sectors · Bases · Yard	· Heavy metals (lead, copper, zinc) · Oil, gasoline, & grease · Toxic chemicals · Trash & debris · Other fluids (antifreeze, solvents, etc.)

POLICIES AND GUIDANCE

PROHIBITIONS

- Prohibit uncontained blasting, sanding, or painting activities near open water or areas that flow to storm water drains.
- Prohibit blasting, sanding, and spray-painting activities during windy conditions that render containment ineffective.

GENERAL

- Follow storm water control measures in **EPC-SW-16A – BOAT & VESSEL WASHING OPERATIONS** or **EPC-SW-08 – MARINE ENGINE FLUSHING** for boat and vessel washing, engine flushing, and invasive species management.
- Avoid conducting repairs of any kind over water when possible. Conduct maintenance and repair work indoors when possible. If work is performed outdoors, use berms, curbs, etc. to contain storm water run-on and runoff.
- Label all materials and waste storage containers with the type of material contained therein (e.g., paints, solvents). Keep labels visible. Store materials and waste indoors within secondary containment and away from doors and drains.
- Ensure that employees are trained in fueling procedures, paint and blasting procedures, used oil management, spent solvent management, disposal of spent abrasives, and wash water disposal.

BILGE AND BALLAST WATER

- Collect and dispose of bilge and ballast waters, which may contain oils, solvents, and detergents, to a licensed waste disposal company in accordance with current regulations and applicable permits. Discharge of bilge and ballast water, pressure wash water, sanitary wastes, and cooling water originating from vessels are prohibited under the general industrial storm water permit. These discharges would require coverage under a separate NPDES permit if discharging to receiving waters or through a municipal separate storm sewer system.

ENGINE AND PARTS STORAGE

- Store covered on an impervious surface to minimize exposure to storm water and use drip pans and absorbent pads to capture/contain oil and grease leaks.

DRYDOCK OPERATIONS

- Use plastic barriers beneath the hull, between the drydock walls, and around the vessel for containment.
- Sweep accessible areas of the drydock to remove and properly dispose of debris and spent sandblasting material prior to flooding. Do not hose debris or solids into surface waters. Clean and maintain the drydock on a regular basis to minimize the potential for pollutants to encounter surface waters. Have absorbent materials and oil containment booms readily available (within 20 feet) to clean and contain spills during drydock operations.
- Collect wash water for treatment and disposal. Remove solids and metals for proper disposal.

SURFACE PREPARATION, SANDING AND PAINT REMOVAL

- Minimize, to the extent possible and practicable, the potential for spent abrasives, paint chips, and overspray to discharge to surface waters or the storm sewer system by confining all sanding, blasting, and painting activities to an enclosed area outside of drainage pathways and away from surface waters. Collect paint chips, blasting debris, and spent abrasives for proper characterization and disposal.
- Use hanging plastic barriers if a dedicated indoor facility is not available.
- Cover drains, trenches, and drainage channels to the storm sewer system to prevent debris from entering.
- Regularly sweep/vacuum the immediate area to remove and properly dispose of debris to minimize exposure to storm/surface waters.

PAINTING

- Disburse paint in small containers in a designated area.
- Spraying outdoors is prohibited. Painting can be done in non-barrier areas where use of roller or brush only is allowed and a drop cloth is used to catch paint drips.
- Have absorbent and cleanup materials readily available (within 20 ft.) for immediate cleanup of spills.
- Use drop cloths under painting areas.

- Store waste paint, solvents, and rags in sealed containers to prevent evaporation.
- Use only solvents with low volatility and coatings with low volatile organic compound (VOC) content.

MATERIAL DISBURSING/MIXING AREAS

- Mix paints and solvents in a designated area away from drains and surface waters, preferably indoors.

IF A SPILL OCCURS:

- þ STOP THE SOURCE OF THE SPILL IMMEDIATELY.
- þ CONTAIN THE SPILL UNTIL CLEANUP IS COMPLETE.
- þ USE OIL CONTAINMENT BOOMS IF THE SPILL MAY REACH SURFACE WATER.
- þ USE ABSORBENT MATERIALS TO CLEAN UP THE SPILL.
- þ DISPOSE OF CLEANUP MATERIALS IN THE SAME MANNER AS THE SPILLED MATERIAL. FOR SPILLED PAINT, DISPOSE OF CLEANUP MATERIAL AS "PAINT WASTE" AND SPILLED MATERIAL AS "WASTE PAINT."
- þ DO NOT USE AN EMULSIFIER OR DISPERSANT.

BUOY OPERATIONS

Facilities that conduct buoy operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from buoy operations and storage from entering the storm drainage system or streams, rivers, and lakes	· Bases · Sectors · Stations · ANTs	· Heavy metals (lead, copper, zinc) · Toxic chemicals · Paint

POLICIES AND GUIDANCE

PROHIBITIONS

- Prohibit uncontained blasting, sanding, or painting activities near open water or areas that flow to storm water drains.
- Prohibit blasting, sanding, and spray-painting activities during windy conditions that render containment ineffective.

GENERAL

- Minimize the potential for spent abrasives, paint chips, and overspray to discharge to surface waters or the storm sewer system by confining all sanding, blasting, and painting activities to an enclosed area outside of drainage pathways and away from surface waters. Use hanging plastic barriers if a dedicated indoor facility is not available. Be aware that air emissions restrictions also exist; consult permits where applicable.
- Collect paint chips, blasting debris, and spent abrasives for proper characterization and disposal; label and store under cover while awaiting disposal.
- Collect spent abrasives from repair work where paint is removed; contain, label, and store spent abrasives under cover to await proper disposal by a licensed waste hauler.
- Collect and properly dispose of wastewater from wet abrasive blasting.
- Avoid rinsing excess concrete forming materials into storm drains or to surface waters.

SURFACE PREPARATION, SANDING, AND PAINT REMOVAL

- Minimize, to the extent possible and practicable, the potential for spent abrasives, paint chips, and overspray to discharge to surface waters or the storm sewer system by confining all sanding, blasting, and painting activities to an enclosed area outside of drainage pathways and away from surface waters. Collect paint chips, blasting debris, and spent abrasives for proper characterization and disposal.
- Use hanging plastic barriers if a dedicated indoor facility is not available.

- Cover drains, trenches, and drainage channels to the storm sewer system to prevent debris from entering.
- Regularly sweep/vacuum the immediate area to remove and properly dispose of debris to minimize exposure to storm/surface waters.

PAINTING

- Use spray equipment that minimizes overspray.
- Have absorbent and cleanup materials readily available (within 20 ft) for immediate cleanup of spills.
- Store waste paint, solvents, and rags in sealed containers to prevent evaporation.
- Use solvents with low volatility and coatings with low volatile organic compound (VOC) content.
- Don't paint outside with high chance of rain.

MATERIAL MIXING AREAS

- Mix paints and solvents in a designated area away from drains and surface waters, preferably indoors.

BUOY CHAIN STORAGE

- Buoy chain must be stored away from surface waters. Store buoy chain within containment (e.g., metal bin) or on raised surface (e.g., wooden pallet) and provide cover (e.g., lid or tarp). Storing chain on pallets will help eliminate additional corrosion and runoff/discharge of rust. If rust debris is present, collect and dispose of properly to prevent stormwater contamination.
- If buoy chain can't be stored under cover, store on a pervious surface away from storm drains to allow for infiltration to ground rather than allowing runoff to discharge to the storm sewer system.
- When possible, provide a grass buffer between chain storage and closest drainage pathway or surface water.

IF A SPILL OCCURS:

- þ STOP THE SOURCE OF THE SPILL IMMEDIATELY.
- þ CONTAIN THE SPILL UNTIL CLEANUP IS COMPLETE.
- þ USE OIL CONTAINMENT BOOMS IF THE SPILL MAY REACH SURFACE WATER.
- þ USE ABSORBENT MATERIALS TO CLEAN UP THE SPILL.
- þ DISPOSE OF CLEANUP MATERIALS IN THE SAME MANNER AS THE SPILLED MATERIAL. FOR SPILLED PAINT, DISPOSE OF CLEANUP MATERIAL AS "PAINT WASTE" AND SPILLED MATERIAL AS "WASTE PAINT."
- þ DO NOT USE AN EMULSIFIER OR DISPERSANT.

COMPRESSOR BLOWDOWN DISCHARGE ACTIVITIES

Facilities that conduct air compressor blowdown discharges shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. Air compressor blowdown water commonly contains lubricating oil or other potential pollutants. These hydrocarbons can contaminate surface and groundwater when improperly managed.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent hydrocarbons from compressor blowdown from entering the storm drainage system or streams, rivers, and lakes	· Stations · Bases · Sectors	· Hydrocarbons (lubricating oil) · Heavy metals (lead, copper, zinc) · Other fluids (antifreeze, solvents, etc.)

POLICIES AND GUIDANCE

Compressor blowdown discharges are considered a non-storm water discharge and are NOT allowed to be put into storm water drainage systems. These discharges may require coverage under a separate NPDES permit if discharging to receiving waters or through a municipal separate storm sewer system. The following storm water management measures should be adhered to:

- Connect blowdown discharge to a floor drain system that is known to discharge to the sanitary sewer system.
- If a sanitary sewer connection is not available:
    - discharge to a containment basin and remove/retain oil prior to discharge to a grassy area away from storm drains, drainage pathways, and surface waters; or
    - discharge to a holding tank for oily waste disposal.
- Waste compressor oil must be managed as used oil.
- Consider the use of an oil/water separator (OWS) to capture compressor oil contained within blowdown discharge.
- If possible, use a dehumidifying system (dryer) in the air compressor to reduce the moisture content of the compressed air. This practice may also prolong the life of the compressor by reducing loss of lubrication and rusting.
- Visually inspect the exterior of compressor equipment for the presence of oil leaks on a regular basis.
- Establish a preventative maintenance program that includes a schedule for cleaning parts, replacing oil, and replacing filters as recommended in the manufacturer's specifications.

CONSTRUCTION SITES FOR SMALL PROJECTS

The pollution prevention practice guidance provided herein is intended for minor construction projects only, such as minor building repairs, repaving, remodeling, or the construction of small pavement areas or buildings. Larger projects typically require permits, and therefore are subject to more significant pollution prevention actions. By following the pollution prevention practices detailed in this sheet, basic grounds maintenance and construction activities are expected to have minimal impact on water quality. General recommendations for hydrostatic testing and excavation dewatering are also provided in this EPC.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent the discharge of pollutants from minor construction or maintenance activities to the storm water drainage system or local waterbodies.	· Minor remodel, repair, and demolition activities · Landscaping and painting activities · Repaving and pavement maintenance · Facilities repair activities	· Paints, solvents, & finishing residues · Asphalt & concrete materials · Other toxic materials · Trash & debris · Sediment

POLICIES AND GUIDANCE

Construction activity occurs at all USCG facilities. Often the smaller construction sites are not subject to NPDES permitting because of the overall size of the project. These larger permitted sites will contain a construction Stormwater Pollution Prevention Plan (SWPPP) dictating erosion prevention and pollution prevention measures. The following best practices should be followed for small sites.

PREVENTATIVE ACTIONS

- Maintain organized and safe working conditions. Keep worksite clean and free of debris, wash water, and sediment.
- To prevent exposure to rainfall, store construction and maintenance materials and equipment indoors or under a sturdy roof (Figure 1).
- Temporarily redirect roof downspouts to discharge away from equipment storage and work areas.
- Store equipment and materials on impervious (e.g., paved) surfaces and temporarily block or otherwise protect storm drains from pollutants in work areas.
- Store spill cleanup materials and equipment close (within 20 feet) to the construction/maintenance areas for easy access.
- Store reactive/ignitable/flammable liquids (e.g., diesel tanks) in compliance with all safety regulations and fire codes.
- Inform service providers, like garbage collection or material delivery persons, about construction and maintenance activities and how to avoid them during their visits to the site.
- Use soil erosion control techniques when bare soil is exposed to prevent direct contact of pollutants with the ground and water.

- Do not remove original product labels from containers, as they contain important safety and disposal information.
- Maintain accurate and up-to-date Safety Data Sheets and ensure that they are readily available, along with spill containment equipment and cleanup materials, in the event of a spill.
- Inspect work and storage areas daily, checking equipment for leaks and spills, particularly at valves, flanges, seams and welds, gaskets, and other connections.
- Limit exposure of equipment to rain, snow, and storm water runoff. Always perform construction and maintenance activities in dry weather. Clean work areas, cover materials and trash stockpiles, and return equipment to safe storage areas before wet weather occurs.
- Regularly inspect and clean the storm water drainage system to avoid escaped pollutants from traveling further during wet weather.
- Train employees on proper usage, spill containment, and cleanup procedures.
- Be conscious of fumes during construction/maintenance operations, which may be indicative of a spill or leak.
- Minimize disturbance; preserve natural vegetation to the extent practicable.
- Implement good housekeeping practices
    - General waste management
    - Spill prevention and control
    - Vehicle maintenance and washing areas

EROSION PROTECTION AND SEDIMENT CONTROL CONSIDERATIONS

Below is a list of some erosion/sediment control options that may be useful to prevent sediment-laden runoff from leaving the site. Refer to Attachment E, Page 7 for advanced structural best management practices (BMPs) that also may be applicable.

- Chemical stabilization
- Dust control
- Mulch
- Riprap
- Seeding
- Sodding
- Stockpile covers
- Temporary slope drains
- Check dams
- Grass-lined channels
- Permanent slope diversions
- Temporary diversion dikes
- Brush barriers
- Filter berms/rolls/socks
- Silt fences
- Inlet protection
- Stabilized construction entrances
- Sediment traps
- Straw or hay bales
- Vegetative buffers

HYDROSTATIC TESTING

Discharging of hydrostatic test water will likely require a general or individual permit from the State in which the discharge is located within. States have varying approaches to permitting this discharge so consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD to determine the best way forward.

The following BMPs are recommended to manage hydrostatic test water discharge from boilers, pipelines, storage tanks, flowlines, etc., that have been used for transportation or storage of natural gas, crude oil, or liquid or gaseous petroleum hydrocarbons, or other substances. Hydrostatic testing has the potential to degrade water quality.

- All discharges must be free from:
  - Debris, oil, scum, and other floating materials other than in trace amounts
  - Eroding soils and other materials that will settle and deposit in receiving waters
  - Suspended solids, turbidity and color at levels inconsistent with the receiving waters
  - Chemicals in concentrations that would cause violation water quality regulations
- Prior to hydrostatic testing, the boiler, pipe, tank, line, etc. must free from pollutants that could be discharged along with the test water. Precleaning is recommended. New pipelines should be relatively free of pollutants that could be discharged along with the hydrostatic test water.
- Discharge of any sludge generated in the precleaning, or any rinsing solutions used in the pre-cleaning of the pipelines is prohibited.
- Discharge of hydrostatic test water to which treatment chemicals, corrosion inhibitors or biocides have been added is also prohibited.
- Discharge flow must not cause erosion or sedimentation to receiving water. If necessary, flow dissipation devices or rip rap must be installed to reduce/control erosion
- Prevent receiving stream degradation avoiding discharge of lower quality water into a higher quality water body. Do not discharge brackish water into fresh water.

EXCAVATION DEWATERING

EPA's Construction General Permit (CGP) defines dewatering as "the act of draining accumulated stormwater from building foundations, vaults, and trenches, or other similar points of accumulation" (CGP, Appendix A). The CGP includes uncontaminated construction dewatering water as an authorized (i.e., allowed) non-stormwater discharge, provided the operator satisfies the conditions specified in Part 2.4 of the CGP.

Dewatering will likely require a general or individual permit from the authorized State or EPA. States have varying approaches to permitting this discharge so consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD to determine the best way forward.

If coverage under a permit is determined to be necessary, the EPA Inspection and Monitoring Guide provided at this link may be helpful: https://www.epa.gov/system/files/documents/2022-01/cgp-inspection-and-monitoring-guide-for-dewatering.pdf

Use the following recommendations to ensure that dewatering activities at your construction site do not threaten water resources.

Figure 1: Energy Dissipation and Sediment Bag (Photo source: EPA)

- Contractor must verify if permit is required based on amount of discharge and locally specific permit requirements.
- Discharge flow must not cause erosion or sedimentation. If necessary, flow dissipation devices or rip rap must be installed to reduce/control erosion.
- Do not directly dispose of dewatering material into wetlands, lakes, creeks, rivers, marshes, or the storm drain. Sediment and any other pollutants must be removed prior to disposal!
- Protect storm sewer system (storm drains, catch basins, inlets, etc.) by using filters and inlet covers. Refer to Attachment E – Advanced Structural BMPs for more information on these devices.
- Avoid damaging silt fencing, tree protection or other construction site measures to install dewatering hose.
- Pump sediment-laden water to:
    - A sediment control structure
    - An infiltration trench
    - A buffer strip or zone
    - Sediment Bag

DUMPSTERS AND SOLID WASTE MANAGEMENT

Solid waste management includes the collection, storage, transfer, and disposal of garbage. These activities often involve dumpsters, rolling containers, grease storage containers, roll-off dumpsters, and outdoor trash cans, which can be a major source of storm water pollution. Improper handling of waste creates a risk for discharging pollutants into the storm water system. When not covered or managed properly, dumpster/trash containers can overflow with trash or leak the liquids and solids contained within them. Special attention to dumpster/trash container selection, location, and design, along with regular emptying, inspection, and maintenance can eliminate these problems. Best practices herein outline preventative techniques to prevent storm water pollution from these activities.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent trash and dumpster pollutants from reaching the storm water drainage system and local streams, rivers, and lakes	· Any unit activity with trash management · Units with kitchens often utilize grease storage/disposal	· Litter · Bacteria · Fats, oils, & grease · Sediment · Nitrogen & Phosphorus · Organic matter · Heavy metals & toxic chemicals

POLICIES AND GUIDANCE

DESIGN & CONSTRUCTION

- Locate dumpsters on impervious (paved) pads and in an area that is not prone to flooding/ponding. Permeable surfaces (like gravel) are NOT suitable for use as a dumpster pad.
- Locate dumpsters at least 50 feet away from concentrated storm water flows, drainage ways, and storm drains.
- Storm drain inlets are prohibited within the dumpster pad.
- Dumpsters and trash compactors MUST be fully enclosed to prevent the discharge of solid and liquid waste into the storm water system. If this isn’t possible, the dumpster must be tied to a sanitary sewer, where appropriate.
- Ensure dumpsters/trash cans have leak-proof lids or covers to prevent rain/snow from entering the container. Cover the area with a roof or canopy, when possible, to prevent its exposure to rain and snow (*Figure 1*).
- Elevate the dumpster pad 2 to 3 inches above the surrounding area or install berms to prevent storm water entry onto the pad (*Figure 1*).
- To prevent trash overflow, ensure the number and capacity of dumpsters and trash containers at the site are adequate for the expected waste stream of the business or industry.
- Do not install a water supply to or near the dumpster pad without also installing a sanitary sewer inlet to capture water used within or near the dumpster.

Figure 1: Examples of multi-pronged pollution prevention measures for dumpsters.

Left - The dumpster is located inside the building on a slightly raised concrete pad. (*Copyright: CC BY-SA-NC*)
Middle - Dumpster with a secure, leak-proof lid located within a pre-cast dumpster enclosure. (*Photo source: Easi-Set Buildings*)
Right - All three dumpsters have leak-proof lids and are in an enclosed and covered area. (*Photo source: Alameda County, CA*)

PREVENTATIVE ACTIONS

Figure 2: Examples of poor dumpster management.

Left - The dumpster is rusted and leaky and is improperly placed near a storm drain.
Right - Evidence of significant leaks of grease and other materials on a pad for a roll-off dumpster. Immediate cleaning is needed to prevent discharges to the storm drain.

- Inspect dumpsters and trash cans weekly for overflow, leaks, pavement stains, damage, or evidence of litter or other wastes exiting the dumpster (*Figure 2*).
- Regularly empty dumpsters, trash containers, and spill containment trays to prevent overflow. If needed, increase the frequency of dumpster and trash pickups.
- Clean up all litter, liquids, and other materials immediately to prevent them from entering storm drains.
- Keep dumpsters, dumpster pads, trash containers, and surrounding areas clean; pick up trash and litter; sweep areas and dispose of waste properly. Hoses or pressure washers can be used only if the water discharges to, or is contained and disposed of into, the sanitary sewer.
- Repair or replace deteriorating trash containers, dumpsters, and spill containment trays (*Figure 2*).
- Always keep dumpster/trash container lids closed with drain plugs in place.
- Use dumpsters and trash containers for dry, non-hazardous wastes only. Appropriate containers should be provided for the disposal of other hazardous wastes. Signs can prevent improper waste disposal (*Figure 3*).

Figure 3: Signs on dumpsters can prevent the disposal of inappropriate wastes.

- Do not dispose of leaky bags or liquid wastes, especially fats, oils, and greases, in dumpsters and trash containers.
- Grease receptacles often are areas of high likelihood for spills and leaks. A containment pad is recommended for placement beneath these containers (*Figure 4*).
- Do not set grease bins near storm drain grates or inlets. Inspect grease bins daily and clean up spills and drips immediately.
- If temporary waste piles are onsite, cover with a waterproof tarp/cover.
- Recycle as much waste as possible.

Figure 4: Spill containment tray
(Photo source: SpillContainment.com)

ENGINE FLUSHING

Facilities that conduct engine flushing activities shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants and prevent invasive species and other pollutants from entering storm water runoff from the site of engine flushing or being discharged directly to local surface water.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent contaminated wash water and pollutants from discharging to the storm drainage system or streams, rivers, and lakes	• Stations • Sectors • Bases	• Sediment & dirt • Soaps, detergents, solvents, & other chemicals • Paint chips & other toxic materials • Grease & oils • Trash & debris • Hot water (thermal pollution) • Invasive species

POLICIES AND GUIDANCE

Best practices for the following activities are addressed in this EPC:

- ***INVASIVE SPECIES MANAGEMENT DURING ENGINE FLUSHING***
- ***SMALL BOAT ENGINE FLUSHING***
- ***AIRCRAFT ENGINE FLUSHING***

Prior to any marine engine flushing, the following invasive species inspection and decontamination procedures should be followed.

INVASIVE SPECIES INSPECTION AND DECONTAMINATION PROCEDURES FOR BOATS & VESSELS

FOR ADDITIONAL INFORMATION, THERE ARE MANY ONLINE RESOURCES THAT PROVIDE INSPECTION GUIDELINES, SUCH AS

Protect Your Waters
(https://stopaquatichitchhikers.org/)

1. Remove the boat, trailer, and/or asset from the water and away from the launch ramp for inspection and cleaning.
2. Thoroughly inspect all exposed surfaces on your vessel, trailer, or asset. If you find any mussels, scrape them off and kill them by crushing them. Dispose of the remains in the trash.
3. Remove all plants and mud from your boat, trailer, and/or asset. Dispose of all material in the trash.
4. Carefully feel your boat's hull and all equipment for any rough or gritty spots, which may be young mussels that have settled on your vessel and cannot be seen. Microscopic quagga and zebra mussels will feel like sandpaper.

5. Wash the boat's hull, trailer, equipment, bilge, and any other exposed surfaces with high-pressure, hot water. When possible, use water at a temperature of 140° F (60° C) at the hull – or about 155° (68° C) at the nozzle – which will kill the mussels. Dry the boat as much as possible. **Pressure washing must take place in a designated wash rack.** Refer to EPC – SW – PRESSURE WASHING ACTIVITIES for storm water management measures regarding pressure washing.

Photo Credit: Stopaquatichitchhikers.org

6. Drain all water from your boat and dry all areas, including the motor exterior, motor cooling system, live wells, ballast tanks, bladders, bilges, and lower outboard units. Make sure that all life jackets, equipment, or other items that have been in the water, including anchors, ropes, rafts, etc., are inspected, cleaned, and dried.

7. Clean and dry personal belongings, clothing, and footwear that have come in contact with the water.

8. Keep the boat, trailer, and/or asset dry for at least 5 days in warm, dry weather and up to 30 days in cool, moist weather before launching into a freshwater.

SPECIAL INVASIVE SPECIES CONSIDERATIONS

- Chemicals, such as bleach, have not been proven to be effective in disinfecting all aquatic invasive species and may damage your watercraft and equipment, so caution should be taken.
- Vinegar can be an effective method to decontaminate areas that do not drain completely (e.g., bilge areas of small boats). Dipping equipment into 100% vinegar for 20 minutes will help to kill harmful aquatic hitchhiker species. Over extended periods of time, vinegar may be corrosive to metal and is toxic to fish at this concentration, so thoroughly rinse with tap water. Ensure that the solution does not runoff directly to storm drains, drainage pathways or surface waters.

Hydrilla on Propellor
(Photo Credit - West Marine)

- A 1 percent table salt solution for 24 hours can replace the vinegar dip. The following table provides correct mixtures for the 1 percent salt solution in water:

GALLONS OF WATER	CUPS OF SALT
5	2/3
10	1 1/4
25	3
50	6 1/4
100	12 1/3

- If hot water is not available, spray equipment such as boats, motors, trailers, anchors, buoys, and chains, with high-pressure water, and always allow enough time to DRY equipment. **Pressure washing must take place in a designated wash rack.**

SMALL BOAT ENGINE FLUSHING

- DO NOT discharge engine flushing water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Non-wash rack washing operations should not be performed during wet weather conditions.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper engine flushing procedures.
- *2-Cycle Engine Flushing* - Engines must be flushed within a contained system. Flush water must be inspected for sheen. If no sheen exists, flush water should be discharged to a grassy area away from drainage pathways and surface waters. All discharges to grassy areas must be under the constant supervision of the Unit Environmental Coordinator with at least a 15-foot buffer around the discharge area and discharged waters are NOT allowed to flow directly to surface waters or a storm water catch basin. If sheen is present, flush water must be disposed of as oily water.

- *4-Cycle or Inboard/Outboard Engine Flushing*
  - **Option 1** **(Best)** = Perform engine flushing in a wash rack. A wash rack is defined as a designated area that drains to an oil/water separator (OWS) set to discharge to a sanitary sewer system and not to the storm sewer system.
  - **Option 2** **(Better)** = Perform engine flushing over a grassy area away from drainage pathways and surface waters with at least a 15-foot buffer around the perimeter. Only freshwater may be used for flushing outside of a wash rack.

AIRCRAFT ENGINE FLUSHING

- **Option 1** **(Best)** = Perform engine flushing in a wash rack. A wash rack is defined as a designated area that drains to an OWS set to discharge to a sanitary sewer system and not to the storm sewer system.

- **Option 2** **(Better)** = Perform engine flushing away from drainage pathways and surface waters. Depending on known engine flush water characterization, flush water may need to be collected, characterized, and properly disposed of in accordance with waste characterization of known constituents.
- DO NOT discharge engine flushing water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Non-wash rack washing operations should not be performed during wet weather conditions.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper engine flushing procedures.

FUELING, DEFUELING, AND FUEL STORAGE ACTIVITIES

Facilities that conduct fueling, defueling, and fuel storage operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. Further guidance pertaining to mobile refueling and bulk fuel farms can be found on EPC-SW-09A – Mobile Refueling and EPC-SW-09B – Bulk Fuel Farms.

EMERGENCIES
EPA
Report oil or chemical spills at:
800-424-8802

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from fueling activities from entering the storm drainage system or streams, rivers, and lakes	· Applicable to most units fueling boats, aircraft, vehicles, and support equipment	· Benzene and other harmful chemicals · Hydrocarbons · Other fluids (lubricants, antifreeze, etc.)

POLICIES AND GUIDANCE

GENERAL

- Use drip pans where leaks or spills can occur and where making or breaking hose connections.
- Use fueling hoses with check valves to prevent dripping after fueling.
- Store fueling hoses within secondary containment.
- Provide curbing or bollards around fuel pumps to prevent vehicle collision.
- DO NOT top off tanks. Tanks should only be filled to 95% capacity, even in the event of impending storm conditions.
- Use dry cleanup methods. Do not wash down fueling areas.

- Minimize exposure to fueling areas. Conduct fueling operations (including fuel transfer) on a contained concrete pad and under a roof when possible. Minimize run-on by using berms, curbing, or surface grading.
- Aboveground storage tanks (ASTs) should be double-walled with overflow protection. If an AST is single-walled, then you must provide secondary containment such as a concrete or earthen berm. Ensure that the height of the berm is sufficient to contain a spill and any accumulated precipitation. All fuel line systems should also be provided with secondary containment.

- Interstitial space within double-walled tanks should be monitored in accordance with spill prevention, control, and countermeasure (SPCC) plans. If a SPCC plan is not required, interstitial space will be monitored in accordance with 40 C.F.R. Part 112 criteria or the manufacturer’s tank inspection certification criteria, whichever is more stringent.
- Inspect fuel storage areas for leaks and spills, quarterly at a minimum or more regularly as needed, to render these areas free of liquids and debris that may add to the quantity of waste, should there be a spill or leak. (A SPCC may dictate frequency of inspections.)
- Valves to drain secondary containment areas must remain closed and locked except when draining storm water from containment. A visual determination of NO sheen must be made prior to draining the secondary containment area.

- If a sheen is detected, storm water runoff should be collected from the fueling area and secondary containment area and directed to a treatment system (e.g., oil/water separator to sanitary sewer system). If no sheen is visible, uncontaminated storm water may be discharged to a grassy area with a 15-foot buffer around the perimeter under the constant supervision of the Unit Environmental Coordinator.
- 55-gallon drums must be stored under a roof or other form of cover and within secondary containment (e.g., spill pallet, clam shell). Clearly label drums with contents and keep labels visible.
- Use spill/overflow protection and cleanup equipment. Spill cleanup materials must be readily available (within 20 ft) in all fueling locations and storage areas and used immediately in the event of a spill.
- Do not drain/rinse/clean fuel canisters or related materials in a sink that discharges to ground or the storm sewer system. Sinks and deep sinks should not discharge to ground or soil and must be piped to the sanitary sewer system.
- Train personnel on proper fueling/defueling, spill prevention, control, cleanup, and transfer techniques.
- Develop and implement SPCC plans, if required, in accordance with 40 C.F.R. Part 112 criteria.
- Post the proper spill notification phone number and directions for notification in or on the mobile fueling vehicle. The National Response Center phone number is 800-424-8802.

PETROLEUM LOADING/UNLOADING

- Confine loading/unloading activities to designated areas, preferably in a covered area away from surface waters.
- Provide diversion berms, curbs, or grassed swales around the loading/unloading areas to prevent run-on and runoff.
- Avoid loading/unloading materials during wet weather conditions if possible or provide cover.

- Ensure hose connection points are inside a containment area or use drip pans when not practical.
- Use dry cleanup methods. Do not wash down the loading areas.
- If available, a floating containment boom should be deployed for oil transfer operations pier-side or over water involving refueling, internal transfers, fuel offloading, oily waste handling, bilge dewatering and lube oil handling. If this option is not available to the unit, or the activity is not an emergency, then these activities should be delayed until precipitation is not a factor.
- A spill kit should be readily available (within 20 feet) during oil transfer operations, and pier-side spill containers should be unlocked.
- Drums awaiting loading or disposal on a pier outside of normal working hours should be provided with secondary containment.

EMERGENCIES
EPA
Report oil or chemical spills at:
800-424-8802

MOBILE REFUELING ACTIVITIES

Facilities that conduct land-based mobile refueling operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from fueling activities from entering the storm drainage system or streams, rivers, and lakes	· Applicable to most units fueling boats, aircraft, vehicles, and support equipment	· Benzene and other harmful chemicals · Hydrocarbons · Other fluids (lubricants, antifreeze, etc.)

POLICIES AND GUIDANCE

GENERAL

- For in-water fueling, ensure that hoses are long enough to safely move with the vessel secured to the dock without straining, and that each hose is adequately supported along its length.
- All equipment should be maintained in good condition and free from leaks or damage that could result in leaks.
- Ensure that each part of the transfer system not necessary for the transfer is securely blanked or shut off.
- If fueling on a pier, ensure drain scuppers or other spill pathways have been closed by mechanical means or secured to prevent releases.
- Welding, hot work operations and smoking are prohibited on or near vehicle or vessel during fuel transfer.
- After each fuel transfer, steps must be taken to secure the equipment to prevent accidental releases:
  - Close all transfer system valves.
  - Drain all hoses into a proper receptacle before disconnection.
  - Blank transfer hoses upon disconnection and before movement out of secondary containment area.
  - Blank transfer header flanges and manifolds immediately after hose removal.
- Store all hoses and equipment where they are protected from the weather.
- Secure storage areas (e.g., store piping, pumps, valves, and manifolds in an area where they will not be tampered with, used for the wrong purpose, and/or damaged by production operations).

- Clean up work area and ensure that all associated equipment is stowed.
- Remove or disconnect electrical power and/or compressed air to transfer equipment.
- DO NOT top off tanks. Tanks should only be filled to 95% capacity to prevent over-filling.
- Minimize exposure to fueling areas. Conduct fueling operations (including fuel transfer) on a contained concrete pad and under a roof when possible. Minimize run-on by using berms, curbing, or surface grading.
- Inspect fuel storage areas for leaks and spills, quarterly at a minimum or more regularly as needed, to render these areas free of liquids and debris that may add to the quantity of waste, should there be a spill or leak. (A SPCC may dictate frequency of inspections.)
- Valves to drain secondary containment areas must remain closed and locked except when draining storm water from containment. A visual determination of NO sheen must be made prior to draining the secondary containment area.
- If a sheen is detected in containment area, storm water runoff should be collected from the fueling area and secondary containment area and directed to a treatment system (e.g., oil/water separator to sanitary sewer system). If no sheen is visible, uncontaminated storm water may be discharged to a grassy area with a 15-foot buffer around the perimeter under the constant supervision of the Unit Environmental Coordinator.
- Use spill/overflow protection and cleanup equipment. Spill cleanup materials must be readily available (within 20 ft) in all fueling locations and storage areas and used immediately in the event of a spill. Train personnel on proper fueling/defueling, spill prevention, control, cleanup, and transfer techniques.
- Develop and implement SPCC plans, if required, in accordance with 40 C.F.R. Part 112 criteria.

ADDITIONAL SPILL PREVENTION MEASURES

- Centralize fueling activities in one location designed with spill protection and containment features.
- Equip the mobile fueling vehicle with adequate spill kit materials, including but not limited to drain covers and absorbent material in case of an accidental spill or hose breakage.
- Place storm drain/catch basin mat covers over nearby catch basins during fueling.
- Train employees and contractors on the proper methods of fueling, spill response, and equipment emergency shutdown procedures.
- If contractors are responsible for refueling, ensure the contractor has a spill prevention and response plan, including an on-call environmental cleanup contractor available to respond in the event of a large fuel spill.

- Routinely check secondary containment structures and fueling equipment for leaks or degraded components.

FUELING OPERATION

- Ensure employees or contractors performing fueling activities always remain with the vehicle being fueled.
- Use drip pans or absorbent pads under the nozzle when fueling and drip tubes for fuel nozzle storage.
- Equip fueling nozzles with an automatic shut-off mechanism.

LEAKS AND DRIPS OR IN CASE OF SPILL

- Clean leaks and drips immediately and properly dispose of used absorbent material. Use dry cleanup methods. Do not wash down or rinse spill area with water.
- Post the proper spill notification phone number and directions for notification in or on the mobile fueling vehicle. The National Response Center phone number is 800-424-8802.

BULK FUEL FARM ACTIVITIES

Facilities that conduct fueling, defueling, and fuel storage operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. Fuel farms are often equipped with large secondary containment areas (SCA) impacted by storm water and are sometimes connected to an OWS which discharges to wastewater or surface water.

EMERGENCIES
EPA
Report oil or chemical spills at:
800-424-8802

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from fueling activities from entering the storm drainage system or streams, rivers, and lakes	· Applicable to a few units requiring large amounts of fuel for aircraft, large cutters, vehicles, and support equipment	· Benzene and other harmful chemicals · Hydrocarbons

POLICIES AND GUIDANCE

GENERAL

- Tanks should be built and tested in accordance with industry standards (concrete and steel).
- Tanks (and piping/valves) should be painted regularly and be rust free.
- Adherence to SPCC regulations with appropriate inspections and documentation is required.
- Provide curbing or bollards around fuel off-loading stations to prevent vehicle collision.
- A Facility Response Plan is required for those tanks in which the storage capacity is greater than 1 million gallons and could cause injury to fish and wildlife or public water supply if discharged (distance calculations from tanks to water body required).

Nyman Peninsula Fuel Farm in Kodiak, AK

- Tanks should have containment such as a concrete or earthen berm. Ensure that the height of the berm is sufficient to contain a spill and any accumulated precipitation. All fuel line systems should also be provided with secondary containment.

Valves for Secondary Containment Discharge in Kodiak, AK

- Valves to drain secondary containment areas must remain closed and locked except when draining storm water from containment. A visual determination of NO sheen must be made prior to draining the secondary containment area.
- If a sheen is detected, storm water runoff should be collected from the fueling area and secondary containment area and directed to a treatment system (e.g., oil/water separator to sanitary sewer system). If no sheen is visible, uncontaminated storm water may be discharged to a grassy area with a 15-foot buffer around the perimeter under the constant supervision of the Unit Environmental Coordinator.
- Use spill/overflow protection and cleanup equipment. Spill cleanup materials must be readily available (within 20 ft) in all fueling locations and storage areas and used immediately in the event of a spill.
- Post the proper spill notification phone number and directions for notification in or on the mobile fueling vehicle. The National Response Center phone number is 800-424-8802.
- Train personnel on proper fueling/defueling, spill prevention, control, cleanup, and transfer techniques.

PETROLEUM LOADING/UNLOADING

- Confine loading/unloading activities to designated areas, preferably in a covered area away from surface waters.
- Provide diversion berms, curbs, or grassed swales around the loading/unloading areas to prevent run-on and runoff.
- Avoid loading/unloading materials during wet weather conditions if possible or provide cover.
- Ensure hose connection points are inside a containment area or use drip pans when not practical.
- Use dry cleanup methods. Do not wash down the loading areas.
- A spill kit should be readily available (within 20 feet) during oil transfer operations, and pier-side spill containers should be unlocked.

GROUNDS MAINTENANCE

This EPC is intended for grounds maintenance projects only, such as landscaping maintenance, minor building repairs, repaving, remodeling, and/or pest control. Larger projects typically require permits, and therefore are subject to more significant pollution prevention actions. By following the pollution prevention practices detailed in this sheet, basic grounds maintenance and construction activities are expected to have minimal impact on water quality.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent the discharge of pollutants from minor grounds maintenance or facility maintenance activities to the storm water drainage system	· Minor remodel, repair, and demolition activities · Landscaping and painting activities · Repaving and pavement maintenance · Facilities and grounds maintenance and repair activities	· Pesticides & herbicides · Fertilizers · Paints, solvents, & finishing residues · Asphalt & concrete materials · Other toxic materials · Trash & debris · Sediment

POLICIES AND GUIDANCE

PREVENTATIVE ACTIONS

- Maintain organized and safe working conditions. Keep worksite clean and free of debris, wash water, and sediment.
- To prevent exposure to rainfall, store maintenance materials and equipment indoors or under roof (*Figure 1*).
- Temporarily redirect roof downspouts to discharge away from equipment storage and work areas.
- Store equipment and materials on impervious (e.g., paved) surfaces and temporarily block or otherwise protect storm drains from pollutants in work areas.
- Store spill cleanup materials and equipment close to the construction/maintenance areas for easy access.
- Store reactive/ignitable/flammable liquids in compliance with all safety regulations and fire codes.
- Educate employees about the importance of keeping pollutants out of the storm water system.
- Inform service workers, like garbage collection or material delivery persons, about ground construction and maintenance activities and how to avoid them during their visits to the site.

Figure 1: Indoor Storage Area

- Use soil erosion control techniques when bare soil is exposed to prevent direct contact of pollutants (sediment) with the ground and water. Refer to Attachment E, Page 7 for advanced structural best management practices (BMPs) that may be applicable. Consider using storm drain protection such as filter socks or drain covers to prevent sediment from leaving site and entering storm water system.
- Do not remove original product labels from containers, as they contain important safety and disposal information.
- Maintain accurate and up-to-date Safety Data Sheets and ensure that they are readily available, along with spill containment equipment and cleanup materials, in the event of a spill.
- Inspect work and storage areas daily, checking equipment for leaks and spills, particularly at valves, flanges, seams and welds, gaskets, and other connections.
- Limit exposure of equipment to rain, snow, and storm water runoff. Always perform construction and maintenance activities in dry weather (*Figure 2*). Clean work areas, cover materials and trash stockpiles, and return equipment to safe storage areas before wet weather occurs.

Figure 2: Maintenance activities should be performed during dry weather
(Photo source: Espina Paving)

- Regularly inspect and clean the storm water drainage system to avoid escaped pollutants from traveling further during wet weather.
- Train employees and contract staff (mowing, pest control, garbage collection, etc.) on proper usage, spill containment, and cleanup procedures.

LANDSCAPING AND GROUNDS MAINTENANCE

- Purchase only pesticides needed per season, use what is mixed and minimize disposal as much as possible. Rinse containers and use rinse water as product. Dispose of unused pesticide as hazardous waste. Dispose of empty pesticide containers according to the instructions on the container label.
- Dispose of grass clippings, leaves, sticks, or other collected vegetation as garbage, or by composting. Do not dispose of collected vegetation into waterways or storm drainage systems.
- Use mulch or other erosion control measures on exposed soils.
- Do not use pesticides if rain is expected.
- Do not mix or prepare pesticides near storm drains.
- Check irrigation schedules so pesticides will not be washed away and to minimize non-storm water discharge.
- Work fertilizers into the soil when possible, using aerators, liquid application, etc. rather than dumping or dispersing them onto the surface.
- Do not apply insecticides within 100 feet of surface waters such as lakes, ponds, wetlands, and streams.

- Provide secondary containment for pesticide, insecticide, herbicide, and fertilizer storage.
- Educate and train employees on use of pesticides and in pesticide application techniques to prevent pollution. Consult existing permits as well as state and local regulatory authorities regarding training and certification requirements in applicable areas.

CLEANUP AND RESPONSE ACTIONS

- Follow the safety and disposal instructions identified for each product and attempt to use the entire product before disposing of its container.
- Thoroughly dry latex paint and paint cans, used brushes, rags, absorbent material, drop cloths, etc. before disposing of them with other construction debris.
- Keep a stockpile of spill cleanup materials readily accessible.
- Contain leaks and spills that may occur.
- Develop and follow standard operating procedures (SOPs) for spill prevention and cleanup.
- Develop a spill prevention, control, and countermeasure (SPCC) plan and keep it up to date and readily available for use.

Figure 3: Protecting a storm drain from potential pollution
(Photo source: Valley Industrial Trucks, Inc.)

- Maintain Safety Data Sheets in an accessible location for all grounds maintenance chemicals that are used, and follow the relevant instructions related to the proper cleanup of spilled materials and proper disposal of cleanup materials.
- Protect any nearby storm drains or streams to prevent access to spilled materials (*Figure 3*).
- Clean up leaks and spills immediately using dry methods when possible; use a rag for small spills, a damp mop for general cleanup, and absorbent material for larger spills.

EPC-SW-11 – HAZARDOUS MATERIALS & WASTE STORAGE

HAZARDOUS MATERIALS AND WASTE STORAGE ACTIVITIES

Facilities that conduct hazardous materials use and waste storage activities shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

EMERGENCIES
EPA
Report oil or chemical spills at:
800-424-8802

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent harmful chemicals from entering the storm drainage system or streams, rivers, and lakes	· All USCG Units	· Toxic chemicals · Other substances (antifreeze, solvents, granules, powders, etc.)

POLICIES AND GUIDANCE

GENERAL

- Hazardous materials and wastes must be secured undamaged, fully enclosed, and appropriately marked containers (e.g., 55-gallon drums). Provide secondary containment for chemicals that will not be loaded or disposed within immediate (i.e., today's) working hours.
- Place hazardous material and waste containers away from open water while awaiting loading or disposal.
- Minimize exposure. Storage areas must be contained under a roof or other form of permanent cover. It is preferable that the storage area is also enclosed on each side to prevent rainwater from collecting within the storage area.
- Label all waste containers with content information and the accumulation start date to ensure proper removal within the regulatory timeframe.
- Aboveground storage tanks (ASTs) should be double-walled with overflow protection. If an AST is single-walled, then you must provide secondary containment such as a concrete or earthen berm. Ensure that the height of the berm is sufficient to contain a spill and any accumulated precipitation. All fuel line systems should also be provided with secondary containment.
- Valves to drain secondary containment areas must normally remain closed and locked. A visual determination of NO sheen must be made prior to draining the secondary containment area.
- Inspections of tank systems and containers shall be performed and documented weekly or in accordance with the spill prevention, control, and countermeasure (SPCC) plan.
- Train employees on proper storage, transfer, and loading/unloading techniques.

- Post the proper spill notification phone number and directions for notification in or on the mobile fueling vehicle. The National Response Center phone number is 800-424-8802.

LOADING/UNLOADING

- Confine loading/unloading activities to a designated area, preferably indoors or in a covered area.
- Close drains using drain seals, mats/covers, guards, plugs, etc. during loading/unloading activities.
- Avoid loading/unloading materials during wet weather conditions, if possible.
- Inspect all containers prior to loading/unloading.
- Berm, curb, or dike outside loading/unloading areas.
- Use dry cleanup methods. Do not wash down areas.

STORAGE

- Store materials away from high traffic areas and open water.
- Berm, curb, or dike outside storage areas.
- Ensure sufficient aisle space to ease inspections and handling.
- Inspect all containers prior to placing in storage area.
- If you must stack materials or wastes, do so with caution and no more than two levels high.
- Keep container labels visible.
- Storage spaces and containers should be routinely inspected for leaks, signs of cracks or deterioration, or any other signs of leakage.
- Use dry cleanup methods and dispose of cleaning materials immediately and properly. Do not wash down areas with water or cleaning fluids.

OUTDOOR STORAGE OF BULK MATERIALS

Facilities that store bulk materials outdoors shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals from materials stored outdoors from entering the storm drainage system or local waterbodies.	· Bases · Sectors · Stations · ANTs	· Heavy metals (lead, copper, zinc) · Toxic chemicals · Paint · Soil/Sand/Gravel/Sediment

POLICIES AND GUIDANCE

Outdoor storage of bulk materials BMPs are used to prevent pollution of local waterbodies resulting from spills, leaks and other contaminants carried away by storm water. These include secondary containment, corrosion protection, protection from natural elements, and leak detection. Outdoor storage is used when storing materials indoors is not feasible. It is important that these materials are stored and maintained properly so that pollutants do not migrate by stormwater into our natural waterways.

GENERAL

- Outdoor storage of bulk materials should only occur when indoor storage is not feasible and should only be allowed in designated areas that are not prone to flooding/ponding. Permeable surfaces (like gravel) are NOT suitable for use as outdoor storage area.
- All materials should be covered where feasible. Covers may consist of roofs, lids, or waterproof tarps.
- Store materials on paved or impervious surfaces that are free of cracks or gaps.
- When possible, store on elevated surface such as pallets or concrete block/pad.
- Inspect outdoor storage area frequently. Address any spills or leaks immediately.
- Spill cleanup materials must be readily available (within 20 ft) in all storage areas and used immediately in the event of a spill.

BUOY CHAIN STORAGE

- Buoy chain must be stored away from surface waters. Store buoy chain within containment (e.g., metal bin) or on raised surface (e.g., wooden pallet) and provide cover (e.g., lid or tarp). If buoy chain can't be stored under cover, store on a pervious surface away from storm drains to allow for infiltration to ground rather than allowing runoff to discharge to the storm sewer system.
- When possible, provide a grass buffer between chain storage and closest drainage pathway or surface water.

SMALL ARMS FIRING RANGES

Facilities that conduct operations associated with small arms firing ranges (SAFRs), particularly outdoor or open-air ranges, shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent pollutants from firing ranges from entering the storm drainage system or streams, rivers, and lakes	· Small Arms Firing Ranges	· Heavy metals (lead, copper, zinc)

POLICIES AND GUIDANCE

INDOOR/OUTDOOR RANGE

- Non-leaded ammunition is recommended.
- The ventilation system, if installed, should be operating at all times while the range is in use and during cleanup. Provide routine inspection and maintenance of the ventilation system.
- Air filter change-out should be performed at least quarterly or in accordance with static/differential pressure guidelines provided by the manufacturer. Filter change-out should be performed by personnel trained/certified in lead safety.
- Keep the range area uncluttered. Airflow patterns (i.e., ventilation) can be disrupted by obstacles.
- Carpeting should not be used anywhere inside a firing range or in rooms adjacent to the range.
- Use vacuum cleaner (HEPA to eliminate dust) or brass collection device after each use to collect cartridges and projectiles.
- After use, the floor of the firing range should be thoroughly cleaned by a combination of vacuuming with an explosion-proof HEPA vacuum cleaner designed to collect lead dust and washing if needed.
- Washing liquids and vacuumed wastes should be contained, characterized, and properly disposed of in accordance with Resource Conservation and Recovery Act (RCRA) site waste characterization regulations.
- Dry sweeping should never be used in a lead range. Pick up casings by hand.
- Change lead collection buckets on a regular basis, if so equipped.
- Use step clean-off pads at the exit or shoe coverings to reduce lead tracked out of the range.

- Range washing machine discharges must be piped to the sanitary sewer system. Clothes washers/sinks shall not discharge to soil.
- Adhere to all applicable hazardous waste management requirements (for waste disposal) and personal protective safety requirements.
- Storm water runoff from an outdoor firing range should not leave the footprint of the range without some type of filtration/treatment; runoff should be collected, characterized, and properly disposed of.

OPEN AIR SAFR

- Use vacuum cleaner (HEPA to eliminate dust) or brass collection device after each use to collect cartridges and projectiles.

GENERAL

- Prevent lead migration by:
  - Monitoring and adjusting soil pH through line spreading
  - Immobilizing lead through phosphate spreading

- Prevent lead laden runoff by installing:
  - Vegetative ground cover (e.g., grass)
  - Organic surface cover (e.g., mulch, compost)
  - Filter beds (e.g., detention ponds)
  - Water/sediment traps (e.g., settling wells)
  - Dams and dikes
  - Ground contouring

- Avoid uncontained bullets using the following:
  - Earthen backstop
  - Sand trap
  - Wet passive bullet trap (steel)
  - Lamella trap
  - Rubber granule
  - Shock absorbing concrete

- Prevent lead dissolution by:
  - Hand raking and sifting
  - Screening
  - Vacuuming
  - Soil washing

SNOW STORAGE MANAGEMENT

Facilities that conduct snow storage activities shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent the pollutants that are collected from snow collection services from reaching the storm drainage system or streams, rivers, and lakes	· Northern climate locations that have snow plowing operations	· Sediment · Metals · Oil and grease · Chlorides · Trash and debris

POLICIES AND GUIDANCE

GENERAL

The key to effective snow removal management is to select snow storage sites that are located adjacent to or on pervious surfaces in upland areas away from water resources and wells. At these locations, the snow meltwater can filter into the soil, leaving behind sand and debris which can then be removed in the springtime. The purpose is to remove direct discharge locations that might pollute local streams/waterbodies. The following areas should be avoided:

- Placing removed snow directly into surface waters is prohibited. Avoid dumping snow into any water body, including rivers, the ocean, reservoirs, ponds, or wetlands. In addition to water quality impacts and flooding, snow disposed of in open water can cause navigational hazards when it freezes into ice blocks.
- Do not store snow where trucks may cause shoreline damage or erosion.
- Do not store snow within a wellhead protection area (WPA) of a public water supply well or within 75 feet of a private well, where road salt may contaminate water supplies.
- Avoid storing snow in sanitary landfills and gravel pits. Snow meltwater could create more contaminated leachate in landfills posing a greater risk to groundwater. In gravel pits, there is little opportunity for pollutants to be filtered out of the meltwater because groundwater is close to the land surface.
- Avoid storing snow on top or in the vicinity of storm drain catch basins or in storm water drainage swales or ditches. Snow combined with sand and debris may block a storm drainage system, causing localized flooding. A high volume of sand, sediment, and litter released from melting snow also may be quickly transported through the system into surface water.

SITE SELECTION

The following steps should be taken when selecting a snow storage site:

- Identify sites that could potentially be used for snow storage such as open space (e.g., parking lots or parks).

- Sites located in upland locations that are not likely to impact sensitive environmental resources should be selected first.
- If more storage space is still needed, prioritize the sites with the least environmental impact.

SITE PREPARATION AND MAINTENANCE

In addition to carefully selecting storage sites before the winter begins, it is important to prepare and maintain these sites to maximize their effectiveness. The following maintenance measures should be undertaken for all snow disposal sites:

- A silt fence or equivalent barrier should be placed securely on the downgradient side of the snow disposal site.
- To filter pollutants out of the meltwater, a 50-foot vegetative buffer strip should be maintained during the growth season between the disposal site and adjacent water bodies.
- Debris should be cleared from the site prior to using the site for snow storage.
- Remove accumulated trash and debris from the snow storage area as they become visible. Debris in surface water is a water quality violation. Waste and litter that become uncovered as the snow melts need to be picked up before off-site migration of the waste becomes a problem.
- After the snow has melted, debris should be cleared from the site, analyzed, and properly disposed of at the end of the snow season and no later than May 15 (date from Massachusetts DEP but may vary).

EMERGENCY SNOW MANAGEMENT

If upland storage sites have been exhausted, snow may be stored near water bodies. A vegetated buffer (ideally comprised of dense turf or native grass) of at least 50 feet should still be maintained between the site and the water body in these situations. Furthermore, it is essential that local municipal/state guidelines and any other receiving stream restrictions are adhered to when preparing and maintaining snow storage sites be followed to minimize the threat to adjacent water bodies. Under extraordinary conditions, when all land-based snow storage options are exhausted, placement of snow that is not obviously contaminated with road salt, sand, and other pollutants may be allowed in certain water bodies under certain conditions. Use the following guidelines in these emergency situations:

- Place snow in open water with adequate flow and mixing to prevent ice dams from forming.
- DO NOT store, stage, or place snow in salt marshes, vegetated wetlands, certified vernal pools, shellfish beds, mudflats, drinking water reservoirs and their tributaries, WPAs of public water supply wells, Outstanding National Resource Waters, or Areas of Critical Environmental Concern.
- DO NOT store, stage, or place snow where trucks may cause shoreline damage or erosion.
- Consult with municipal/city officials to ensure that snow storage near or in open water complies with local ordinances and laws.

SWIMMING POOLS

Facilities that operate and maintain swimming pools shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. Some municipalities have strict regulations regarding swimming pool operations that may not be covered in this guidance. Consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD if you have questions regarding swimming pools and hot tubs.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent chemicals used in swimming pools from reaching the storm drainage system or streams, rivers, and lakes	· Units with swimming pools	· Sanitizing chemicals like chlorine, bromine, or ozone · Acids and calcium chloride · Bacteria (fecal coliform) · Trash and debris

POLICIES AND GUIDANCE

GENERAL

- To prevent overflows from storms, splashing, and waves, do not overfill pools.
- Prevent "green water" (stagnant, algae-laden) to enter a storm drain or nearby stream, river, or lake.
- Do not drain pool water (whether for cleaning or repair, winterization, or filter backwater) directly to the storm drainage system or a nearby stream, river, or lake. Drain the water slowly. Use it to water the lawn or landscape, or allow it to run overland, preferably through a dense grassy area, for a long distance. This allows the water to infiltrate, evaporate, and naturally filter or separate the chlorine out of the water.

- Maintain proper chlorine, pH, and water hardness levels to lessen the need to drain the pool for repairs and larger maintenance issues (*Figure 1*).
- Avoid using copper-based algaecides to kill or prevent algae.
- Develop a regular maintenance and cleaning schedule. Train employees on proper maintenance and pollution prevention, usage of chemicals, chemical spill prevention and containment, and spill cleanup procedures.

Figure 1: Regularly check chemical levels to avoid algae growth and draining the pool for repairs.
(Photo source: homedepot.com)

- Maintain accurate and up-to-date Safety Data Sheets and ensure that they are readily available in the event of a spill, along with spill containment equipment and cleanup materials.
- Keep pool chemicals organized and stored in a covered or indoor space (*Figure 2*).
- Regularly inspect the facility for damage, leaks, and/or unexpected overflows. Repair any problems immediately.
- Provide an ample number of undamaged trash cans with lids for use by facility users. Trash cans without lids should be located indoors or under a covered area.
- Check trash cans regularly and empty them as often as necessary to prevent overflow.
- Inspect user areas regularly and pick up/dispose of trash.

Figure 2: Pool chemicals must be stored indoors or in a covered area, protected from rainfall.
(Photo source: barracuda.com)

OPTIONS FOR POOL FILTER BACKWASH DISPOSAL

Option 1 **(Best)** = Discharge backwash to the sanitary sewer system with prior approval from the local water/wastewater utility:

- The facility must apply for and receive a discharge permit or other written authorization to discharge to the wastewater treatment plant (WWTP) prior to discharge.
- The facility must determine and use an appropriate discharge flow rate to avoid any back-ups or overflows in the sanitary sewer system.
- The facility must prevent the discharge of soil, sediment, rock, debris, or other solid material during the discharge.

Option 2 **(Better)** = Discharge backwash to a vegetated area within the facility property:

- The area should be large enough to contain the discharge without any subsequent runoff or negative effect to neighboring properties.
- Care should be taken not to create puddles of standing water, which could potentially provide breeding grounds for mosquitoes.
- Use caution when backwashing diatomaceous earth filter material as it will harden to cement-like consistency over time, possibly damaging soil and vegetation.
- Saline pool backwash water can significantly impact vegetation and soils due to the high salt content.
- High chlorine levels in backwash water can harm certain sensitive plants.
- If you are located in an aquifer recharge zone, contact the local/state environmental office prior to discharge.

OPTIONS FOR CHLORINATED POOL WATER DISPOSAL

Option 1 (Best) = Discharge pool water to the sanitary sewer system with prior approval from the local water/wastewater utility:

- The facility must apply for and receive a discharge permit or other written authorization prior to discharge to the WWTP.
- The facility must determine and use an appropriate discharge flow rate to avoid any back-ups or overflows in the sanitary sewer system.
- The facility must prevent the discharge of soil, sediment, rock, debris, or other solid material during the discharge.

Figure 3: If water is dechlorinated, it can be released slowly to a grassy area to soak into the soil.
(Photo source: The Home Depot)

- Never discharge to an individual sewer system or septic system.

Option 2 (Better) = Discharge pool water to a vegetated area within the facility property:

- The area should be large enough to contain the discharge without any subsequent runoff or negative effect to neighboring properties.
- Care should be taken not to create puddles of standing water, which could potentially provide breeding grounds for mosquitoes.
- It is suggested that the pool water be dechlorinated to avoid damage to vegetation. This can be done by letting it sit for a sufficient period of time (~ 3 days) before discharging. If a pool has been treated with cyanuric acid, other dechlorination techniques will be needed (e.g., dechlorination unit). A home pool test kit (chlorine/pH) can help determine presence/absence of chlorine prior to discharge.

Option 3 (Good) = Discharge pool water to the storm sewer system if the discharge meets the following requirements:

- The chlorine residual is less than 0.01 milligrams per liter (mg/L).
- The pH is between 6.0 and 10.5.
- The water is clear (no algae, sediments, or other pollutants).
- The pool is not a saltwater swimming pool.
- No discharge directly to an open waterbody.

WASHING OPERATIONS

Facilities that conduct boat and vessel washing operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent contaminated wash water and pollutants from discharging to the storm drainage system or streams, rivers, and lakes	· Most USCG Units	· Sediment & dirt · Soaps, detergents, solvents, & other chemicals · Paint chips & other toxic materials · Grease & oils · Trash & debris · Hot water (thermal pollution) · Invasive Species

POLICIES AND GUIDANCE

Best practices for the boat and vessel washing activities along with special consideration for invasive species are covered in this EPC:

<u>WASHING OVER LAND (PREFERRED)</u>

- **Option 1** **(Best)** - Perform all cleaning operations in a wash rack. A wash rack is defined as a designated area that drains to an oil/water separator (OWS) or other type of treatment device set to discharge to a sanitary sewer system but not to the storm sewer system. The use of phosphate detergents is prohibited. If you must use a detergent, use phosphate-free and biodegradable detergent. Pressure washing is allowed within a wash rack up to 3,500 pounds per square inch (psi). Steam cleaning is approved up to 140^{o}F.

- **Option 2** **(Better)** - If a wash rack is not available, wash trailer-able boats off base at a commercial car wash.

- **Option 3** **(Good)** - If neither the wash rack nor the commercial car wash options are available, perform cleaning operation in a grassy area away from drainage pathways and surface waters with at least a 15-foot buffer around the perimeter. Spot clean small areas affected by grease and oil with a spray bottle and rag prior to washing. The use of phosphate detergents is prohibited. Spot cleaning detergents must be phosphate-free and biodegradable. Only freshwater rinse (low pressure) with a nozzle to limit flow is allowed. Pressure washing and steam cleaning are not allowed outside of a designated wash rack.

- **Option 4 (Not preferred)** - If washing must be performed on a marine travel lift (not preferred), confine washing activities to a designated area away from drainage pathways and surface waters. The use of all detergents is prohibited if wash water cannot be contained. Wash water with detergents must be treated by one of the following options:
  - Wash water must be completely contained, collected, characterized, and properly disposed of.
  - Wash water must be contained and allowed to evaporate. The area must then be swept, and any debris disposed of based on current waste characterization and/or generator knowledge.
  - Wash water may not be discharged to the storm sewer system or surface water unless an Individual NPDES Permit has been obtained.

WASHING OVER WATER (ONLY WHEN NO OTHER OPTION AVAILABLE)

Only spot cleaning of small areas affected by grease and oil is allowed with a spray bottle and rag. Spot cleaning detergents must be phosphate-free and biodegradable. Only freshwater rinse (low pressure) with a nozzle to limit flow is allowed. Pressure washing is not allowed over water.

- Boats MAY NOT be washed on or at the top of a boat ramp when detergents are used.
- Avoid in-water hull scraping unless necessary to the mission and only in areas necessary for the purposes of emergency repair.
- DO NOT discharge wash water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Non-wash rack washing operations should not be performed during wet weather conditions.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper washing procedures.

INVASIVE SPECIES INSPECTION AND DECONTAMINATION PROCEDURES FOR BOATS & VESSELS

FOR ADDITIONAL INFORMATION, THERE ARE MANY ONLINE RESOURCES THAT PROVIDE INSPECTION GUIDELINES, SUCH AS

Protect Your Waters
(https://stopaquatichitchhikers.org/)

1. Remove the boat, trailer, and/or asset from the water and away from the launch ramp for inspection and cleaning.
2. Thoroughly inspect all exposed surfaces on your vessel, trailer, or asset. If you find any mussels, scrape them off and kill them by crushing them. Dispose of the remains in the trash.
3. Remove all plants and mud from your boat, trailer, and/or asset. Dispose of all material in the trash.
4. Carefully feel your boat's hull and all equipment for any rough or gritty spots, which may be young mussels that have settled on your vessel and cannot be seen. Microscopic quagga and zebra mussels will feel like sandpaper.
5. Wash the boat's hull, trailer, equipment, bilge, and any other exposed surfaces with high-pressure, hot water. When possible, use water at a temperature of 140° F (60° C) at the hull – or about 155° (68° C) at the nozzle – which will kill the mussels. Dry the boat as much as possible. **Pressure washing must take place in a designated wash rack.** Refer to EPC – SW – PRESSURE WASHING ACTIVITIES for storm water management measures regarding pressure washing.
6. Drain all water from your boat and dry all areas, including the motor exterior, motor cooling system, live wells, ballast tanks, bladders, bilges, and lower outboard units. Make sure that all life jackets, equipment, or other items that have been in the water, including anchors, ropes, rafts, etc., are inspected, cleaned, and dried.
7. Clean and dry personal belongings, clothing, and footwear that have come in contact with the water.
8. Keep the boat, trailer, and/or asset dry for at least 5 days in warm, dry weather and up to 30 days in cool, moist weather before launching into a freshwater.

Photo Credit: Stopaquatichitchhikers.org

SPECIAL INVASIVE SPECIES CONSIDERATIONS

- Chemicals, such as bleach, have not been proven to be effective in disinfecting all aquatic invasive species and may damage your watercraft and equipment, so caution should be taken.
- Vinegar can be an effective method to decontaminate areas that do not drain completely (e.g., bilge areas of small boats). Dipping equipment into 100% vinegar for 20 minutes will help to kill harmful aquatic hitchhiker species. Over extended periods of time, vinegar may be corrosive to metal and is toxic to fish at this concentration, so thoroughly rinse with tap water. Ensure that the solution does not runoff directly to storm drains, drainage pathways or surface waters.

Hydrilla on Propellor *(Photo Credit - West Marine)*

- A 1 percent table salt solution for 24 hours can replace the vinegar dip. The following table provides correct mixtures for the 1 percent salt solution in water:

GALLONS OF WATER	CUPS OF SALT
5	2/3
10	1 1/4
25	3
50	6 1/4
100	12 1/3

- If hot water is not available, spray equipment such as boats, motors, trailers, anchors, buoys, and chains, with high-pressure water, and always allow enough time to DRY equipment. **Pressure washing must take place in a designated wash rack.**

AIRCRAFT WASHING OPERATIONS

Facilities that conduct aircraft washing operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent contaminated wash water and pollutants from discharging to the storm drainage system or streams, rivers, and lakes	· Air Stations · Bases (with flying operations)	· Sediment & dirt · Soaps, detergents, solvents, & other chemicals · Paint chips & other toxic materials · Grease & oils · Trash & debris · Hot water (thermal pollution)

POLICIES AND GUIDANCE

- **Option 1** **(Best)** = Perform all cleaning operations in a wash rack. A wash rack is defined as a designated area that drains to an OWS set to discharge to a sanitary sewer system and not to the storm sewer system. The use of detergent is not recommended but may be required depending on current maintenance procedures. If you must use a detergent, use phosphate-free and biodegradable detergent. Pressure washing is allowed within a wash rack up to 3,500 psi. Steam cleaning is approved up to 140°F.
- **Option 2** **(Better)** = If a wash rack set to discharge to an OWS is not available, only freshwater rinsing (low pressure) without the use of detergents is allowed. Perform rinsing operations away from drainage pathways and surface waters with at least a 15-foot buffer from the nearest storm water system drain or intake point. If no sheen is detected, rinse waters should be allowed to evaporate or discharge to a grassy area away from drainage pathways and surface waters with a 15-foot buffer from the nearest storm water system drain or intake point. Any rinse water containing petroleum, oils or lubricants (POLs) must be contained, collected, and disposed of properly.

GENERAL

- DO NOT discharge wash water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Non-wash rack washing operations should not be performed during wet weather conditions.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper washing procedures.

VEHICLE & EQUIPMENT WASHING OPERATIONS

Facilities that conduct vehicle and equipment washing operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent contaminated wash water and pollutants from discharging to the storm drainage system or streams, rivers, and lakes	· Most USCG Units	· Sediment & dirt · Soaps, detergents, solvents, & other chemicals · Paint chips & other toxic materials · Grease & oils · Trash & debris

POLICIES AND GUIDANCE

Best practices options for the vehicle and equipment washing activities are covered in this EPC.

- **Option 1** **(Best)** = Wash vehicles off base at a commercial car wash.
- **Option 2** **(Better)** = Perform all cleaning operations in a wash rack. A wash rack is defined as a designated area that drains to an OWS set to discharge to a sanitary sewer system and not to the storm sewer system. The use of detergent is not recommended. If you must use a detergent, use phosphate-free and biodegradable detergent. Pressure washing is allowed within a wash rack up to 3,500 psi. Steam cleaning is approved up to 140^{o}F. Refer to Attachment E – Advanced Structural BMPS in this Guide for additional details on wash racks.
- **Option 3** **(Good)** = If a wash rack is not available, perform cleaning operation in a grassy area away from drainage pathways and surface waters with at least a 15-foot buffer around the perimeter. If you must use a detergent, use phosphate-free and biodegradable detergent. Only freshwater rinse (low pressure) with a nozzle to limit flow is allowed. Pressure washing and steam cleaning are not allowed outside of a designated wash rack.

GENERAL

- DO NOT discharge wash water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Non-wash rack washing operations should not be performed during wet weather conditions.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper washing procedures.

PRESSURE WASHING ACTIVITIES

Facilities that conduct pressure washing activities shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent contaminated wash water and pollutants from discharging to the storm drainage system or streams, rivers, and lakes	· Most USCG Units	· Sediment & dirt · Soaps, detergents, solvents, & other chemicals · Paint chips & other toxic materials · Grease & oils · Trash & debris · Hot water (thermal pollution)

POLICIES AND GUIDANCE

PRESSURE WASHING OF BUOYS, SINKER CHAIN, AND OTHER LARGE OBJECTS

- If soaps or detergents are used and the surrounding area is paved, pressure wash waters and associated solids (e.g., paint solids, debris) must be collected, characterized, and disposed of properly. A wet vacuum or similarly effective device must be used to collect the runoff and loose materials.
- Pressure washing and steam cleaning are not allowed outside of a designated wash rack or station (pictured right).
- DO NOT discharge detergent or debris laden wash water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.

Pressure Washing Station
(Photo source: Water2Water)

- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper washing procedures.

PRESSURE WASHING OF BUILDINGS, ROOFTOPS, AND BOAT RAMPS

- If soaps or detergents are used and the surrounding area is paved, pressure wash waters and associated solids (e.g., paint solids, debris) must be collected, characterized, and disposed of properly. A wet vacuum or similarly effective device must be used to collect the runoff and loose materials.
- If soaps or detergents are not used, and the surrounding area is paved, wash water runoff does not have to be collected but should be screened prior to entering the storm sewer system. Use filter socks

(*pictured below*) or other type of screen on the ground and in the catch basin to trap solids in wash water runoff.

- Pavement wash waters must be free of spills or leaks of hazardous materials (e.g., POLs).
- DO NOT discharge detergent or debris laden wash water to a storm drain or surface water unless a separate Individual NPDES Permit has been obtained for this type of discharge.
- Inspect cleaning area regularly to ensure control measures are implemented and maintained.
- Train employees on proper washing procedures.

Filter Sock at Storm Drain
(Photo source: Eugene, OR Government)

WASTEWATER TREATMENT PLANT OPERATIONS

Facilities that conduct wastewater treatment plant (WWTP) operations shall establish procedures that incorporate the following storm water management measures to prevent or minimize pollutants from entering storm water runoff from the site. In most cases, USCG WWTPs are managed by contractors, but some units do have small pre-manufactured treatment facilities (package plants) which are used to treat wastewater when other options are not available.

Pollution Prevention Goal	Common Applications	Potential Pollutants
Prevent process chemicals, nitrogen, and phosphorus associated with wastewater treatment activities from entering the storm drainage system or streams, rivers, and lakes	· Operations that have small on-site package plants.	· Nitrogen · Phosphorus · Toxic chemicals

POLICIES AND GUIDANCE

Facilities with wastewater treatment package plants are generally required to have permit coverage under the National Pollutant Discharge Elimination System (NPDES) program. This requires development of a stormwater pollution prevention plan (SWPPP) and effluent monitoring. Thoughtful development and adherence to the SWPPP is the best safeguard against pollutant discharges from activities performed at the unit. The SWPPP should include the following best practices:

GENERAL

- Store process chemicals indoors. See storm water control measures in **EPC-SW-11 - HAZARDOUS MATERIALS AND WASTE STORAGE ACTIVITIES** for hazardous materials and waste storage.
- Train employees on proper process chemical management and spill prevention and controls.
- Submit NPDES discharge monitoring reports (DMRs) as required by permit. These are often submitted on a quarterly and/or annual basis. Refer to your NPDES permit for site-specific requirements. Consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD if you have questions regarding wastewater treatment package plant operation or monitoring.

ATTACHMENTS

ATTACHMENT A

LIST OF ACRONYMS

LIST OF ACRONYMS

A

ANT	Aids to Navigation Team
AST	Aboveground storage tank

B

BMP	Best management practice
BOD	Biological oxygen demand
BWE	Ballast water exchange

C

CFR	Code of Federal Regulations
CGP	Construction General Permit
COD	Chemical oxygen demand
COMDTINST	Commandant Instruction
CWA	Clean Water Act

D

DHS	Department of Homeland Security
DO	Dissolved oxygen

E

EISA	Energy Independence and Security Act
EMD	Environmental Management Division
EO	Executive Order
EPA	Environmental Protection Agency
EPC	Environmental Procedure Card
EPS	Environmental Protection Specialist

G

GPD	Gallons per day

L

LQG	Large Quantity Generator

M

MGD	Million gallons per day
MSGP	Multi-Sector General Permit
MS4	Municipal Separate Storm Sewer System

N

NAICS	North American Industry Classification System
NESU	Naval Engineering Support Unit
NPDES	National Pollutant Discharge Elimination System

O

OSHA	Occupational Safety and Health Administration
OWS	Oil/water separator

Q

QRP	Qualified Recycling Program

P

POL	Petroleum, oils, and lubricants

R

RCRA	Resource Conservation and Recovery Act
RUSLE	Revised Universal Soil Loss Equation

S

SAFR	Small arms firing range
SIC	Standard Industrial Code
SILC	Shore Infrastructure Logistics Center
SPCC	Spill prevention, control, and countermeasure
STANT	Station Aid to Navigation Team
SWPPP	Stormwater pollution prevention plan
SWPCP	Stormwater pollution control plan

T

TDS	Total dissolved solids
TMDL	Total Maximum Daily Load
TRACEN	Training Center
TSDF	Treatment, storage, and disposal facility
TSS	Total suspended solids

U

USCG	United States Coast Guard
USGS	United States Geological Survey

V

VOC	Volatile organic compound

W

WIP	Watershed implementation plan
WLA	Waste load allocation
WPA	Wellhead protection area
WWTP	Wastewater treatment plant

ATTACHMENT B

PERMIT MATRIX

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Oakland	D-11	Base	Alameda	Alameda	CA	X	X	
Cleveland	D-9	Station	Alexandria Bay	Wellesley Island	NY			X
Cleveland	D-9	Auxiliary Operation	Alpena	Alpena	MI			
Cleveland	D-5	Station	Annapolis	Annapolis	MD			
Honolulu	D-14	Station	Apra Harbor	Santa Rita	Guam			
Cleveland	D-9	Station	Ashtabula	Ashtabula	OH			
Cleveland	D-5	Air Station	Atlantic City	Egg Harbor Town	NJ	X		
Cleveland	D-5	Station	Atlantic City	Atlantic City	NJ			
Honolulu	D-14	Air Station	Barbers Point	Barbers Point	HI	X		
Cleveland	D-5	Station	Barnegat Light	Barnegat Light	NJ			
Miami	D-8	Marine Safety Unit	Baton Rouge	Baton Rouge	LA			
Cleveland	D-9	Station	Bayfield	Bayfield	WI			
Cleveland	D-9	Station	Belle Isle	Belle Isle	MI			
Oakland	D-13	Station	Bellingham	Bellingham	WA			
Oakland	D-11	Station	Bodega Bay	Bodega Bay	CA			
Providence	D-1	Station	Boothbay Harbor	Boothbay Harbor	ME			
Miami	D-7	Air Station	Borinquen	Aguadilla	PR	X	X	
Providence	D-1	Base	Boston	Boston	MA	X		
Providence	D-1	Station	Brant Point	Nantucket	MA			
Providence	D-1	ANT	Bristol	Bristol	RI			
Miami	D-7	Station	Brunswick	Brunswick	GA			
Cleveland	D-9	Sector	Buffalo	Buffalo	NY	X		
Providence	D-1	Station	Burlington	Burlington	VT			
Cleveland	D-9	Station	Calumet Harbor	Calumet City	IL			
Cleveland	D-5	JMTC	Camp Lejeune	Camp Lejeune	NC			
Cleveland	D-5	Station	Cape Charles	Cape Charles	VA			
Providence	D-1	Air Station	Cape Cod	Buzzards Bay	MA	X	X	
Providence	D-1	Station	Cape Cod Canal	Sandwich	MA			
Oakland	D-13	Station	Cape Disappointment	Ilwaco	WA			
Cleveland	D-5	Sector	Cape Hatteras	Cape Hatteras	NC	X		
Juneau	D-17	Lighthouse	Cape Hinchinbrook	Valdez-Cordova	AK			
Cleveland	D-5	TRACEN	Cape May	Cape May	NJ		X	
Providence	D-1	Station	Castle Hill	Newport	RI			
Providence	D-1	Academy	CG Academy	New London	CT		X	X
Juneau	D-17	Mooring	Mooring	Petersburg	AK			
Oakland	D-11	Mooring	Mooring	Eureka	CA			
Oakland	D-11	Mooring	Mooring	Santa Barbara	CA			
Oakland	D-13	Mooring	Mooring	Everett	WA			
Miami	D-8	Mooring	Mooring	Pensacola	FL			
Oakland	D-11	Mooring	Mooring	Crescent City	CA			
Oakland	D-11	Mooring	Mooring	San Diego	CA			
Juneau	D-17	Mooring	Mooring	Petersburg	AK			
Providence	D-1	Mooring	Mooring	Gloucester	MA			
Oakland	D-11	Mooring	Mooring	Marina Del Ray	CA			
Oakland	D-13	Mooring	Mooring	Everett	WA			
Juneau	D-17	Mooring	Mooring	Homer	AK			
Cleveland	D-9	Mooring	Mooring	Port Huron	OH			
Juneau	D-17	Mooring	Mooring	Auke Bay	AK			
Juneau	D-17	Mooring	Mooring	Sitka	AK			
Juneau	D-17	Mooring	Mooring	Seward	AK			
Oakland	D-11	Mooring	Mooring	Corona Del Mar	CA			
Oakland	D-13	Mooring	Mooring	Coos Bay	OR			
Oakland	D-13	Mooring	Mooring	Port Townsend	WA			

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Miami	D-8	Mooring	Mooring	Abbeville	LA			
Juneau	D-17	Mooring	Mooring	Homer	AK			
Oakland	D-11	Mooring	Mooring	San Diego	CA			
Miami	D-8	Mooring	Mooring	Carrabelle	FL			
Juneau	D-17	Mooring	Mooring	Cordova	AK			
Oakland	D-11	Station	Channel Islands Harbor	Oxnard	CA			
Miami	D-7	Air Facility	Charleston	Joint Base Charles	SC	X		
Miami	D-7	Base (Register St.)	Charleston	Charleston	SC	X		
Miami	D-7	Base (Tradd St.)	Charleston	Charleston	SC	X		
Cleveland	D-9	Station	Charlevoix	Charlevoix	MI			
Providence	D-1	Station	Chatham	Chatham	MA			
Miami	D-8	Cutter Support Team	Chattanooga	Chattanooga	TN			
Cleveland	D-5	CAMSLANT	Chesapeake	Chesapeake	VA			
Cleveland	D-5	MSST/MSRT	Chesapeake	Chesapeake	VA			
Oakland	D-13	Station	Chetco River	Harbor	OR			
Miami	D-8	Marine Safety Unit	Cincinnati	Cincinnati	OH			
Miami	D-7	Air Station	Clearwater	Clearwater	FL	X		
Cleveland	D-9	Station	Cleveland Harbor	Cleveland	OH			
Miami	D-8	ANT	Colfax	Colfax	LA			
Oakland	D-13	Base	Astoria	Warrenton	OR	X		
Oakland	D-13	Station	Coos Bay	Charleston	OR			
Juneau	D-17	AVSUPFAC	Cordova	Cordova	AK	X		
Miami	D-8	Air Station	Corpus Christi	Corpus Christi	TX	X		
Miami	D-8	Harbor Facility	Corpus Christi	Corpus Christi	TX			
Miami	D-8	Sector	Corpus Christi	Corpus Christi	TX	X		
Miami	D-7	Station	Cortez	Cortez	FL			
Cleveland	D-5	Station	Crisfield	Crisfield	MD			
Miami	D-8	Station	Dauphin Island	Dauphin Island	AL			
Cleveland	D-5	Sector	Delaware Bay	Philadelphia	PA	X		
Miami		Cutter Support Team	Demopolis	Demopolis	AL			
Oakland	D-13	Station	Depoe Bay	Newport	OR			
Miami	D-8	Station	Destin	Destin	FL			
Cleveland	D-9	Air Station	Detroit	Mt. Clemens	MI	X	X	
Cleveland	D-9	Sector	Detroit	Detroit	MI	X		
Miami	D-8	Cutter Support Team	Dubuque	Dubuque	IA			
Miami	D-8	ANT	Dulac	Dulac	LA			
Cleveland	D-9	Station	Duluth	Duluth	MN			
Cleveland	D-5	Sector Field Office	Eastern Shore	Chincoteague	VA	X		X
Providence	D-1	Station	Eastport	Eastport	ME			
Providence	D-1	Station	Eatons Neck	Northport	NY			X
Cleveland	D-5	Base	Elizabeth City	Elizabeth City	NC	X		
Cleveland	D-5	Station	Emerald Isle	Emerald Isle	NC			
Cleveland	D-9	Station	Erie	Erie	PA			
Miami	D-8	ANT	Eufaula	Eufaula	AL			
Cleveland	D-9	Station	Fairport Harbor	Painesville	OH			
Providence	D-1	Station	Fire Island	Babylon	NY			X
Cleveland	D-5	Atlantic Strike Team	Fort Dix	Fort Dix	NJ			
Miami	D-7	Station	Fort Lauderdale	Dania	FL			
Cleveland	D-5	Sector Field Office	Fort Macon	Atlantic Beach	NC	X	X	X
Miami	D-7	Station	Fort Myers Beach	Fort Myers Beach	FL			
Miami	D-7	Station	Fort Pierce	Fort Pierce	FL			
Cleveland	D-5	Station	Fortescue	Cape May	NJ			
Cleveland	D-9	Station	Frankfort	Frankfort	MI			

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Miami	D-8	Station	Freeport	Freeport	TX			
Miami	D-8	MSST	Galveston	Galveston	TX			
Miami	D-8	Sector Field Office	Galveston	Galveston	TX	X		
Miami	D-8	Station	Galveston	Galveston	TX			
Miami	D-7	Station	Georgetown	Georgetown	SC			
Providence	D-1	Station	Gloucester	Gloucester	MA			
Oakland	D-11	Station	Golden Gate	Sausalito	CA			
Cleveland	D-9	Sector Field Office	Grand Haven	Grand Haven	MI	X		
Cleveland	D-9	Station	Grand Haven	Grand Haven	MI			
Miami	D-8	Station	Grand Isle	Grand Isle	LA			X
Oakland	D-13	Station	Grays Harbor	Westport	WA			
Cleveland	D-5	Station	Great Egg	Atlantic City	NJ			
Miami	D-7	OPBAT	Great Inagua	Great Inagua	BH			
Cleveland	D-5	Auxiliary Operation	Green Bay	Green Bay	WI			
Cleveland	D-9	Station	Green Bay	Green Bay	WI			
Miami		Cutter Support Team	Greenville	Greenville	MS			
Honolulu	D-14	Sector	Guam	Santa Rita	Guam	X		
Miami	D-8	Station	Gulfport	Gulfport	MS			
Cleveland	D-9	Station	Harbor Beach	Harbor Beach	MI			
Cleveland	D-5	Station	Hatteras Inlet	Hatteras Inlet	NC			
Miami	D-8	Cutter Support Team	Hickman	Hickman	KY			
Cleveland	D-5	Station	Hobucken	Hobucken	NC			
Cleveland	D-9	Station	Holland	Holland	MI			
Honolulu	D-14	Base	Honolulu	Honolulu	HI	X		
Honolulu	D-14	COMMSTA	Honolulu	Wahiawa	HI			
Miami	D-8	Marine Safety Unit	Houma	Houma	LA			
Miami	D-8	Air Station	Houston	Houston	TX	X		
Miami	D-8	Sector	Houston	Houston	TX	X		
Oakland	D-11	Air Station	Humboldt Bay	McKinleyville	CA			
Oakland	D-11	Station	Humboldt Bay	Samoa	CA			
Miami	D-8	Marine Safety Unit	Huntington	Huntington	WV			
Cleveland	D-5	Station	Indian River Inlet	Rehoboth Beach	DE			
Cleveland	D-5	Station	Inigoes	St. Inigoes	MD			
Miami	D-7	Station	Islamorada	Islamorada	FL			
Miami	D-7	ANT	Jacksonville	Atlantic Beach	FL			
Miami	D-7	HITRON	Jacksonville	Jacksonville	FL	X		
Miami	D-7	Sector	Jacksonville	Atlantic Beach	FL	X		
Providence	D-1	Station	Jones Beach	Freeport	NY			
Providence	D-1	Station	Jonesport	West Jonesport	ME			
Juneau	D-17	Station	Juneau	Juneau	AK			
Honolulu	D-14	Station	Kauai	Lihue	HI			
Oakland	D-13	ANT	Kennewick	Kennewick	WA			
Cleveland	D-9	Station	Kenosha	Kenosha	WI			
Miami	D-8	Cutter Support Team	Keokuk	Keokuk	IA			
Juneau	D-17	Base	Ketchikan	Ketchikan	AK	X		
Miami	D-7	Sector	Key West	Key West	FL	X		
Miami	D-7	MFPU	Kings Bay	Kings Bay	GA			
Miami	D-7	MSST	Kings Bay	Kings Bay	GA			
Juneau	D-17	Base	Kodiak	Kodiak	AK	X		X
Miami	D-8	Station	Lake Charles	Lake Charles	LA			
Cleveland	D-5	Sector	Lake Michigan	Milwaukee	WI	X		
Oakland	D-11	Station	Lake Tahoe	Tahoe City	CA			
Miami	D-7	Station	Lake Worth Inlet	Riviera Beach	FL			
Cleveland	D-5	Station	Little Creek	Norfolk	VA			

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Providence	D-1	Sector	Long Island Sound	New Haven	CT	X		
Cleveland	D-9	Station/ANT	Lorain	Lorain	OH			
Oakland	D-11	Air Station	Los Angeles	Point Mugu	CA			
Oakland	D-11	Base	Los Angeles/Long Beach	San Pedro	CA	X		
Miami	D-8	Marine Safety Unit	Louisville	Louisville	KY			
Miami	D-8	Sector	Lower Mississippi River	Memphis	TN	X		
Cleveland	D-9	Station	Ludington	Ludington	MI			
Cleveland	D-5	Station	Manasquan Inlet	Point Pleasant Beach	NJ			
Cleveland	D-9	Station	Manistee	Manistee	MI			
Miami	D-7	Station	Marathon	Marathon	FL			
Cleveland	D-9	Station	Marblehead	Marblehead	OH			
Cleveland	D-9	Station	Marquette	Marquette	MI			
Honolulu	D-14	Station	Maui	Wailuku	HI			
Providence	D-1	Station	Menemsha	Chilmark	MA			
Providence	D-1	Station	Merrimack River	Newburyport	MA			
Miami	D-7	Air Station	Miami	Opa Locka	FL	X		
Miami	D-7	Base	Miami	Miami Beach	FL	X		
Miami	D-7	COMMSTA	Miami	Miami	FL			
Miami	D-7	MSST	Miami	Homestead	FL			
Cleveland	D-9	Station	Michigan City	Michigan City	IN			
Cleveland	D-5	Station	Milford Haven	Hudgins	VA			X
Miami	D-8	Aviation Training Center	Mobile	Mobile	AL	X	X	
Miami	D-8	Sector	Mobile	Mobile	AL	X		
Providence	D-1	Station	Montauk	Montauk	NY			X
Oakland	D-11	Station	Monterey	Monterey	CA			
Miami	D-8	ANT	Morgan City	Morgan City	LA			
Providence	D-1	Sector Field Office	Moriches	Moriches	NY	X		X
Oakland	D-11	Station	Morro Bay	Morro Bay	CA			
Cleveland	D-9	Air Facility	Muskegon	Muskegon	MI	X		
Cleveland	D-9	Station/ANT	Muskegon	Muskegon	MI			
Miami	D-8	Marine Safety Unit	Nashville	Nashville	TN			
Miami	D-8	Cutter Support Team	Natchez	Natchez	MS			
Oakland	D-13	Station	Neah Bay	Neah Bay	WA			
Providence	D-1	Station	New London	New London	CT			
Miami	D-8	Air Station	New Orleans	New Orleans	LA	X		
Miami	D-8	Base	New Orleans	New Orleans	LA	X		
Miami	D-8	COMMSTA	New Orleans	New Orleans	LA			X
Miami	D-8	Station	New Orleans	Metairie	LA			
Providence	D-1	ANT	New York	Bayonne	NJ			
Providence	D-1	Sector	New York	Staten Island	NY	X		
Providence	D-1	Station	New York	Staten Island	NY			
Oakland	D-13	Air Facility	Newport	Newport	OR	X		
Providence	D-1	Maintenance Augmentation	Newport	Newport	RI	X		
Cleveland	D-9	Station	Niagara	Youngstown	NY			
Oakland	D-13	Group/Air Station	North Bend	North Bend	OR	X		
Cleveland	D-9	Station	North Superior	Duluth	MN			
Providence	D-1	Sector	Northern New England	South Portland	ME	X		
Oakland	D-11	Station	Noyo River	Fort Bragg	CA			
Cleveland	D-5	Station	Oak Island	Oak Island	NC			
Cleveland	D-5	Station	Ocean City	Ocean City	MD			
Cleveland	D-5	Station	Ocracoke	Ocracoke	NC			

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Miami	D-8	Sector	Ohio Valley	Louisville	KY	X		
Miami	D-8	Cutter Support Team	Omaha	Omaha	NE			
Cleveland	D-5	Station	Oregon Inlet	Nags Head	NC			
Cleveland	D-9	Station	Oswego	Oswego	NY			
Miami	D-8	Cutter Support Team	Owensboro	Owensboro	KY			
Cleveland	D-5	Station	Oxford	Oxford	MD			
Oakland	D-11	Strike Team	Pacific	Novato	CA			
Miami	D-8	Marine Safety Unit	Paducah	Paducah	KY			
Miami	D-8	Station	Panama City	Panama City	FL			
Miami		Cutter Support Team	Paris Landing	Buchanan	TN			
Miami	D-8	Station	Pascagoula	Pascagoula	MS			
Miami	D-8	Station	Pensacola	Pensacola	FL			
Miami	D-8	Cutter Support Team	Peoria	East Peoria	IL			
Oakland	D-11	TRACEN	Petaluma	Petaluma	CA		X	X
Miami	D-7	Cutter Support Team	Pine Bluff	Pine Bluff	AR			
Providence	D-1	Station	Point Allerton	Hull	MA			
Providence	D-1	Station	Point Judith	Narragansett	RI			
Oakland	D-11	CAMSPAC	Point Reyes	Point Reyes	CA			
Miami	D-7	Station	Ponce de Leon Inlet	New Smyrna Beach	FL			
Oakland	D-13	Air Station	Port Angeles	Port Angeles	WA	X		
Miami	D-8	Station	Port Aransas	Port Aransas	TX			
Miami	D-8	Marine Safety Unit	Port Arthur	Port Arthur	TX			
Miami	D-7	Station	Port Canaveral	Cape Canaveral	FL			
Cleveland	D-9	Station	Port Huron	Port Huron	MI			
Miami	D-8	Station	Port O'Connor	Port O'Connor	TX			
Cleveland	D-9	Station	Portage	Dollar Bay	MI			
Oakland	D-13	Marine Safety Unit	Portland	Portland	OR			
Cleveland	D-5	Base	Portsmouth	Portsmouth	VA	X	X	
Providence	D-1	Station	Portsmouth Harbor	New Castle	NH			
Juneau	D-17	VTS	Potato Point	Valdez	AK			
Providence	D-1	Station	Provincetown	Provincetown	MA			
Cleveland	D-5	Port Security Unit	PSU 305	Fort Eustis	VA			
Miami	D-8	Port Security Unit	PSU 308	St. Louis	MO			
Cleveland	D-9	Port Security Unit	PSU 309	Port Clinton	OH			
Cleveland	D-5	CAMSLANT	Pungo	Pungo	VA			
Miami	D-8	MSD	Quad Cities	Rock Island	IL			
Oakland	D-13	Station	Quillayute River	LaPush	WA			
Oakland	D-11	Station	Rio Vista	Rio Vista	CA			
Cleveland	D-9	Station	Rochester	Rochester	NY			
Providence	D-1	Station	Rockland	Rockland	ME			
Miami	D-8	Station	Sabine	Port Arthur	TX			
Oakland	D-11	Air Station	Sacramento	McClellan	CA	X		
Cleveland	D-9	Station	Saginaw River	Essexville	MI			
Miami	D-8	Cutter Support Team	Sallisaw	Sallisaw	OK			X
Oakland	D-11	MSST	San Diego	San Diego	CA			
Oakland	D-11	Sector	San Diego	San Diego	CA	X		
Oakland	D-11	Air Station	San Francisco	San Francisco	CA	X	X	
Oakland	D-11	Sector	San Francisco	San Francisco	CA	X		
Miami	D-7	Sector	San Juan	San Juan	PR	X		
Miami	D-7	Station	Sand Key	Sand Key	FL			
Providence	D-1	Station	Sandy Hook	Sandy Hook	NJ			
Providence	D-1	ANT	Saugerties	Saugerties	NY			
Cleveland	D-9	Station	Sault Ste. Marie	Sault Ste. Marie	MI			

CEU AOR	District Location	Type	Unit	City	State	MSGP	MS4	Individual NPDES
Miami	D-7	Air Station	Savannah	Savannah	GA	X		
Oakland	D-13	Base	Seattle	Seattle	WA	X		
Miami	D-8	Cutter Support Team	Sewickley	Sewickley	PA			
Cleveland	D-5	Station	Shark River	Avon by the Sea	NJ			
Cleveland	D-9	Station	Sheboygan	Sheboygan	WI			
Providence	D-1	Station	Shinnecock	Hampton Bays	NY			X
Juneau	D-17	Air Station	Sitka	Sitka	AK	X		
Oakland	D-13	Station	Siuslaw River	Florence	OR			
Miami	D-8	Station	South Padre Island	South Padre Island	TX			
Providence	D-1	Sector	Southeastern New Eng	Woods Hole	MA	X		
Providence	D-1	Sector Field Office	Southwest Harbor	Southwest Harbor	ME	X		
Cleveland	D-9	Station	St. Clair Shores	St. Clair Shores	MI			
Miami	D-8	Resident Inspection Office	St. Croix	St. Croix	VI			
Cleveland	D-9	Station	St. Ignace	St. Ignace	MI			
Cleveland	D-9	Station	St. Joseph	St. Joseph	MI			
Miami	D-8	Base Detachment	St. Louis	St. Louis	MO			
Miami		MSD	St. Paul					
Miami	D-7	Sector	St. Petersburg	St. Petersburg	FL	X		
Miami	D-7	MSD	St. Thomas	St. Thomas	VI			
Cleveland	D-5	Station	Stillpond	Baltimore	MD			
Cleveland	D-5	Station	Sturgeon Bay	Sturgeon Bay	WI			X
Cleveland	D-9	Station	Tawas	East Tawas	MI			
Oakland	D-13	Station	Tillamook	Garibaldi	OR			
Cleveland	D-9	Station	Toledo	Toledo	OH			
Oakland	D-13	ANT	TONGUE Point	Astoria	OR			
Cleveland	D-5	Station	Townsend Inlet	Cape May	NJ			
Cleveland	D-9	Air Station	Traverse City	Traverse City	MI	X		
Cleveland	D-9	Station	Two Rivers	Two Rivers	WI			
Miami	D-7	Station	Tybee	Tybee Island	GA			
Oakland	D-13	Station	Umpqua River	Winchester Bay	OR			
Miami	D-8	Sector	Upper Mississippi River	St. Louis	MO	X		
Cleveland	D-5	Yard	USCG Yard	Baltimore	MD	X		
Juneau	D-17	Sector Field Office	Valdez	Valdez	AK			
Oakland	D-11	Station	Vallejo	Vallejo	CA			
Miami	D-8	Station	Venice	Venice	FL			X
Miami	D-8	Cutter Support Team	Vicksburg	Vicksburg	MS			
Cleveland	D-5	Station	Wachapreague	Wachapreague	VA			
Cleveland	D-5	ANT	Wanchese	Wanchese	NC			
Cleveland	D-5	Station	Washington	Washington	DC			
Cleveland	D-9	Station	Washington Island	Washington Island	WI			
Cleveland	D-9	Air Facility	Waukegon	Waukegon	IL	X		
Providence	D-1	Industrial Production Detach	Weymouth Buoy Depo	South Weymouth	ME			
Cleveland	D-9	Station	Wilmette Harbor	Wilmette	IL			
Cleveland	D-5	Marine Safety Unit	Wilmington	Wilmington	NC			
Providence	D-1	ANT	Woods Hole	Woods Hole	MA			
Providence	D-1	Station	Woods Hole	Woods Hole	MA			
Cleveland	D-5	Station	Wrightsville Beach	Wrightsville Beach	NC			
Miami	D-7	Station	Yankeetown	Yankeetown	FL			
Oakland	D-13	Station	Yaquina Bay	Newport	OR			
Cleveland	D-5	TRACEN	Yorktown	Yorktown	VA		X	

ATTACHMENT C

INDUSTRIAL MULTI-SECTOR GENERAL PERMIT (MSGP) DETAILS

ATTACHMENT C – INDUSTRIAL MULTI-SECTOR GENERAL PERMIT (MSGP) DETAILS

TABLE OF CONTENTS

1. INTRODUCTION

Industrial activities may require coverage under an NPDES Industrial Multi-Sector General permit (MSGP) for storm water discharges. Most units with applicable industrial facilities will need to obtain NPDES permit coverage through their state. EPA remains the permitting authority in a few states, most territories, and most Indian country.

An individual permit may be required when an industrial activity or pollutants generated from an activity is not included within the activities covered under the MSGP (i.e., industrial facilities with specialized discharges, POTWs, etc.). Since almost all USCG operations fall under general permitting parameters, general permit parameters are be discussed in detail within this attachment

The 11 categories of storm water discharges associated with industrial activity that are required to be covered under an MSGP are identified in 40 C.F.R. § 122.26(b)(14)(i)-(xi) as follows:

- Category One (i): Facilities subject to federal storm water effluent discharge standards in 40 C.F.R. Parts 405-471.
- Category Two (ii): Heavy manufacturing (for example: paper mills, chemical plants, petroleum refineries, and steel mills and foundries).
- Category Three (iii): Coal and mineral mining; oil and gas exploration and processing.
- Category Four (iv): Hazardous waste treatment, storage, or disposal facilities.
- Category Five (v): Landfills, land application sites, and open dumps with industrial wastes.
- Category Six (vi): Metal scrap yards, salvage yards, automobile junkyards, and battery reclaimers.
- Category Seven (vii): Steam electric power generating plants.
- Category Eight (viii): Transportation facilities that have vehicle maintenance, equipment cleaning, or airport de-icing operations.
- Category Nine (ix): Treatment works treating domestic sewage with a design flow of 1 million gallons a day or more.
- Category Ten (x): Covered in Section 2.3 of the Guide, construction sites that disturb 1 acre or more, including smaller sites in a larger common plan of development or sale.
- Category Eleven (xi): Light manufacturing (for example, food processing, printing, and publishing, electronic and other electrical equipment manufacturing, and public warehousing and storage).

A: Timber Products	P: Land Transportation
B: Paper Products	Q: Water Transportation
C: Chemical Products	R: Ship/Boat Building, Repair
D: Asphalt/Roofing	S: Air Transportation
E: Glass, Clay, Cement	T: Treatment Works (WWTPs)
F: Primary Metals	U: Food Products
G: Metal Mining	V: Textile Mills
H: Coal Mines	W: Furniture and Fixtures
I: Oil and Gas	X: Printing, Publishing
J: Mineral Mining	Y: Rubber, Misc. Plastics
K: Hazardous Waste	Z: Leather Tanning/Finishing
L: Landfills	AA: Fabricated Metal Products
M: Auto Salvage Yards	AB: Transportation Equip.
N: Scrap Recycling	AC: Electronic, photo goods
O: Steam Electric Facilities	AD: Non-classified facilities

Sectors of Industrial Activity
(Source: https://www.epa.gov/)

The MSGP has broken down the 11 categories discussed in the previous section even further into Industrial Activity Sectors (*at right)* that are determined mainly by the facility's Standard Industrial Classification (SIC) or North American Industry

Classification System (NAICS) code. For the USCG, this can be confusing since the primary SIC code for the USCG is **9621** which does not require a MSGP permit. The USCG contends that this SIC classification includes the following major categories: **search and rescue, oil spill response, marine safety, security, pollution prevention, and the administration and enforcement of USCG regulations at Title 33 C.F.R., which includes: vessel operation and security, international and inland navigation, aids to navigation, bridges, anchorages, homeland security, maritime security, regattas and parades, waterfront facilities, oil spills, pollution prevention, ports and waterway safety, and boating safety.** Again, the NPDES MSGP permit program does not regulate SIC code 9621 as it is not included in the 11 industrial categories.

Activities undertaken by the USCG that do require MSGP coverage and their respective industrial category, SIC/NAICS code, and Sector are provided in Table 1-1. Examples of these operations and their applicability to the MSGP permit are further discussed by Sector in Section 2 of this Attachment.

Table 1-1 – Summary of Coast Guard Unit Operations Regulated under EPA's Industrial MSGP

USCG Activity	Industrial Category	SIC Code or Activity Code	Sector
Ship and Boat Repair	2	3731	R
	2	3732	R
Hazardous Waste Storage	4	HZ	K
Landfill	5	LF	L
Water Transportation	8	4499	Q
Air Transportation	8	4581	S
Land Transportation	8	5171	P
Sewage Treatment	9	TW	T

In some cases, more than one industrial sector may be associated with a facility. For example, unit activities may include ship and boat repair (Sector R) and Hazardous Waste Storage (Sector K). This would require one permit but would have multiple requirements under that permit for each of the industrial activities and are considered *co-located industrial activities*.

Most USCG facilities have previously determined MSGP permitting requirements and are covered under appropriate permits. If you have questions about permitting at your USCG facility, contact a storm water SME at your servicing CEU, Sector or SILC EMD who will help to determine the correct industrial sector (Air Transportation, Water Transportation, etc.) and establish whether MSGP regulations apply there.

FOR ADDITIONAL INFORMATION OR CLARIFICATION OF REQUIREMENTS OR PROCEDURES DISCUSSED IN THIS GUIDE, UNIT PERSONNEL SHOULD SEEK ASSISTANCE THROUGH THEIR STORM WATER SMES AT SERVICING SECTOR, CEU, OR SILC EMD.

2. SUMMARY OF APPLICABLE INDUSTRIAL MSGP ACTIVITIES

The following is a summary of the Industrial Sectors which are regulated under the MSGP and apply to activities undertaken by the USCG. These SIC and activity codes ("code") should be utilized when considering a facility's primary and co-located industrial activities.

2.1.1 Sector R: Ship and Boat Building and Repairing Yards

- **Major Group 37: Transportation Equipment**
    - Industry Group 373: Ship and Boat Building and Repairing
        - Code 3731: Ship Building and Repairing **(Sector R)**
        - Code 3732: Boat Building and Repairing **(Sector R)**

Code 3731 identifies ship building and repairing and includes establishments primarily engaged in building and repairing ships, barges, and lighters, whether self-propelled or towed by other craft. This industry also includes the conversion and alteration of ships and the manufacture of off-shore oil and gas well drilling and production platforms (whether or not self-propelled). Establishments primarily engaged in fabricating structural assemblies or components for ships, or subcontractors engaged in ship painting, joinery, carpentry work, and electrical wiring installation, are classified in other industries.

Code 3732 identifies boat building and repairing and includes establishments primarily engaged in building and repairing boats. Establishments primarily engaged in manufacturing rubber and non-rigid plastics boats are classified in Major Group 30. Establishments primarily engaged in operating marinas and which perform incidental boat repair are classified in Transportation, Industry 4493; membership yacht clubs are classified in Services, Industry 7997; and those performing outboard motor repair are classified in Services, Industry 7699.

The USCG **does** conduct ship and boat repair and maintenance activities that would fit SIC Codes 3731/3732 at several faciltities.

Qualifying boat and engine repair is typically conducted at Bases, Sectors, and the CG Yard. Therefore, **Stations should not have SIC codes 3731/3732**. Bases, **and** Sectors conducting boat repair should have secondary SIC codes of 3731/3732, as Ship and Boat Building and Repairing is not the primary function of the facility. The following operations would be included as industrial activity for USCG facilities with a SIC code of 3731/3732:

- Ship and boat repair or maintenance
- Engine repair and maintenance
- Bulk liquid storage and containment
- Fueling
- Washing (vessels and vehicles)
- Painting (preparation and application)
- Material handling
- Material storage

The activities, pollutant sources, and pollutants that commonly occur at ship and boat repair and maintenance facilities are summarized in Table 2-1.

Table 2-1 – Common Activities, Pollutant Sources, and Pollutants Associated with Ship and Boat Repair and Maintenance Facilities

Activity	Pollutant Source	Pollutant
Pressure washing/rinsing	Wash water	Detergent, paint solids, heavy metals, suspended solids
Paint removal, sanding/blasting	Sanding, grinding, blasting, paint stripping	Spent abrasives, paint solids, heavy metals, solvents, dust, debris
Painting	Paint and paint thinner spills, overspray, paint stripping, sanding, and paint cleanup	Paint solids, spent solvents, heavy metals, dust, debris
Drydock operation and maintenance	Sanding, grinding, blasting, paint stripping, building materials	Spent abrasives, paint solids, heavy metals, solvents, dust, debris, floatable waste
Engine maintenance and repairs	Parts cleaning; waste disposal; fluid spills	Spent solvents, heavy metals, ethylene glycol, acid/alkaline wastes, detergents, rags, batteries, loose parts
Material handling	Fueling: spills and leaks, area wash down	Fuel, oil, heavy metals
	ASTs: spills and overfills, external corrosion, failure of piping systems	Fuel, oil, heavy metals
	Waste Storage: paint solids, solvents, spent abrasives, petroleum products, outdoor equipment/material storage	Paint solids, heavy metals, spent solvents, oil
Buoy operations	Chain storage, sanding, blasting, paint stripping, painting	Spent abrasives, paint solids, heavy metals, solvents, dust, debris

Based on EPA's sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

2.1.2 Sector K: Hazardous Waste Treatment, Storage, or Disposal Facilities

Activity Description: Hazardous Waste Treatment, Storage, or Disposal Facilities (TSDFs), including those that are operating under interim status or a permit under subtitle C of the Resource Conservation and Recovery Act (RCRA)

- Code HZ: Hazardous Waste Treatment, Storage, or Disposal Facilities **(Sector K)**

Hazardous waste TSDFs, including those that are operating under interim status or a permit under subtitle C of RCRA are required to obtain an NPDES permit for Sector K.

Most USCG units are Very Small Quantity Generators (VSQG) or Small Quantity Generators (SQG). Sector K does not apply to VSQG and SQG if operating within their respective RCRA provisions. In addition, Sector K does not apply to Large Quantity Generators (LQGs) operating in compliance and disposing in less than 90 days.

The activities, pollutant sources, and pollutants commonly occurring at hazardous waste TSDFs are summarized in Table 2-2.

Table 2-2 – Common Activities, Pollutant Sources, and Pollutants Associated with Hazardous Waste Treatment, Storage, or Disposal Facilities

Activity	Pollutant Source	Pollutant
Bulk Liquid/Solid Transfer	Spills during transfer of chemicals between above ground storage tank and drums or other containers	Acids, solvents, ammonia, hydroxides, detergents, fuels
	Spills or leaks of hazardous materials used for operations	Total suspended solids (TSS), chemical oxygen demand (COD) pH, biological oxygen demand (BOD)
	Outdoor storage or handling of chemicals	Organic and inorganic compounds
	Unloading of chemicals and other hazardous materials	
	Leaks and spills of acids or solvents from drums or tanks	
Hazardous Material Storage	Spills or leaks	Organic and inorganic compounds
	Residual hazardous material due to poor housekeeping	
Waste Handling & Disposal	Chemical mixing	Mixed waste which can limit recyclables
Building and Grounds Maintenance	Storage of pesticides and other chemicals	Pesticides, oxygen-demanding substances, sediments, nutrients, organics, and toxicants
	Application of chemicals	

Based on EPA's sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

2.1.3 Sector L: Landfills, Land Application Sites, and Open Dumps with Industrial Wastes

Activity Description: Landfills, land application sites, and open dumps that receive or have received any industrial wastes including those that are subject to regulation under subtitle D of RCRA.

- Code LF: All Landfill, Land Application Sites and Open Dumps **(Sector L)**

All facilities of this type require coverage under an industrial storm water permit. This also includes sites subject to regulation under subtitle D of RCRA including municipal solid waste landfills, industrial solid nonhazardous waste landfills, and industrial waste land application sites.

The activities, pollutant sources, and pollutants commonly occuring at landfills are summarized in Table 2-3.

Table 2-3 – Common Activities, Pollutant Sources, and Pollutants Associated with Landfill Operations

Activity	Pollutant Source	Pollutant
Cover crop management	Applied chemicals	Fertilizers, pesticides, and herbicides
Outdoor chemical storage	Exposure of chemical material storage areas to precipitation	Various chemicals stored
Waste transportation	Waste tracking on-site and haul road, solids transport on wheels and exterior of trucks and equipment	Total suspended solids (TSS), total dissolved solids (TDS), turbidity, floatable
Leachate collection	Commingling of leachate with runoff or run-on	Iron, TSS, biological oxygen demand (BOD), ammonia, alpha terpineol, benzoic acid, p-cresol, phenol, zinc, pH
Landfill operations	Exposure of waste	BOD, TSS, TDS, turbidity
Exposed soil from excavation	Erosion	TSS, TDS, turbidity
Exposed stockpiles of cover material		
Inactive cells not fully stabilized		
Daily or intermediate cover of cells		
Haul roads		

Based on EPA's sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

2.1.4 Sector Q: Water Transportation

- Industry Group 4493: Services Incidental to Water Transportation

Code 4493 identifies facilities primarily engaged in operating marinas. These establishments rent boat slips and store boats, and generally perform a range of other services including cleaning and incidental boat repair. They frequently sell food, fuel, and fishing supplies, and may sell boats. Establishments primarily engaged in building or repairing boats and ships are classified in Manufacturing, Industry Group 373.

The following operations would be considered an industrial activity for USCG facilities with a SIC code of 4493:

- Boat yards, storage, and incidental repair
- Marinas
- Marine basins, operation of
- Yacht basins, operation of

Since the USCG does not operate marinas (either primary or co-located activities) and is not in the business of renting boat slips or selling boats, then **USCG facilities should not be subject to storm water permit requirements associated with this SIC code**. Some states' interpretation of the MSGP views this differently, therefore marina operations was included in this guide. A facility that conducts cleaning, incidental boat repair, and boat yard storage but does not otherwise engage in the operation of a marina is not required to obtain a storm water permit for those industrial activities because they alone are not an activity identified by one of the 11 industrial categories and therefore should not be considered co-located activities. Therefore, USCG facilities should not be regulated under the MSGP for SIC code 4493.

- Industry Group 4499: Services Incidental to Water Transportation

SIC code 4499 identifies facilities that conduct water transportation services that are not elsewhere classified. It includes establishments primarily engaged in furnishing miscellaneous services incidental to water transportation, not elsewhere classified, such as lighterage; boat hiring, except for pleasure; chartering of vessels; canal operation; ship cleaning, except hold cleaning; and steamship leasing.

The following operations would be considered an industrial activity for USCG facilities with a SIC code of 4499:

- Boat cleaning
- Boat hiring, except pleasure
- Boat livery, except pleasure
- Boat rental, commercial
- Canal operation
- Cargo salvaging, from distressed vessels
- Chartering of commercial boats
- Dismantling ships
- Lighterage
- Marine railways for drydocking, operation
- Marine salvaging
- Marine surveyors, except cargo
- Marine wrecking: ships for scrap
- Piloting vessels in and out of harbors
- Ship cleaning, except hold cleaning
- Ship registers: survey and classification of ships and marine
- Steamship leasing

Since the USCG does not conduct water transportation services (either primary or co-located activities) and is not in the business of providing ship cleaning, boat cleaning, or equipment cleaning services, then USCG facilities should not be subject to storm water permit requirements associated with this SIC code. A facility that conducts boat cleaning but does not otherwise engage in water transportation services (e.g., provide boat cleaning services) is not required to obtain a storm water permit for that industrial activity because it is not an activity identified by one of the 11 industrial categories and is therefore not a co-located activity. **Therefore, USCG facilities should not be regulated under the MSGP for SIC code 4499.**

2.1.5 Sector S: Air Transportation Facilities

- Industry Group 458: Airports Flying Fields and Airport Terminal

Industrial code 4581 identifies Airports, Flying Fields, and Airport Terminal Services. This includes establishments primarily engaged in operating and maintaining airports and flying fields; in servicing, repairing (except on a factory basis), maintaining, and storing aircraft; and in furnishing coordinated handling services for airfreight or passengers at airports. In general, USCG Air Stations, Air Facilities, the Air Training Facility, and Bases with airfields **do** conduct aircraft repair and maintenance and maintain the airfield (e.g., tarmac maintenance and tarmac de-icing); therefore, SIC code 4581 would apply to most USCG airfields. The following operations would be considered an industrial activity for USCG facilities with a SIC code of 4581.

- Aircraft Maintenance
- De-icing aircraft, runways, and pads
- Aircraft storage
- Aircraft cleaning
- Bulk liquid storage and containment
- Fueling
- Washing (vessels and vehicles)
- Painting (preparation and application)
- Material handling
- Material storage

The activities, pollutant sources, and pollutants commonly found at air transportation facilities are summarized in Table 2-4.

Table 2-4 – Common Activities, Pollutant Sources, and Pollutants Associated with Air Transportation Facilities

Activity	Pollutant Source	Pollutant
Aircraft, ground vehicle and equipment maintenance and washing	Spills or leaks	Oils, hydraulic/radiator fluids, chemical solvents
	Disposal of waste	Batteries, oil, filters, oily rags
	Wash water	Detergent, TSS, metals, fuel, hydraulic fluid, oil
Aircraft fueling	Spills or leaks, runoff from fueling areas, wash down of fueling areas	Fuel, fuel additives, oil, lubricants, heavy metals
Aircraft servicing	Spills or leaks	Oil, hydraulic fluid, fuel
Runway maintenance	Wash water	Tire rubber, oil, fuel, paint chips, chemical solvents
Runway and aircraft de-icing/anti-icing	Runoff from deicing area	Ethylene/propylene glycol, urea, potassium/sodium acetate, potassium/sodium formate

Based on EPA's sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

2.1.6 Sector P: Petroleum Bulk Stations and Terminals

- Industry Group 517: Petroleum and Petroleum Products

Code 5171 characterizes establishments primarily engaged in the wholesale distribution of crude petroleum and petroleum products, including liquefied petroleum gas, from bulk liquid storage facilities. **Since the USCG does not engage in the wholesale distribution of crude petroleum and petroleum products, then USCG facilities should not subject to storm water permit requirements associated with this SIC code.** The activities, pollutant sources, and pollutants commonly occurring at facilities performing fueling and defueling activities and bulk fuel storage are summarized in Table 2-5.

Table 2-5 – Common Activities, Pollutant Sources, and Pollutants Associated with Fueling Operations

Activity	Pollutant Source	Pollutant
Fueling	Spills and leaks during fuel delivery	Fuel, oil, heavy metals
	Spills caused by “topping off” fuel tanks	
	Storm water run-on into the fuel area	
	Washing down fuel area	
	Leaking storage tanks	
Liquid storage in ASTs	External corrosion and structural failure	Oil, fuel, grease, heavy metals
	Installation problems	
	Spills and overfills	
	Failure of piping systems (pipes, pumps, flanges, couplings, hoses and valves)	
Petroleum loading/unloading	Spills and overfills	Oil, fuel, grease
Vehicle and equipment washing and maintenance	Parts cleaning	Chlorinated solvents, oil, heavy metals, acid/alkaline wastes
	Waste disposal of greasy rags, oil filters, air filters, batteries, hydraulic fluids, transmission fluid, radiator fluids, degreasers	Oil, heavy metals, chlorinated solvents, acid/alkaline wastes, ethylene glycol
	Fluids replacement including oil, hydraulic fluids, transmission fluid, radiator fluids	Oil, arsenic, heavy metals, organics, chlorinated solvents, ethylene glycol
	Washing or stream cleaning	Oil, detergent, heavy metals, chlorinated solvents, phosphorus
	Spills of oil, degreasers, hydraulic fluids, transmission fluid, radiator fluids	Oil, arsenic, heavy metals, organics, chlorinated solvents, ethylene glycol
Outdoor vehicle and equipment storage	Leaking vehicle fluids, brake dust	Oil, hydraulic fluids, arsenic, heavy metals, organics, fuel

Based on EPA’s sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

2.1.7 Sector T: Treatment Works Treating Domestic Sewage

- Activity Code TW: Treatment Works **(Sector T)**

This sector is defined by an activity treating domestic sewage or any other sewage sludge or wastewater treatment device or system. Note that if a unit uses the local municipality publicly owned treatment works (POTW) then this does not apply. This treatment is used in the storage, treatment, recycling, and reclamation of municipal or domestic sewage, including land dedicated to the disposal of sewage sludge that are located within the confines of the facility, with a design flow of 1.0 million gallons per day (MGD) or more. It is further defined as requiring an approved pretreatment program under 40 C.F.R. Part 403. Not included are farmlands, domestic gardens, or lands used for sludge management where sludge is beneficially reused and which are not physically located in the confines of the facility, or areas that are compliant with section 405 of the CWA. The activities, pollutant sources, and pollutants commonly occurring at wastewater treatment plant facilities are summarized in Table 2-6.

Table 2-6 – Common Activities, Pollutant Sources, and Pollutants Associated with Wastewater Treatment Plants

Activity	Pollutant Source	Pollutant
Preparation of chemical, biological and physical treatment processes	Spills and leaks of process chemicals and materials	Disinfectants, polymers and coagulants, alum, ferric chloride, soda ash, lime, sodium aluminate, sodium hypochlorite, caustic soda, chlorine, sodium bisulfite
Soil amending and grass fertilizing	Over-fertilization	Nitrogen, phosphorous, ammonia, aluminum sulfate, liquid chlorine, liquid polymer, fuel oil
Pest control	Application and storage	Diazanon, malathion, amdro, dimethylphthalate, diethyl phthalate, dichlorvos, carbaryl, skeetal, batex, liquid copper
Sludge drying beds or storage piles	Sludge	Nitrate, TDS, TSS, ammonia, fecal pathogens
Sludge transfer	Sludge, vehicles, transfer equipment	Nitrate, TDS, TSS, oil, fuel, hydraulic fluids, ammonia, fecal pathogens
Septage transfer	Solid and liquid sanitary waste, vehicles	Nitrate, TDS, TSS, oil, fuel, hydraulic fluids, ammonia, fecal pathogens
Incineration	Ash impoundments/piles	Heavy metals, TDS, TSS

Based on EPA's sector-specific *Industrial Storm Water Fact Sheet Series* https://www.epa.gov/npdes/industrial-stormwater-fact-sheet-series

3. REQUIRED MONITORING

<u>Benchmark Monitoring</u>

Benchmark monitoring requirements only apply to units in the process of obtaining new permits.

Benchmark monitoring data are primarily for a facility's use to determine the overall effectiveness of the storm water control measures that have been implemented at the site. Units that fall under Sectors K, L, Q, or S (primary or co-located) are subject to benchmark monitoring requirements, which must be conducted quarterly for the first four quarters of permit coverage. Table 3-1 provides a summary of sectors requiring benchmark monitoring.

Units that discover an exceedance of a benchmark concentration specified in their permit should notify their CEU immediately. Units should work with their CEU to review the selection, design, installation, and implementation of their control measures to determine if modifications are necessary. Units must also follow any corrective action reporting requirements specified in their permit.

<u>Effluent Limitation Monitoring</u>

Units that fall under Sectors K and L are subject to effluent limitations monitoring, which must be conducted annually at each outfall containing discharges associated with runoff from these two industrial activities. Units that discover an exceedance of a numeric effluent limit specified in their permit should notify their CEU. Units should work with their CEU to review the selection, design, installation, and implementation of their control measures to determine if modifications are necessary. Units must also follow any corrective action reporting requirements specified in their permit.

Table 3-1 – Applicable Monitoring by Sector

Sector	Sector Description	Benchmark Monitoring Required?	Effluent Monitoring Required?
Sector R	Ship and Boat Building and Repairing Yards	No	No
Sector K	Hazardous Waste Treatment, Storage, or Disposal Facilities	Yes	Yes
Sector L	Landfills, Land Application Sites, and Open Dumps with Industrial Wastes	Yes	Yes
Sector Q	Water Transportation	Yes	No
Sector S	Air Transportation Facilities	Yes	No
Sector P	Petroleum Bulk Stations and Terminals	No	No
Sector T	Treatment Works Treating Domestic Sewage	No	No

4. ENTERPRISE-WIDE DETERMINATIONS

The following facilities will utilize the primary SIC codes and secondary SIC codes based on the primary activities and co-located industrial activities defined herein. All facilities must confirm that actual on-site activities are consistent with the following determinations.

USCG Yard
Primary SIC Code: 3731 and/or 3732
Secondary SIC Code: 9621

USCG Academy
Primary SIC Code: 8221
Secondary SIC Code: None

Training Centers
Primary SIC Code: 8249
Secondary SIC Code: None

The USCG Training Centers (TRACENs) Yorktown, Cape May, and Petaluma will need to verify if they are subject to an NPDES permit for a small MS4 (see **Section 2.3 of Part A** of this Guide for additional information regarding MS4 permitting requirements). The Aviation Training Center (ATC) Mobile would include a secondary SIC Code of 4581.

Base Elizabeth City
Primary SIC Code: 9621
Secondary SIC Code: 4512
Additional Activity Code: HZ

The primary activity at Base Elizabeth City would be the activities associated with SIC code 9621. Base Elizabeth City also operates an airport (secondary SIC code 4581) and holds a RCRA Part B Permit (secondary activity code HZ).

Base Elizabeth City may also need to have an NPDES permit in accordance with Sectors K and S facility requirements, which includes specified effluent limitation guidelines for Sector K and benchmark sampling for Sector S.

Base Kodiak
Primary SIC Code: 9621
Secondary SIC Code: 3731, 3732, 4581
Additional Activity Code: HZ, LF, TW

The primary activity at Base Kodiak would be the activities associated with SIC code 9621. In accordance with current NPDES permit coverage, Base Kodiak functions as a water transportation facility (secondary SIC code 4499/Sector Q), operates an airport (secondary SIC code 4581/Sector S), conducts land transportation activities and bulk fuel storage (secondary SIC code 5171/Sector P), holds a RCRA Part B Permit (secondary activity code HZ/Sector K), operates a landfill (secondary SIC code LF/Sector L), and operates a Wastewater treatment plant (secondary activity code TW/Sector T).

Base Kodiak may also need to have NPDES permit coverage in accordance with Sectors K, L, R, S and T, which includes benchmark sampling and specified effluent limitation guidelines for Sectors K and L and benchmark sampling for Sectors L and S. Coverage under Sectors P and Q is not applicable and should not be required by the permitting authority.

Bases (other)
Primary SIC Code: 9621
Secondary SIC Code: 3731 and/or 3732

The primary activity at Bases would be the activities associated with SIC code 9621. A review of the types of personnel at Bases shows that engineering and maintenance personnel represent an average of 35% of the USCG staffing. In addition, USCG policy indicates that Bases are conducting intermediate-level or depot-level boat repair. Bases conducting intermediate-level and depot-level ship and boat repair will have a secondary SIC Code of 3731 and/or 3732 if the Base (s) has structural best management practices to control discharges.

Under these parameters, Bases (other) are required to have an NPDES permit in accordance with Sector R facility requirements.

Sectors
Primary SIC Code: 9621
Secondary SIC Code: 3731 and/or 3732

The primary activity at Sectors would be the activities associated with SIC code 9621. A review of the types of personnel at Sectors shows that engineering and maintenance personnel represent an average of 23% of staff. In addition, USCG policy indicates that Sectors are conducting intermediate-level or depot-level boat repair. Sectors conducting intermediate-level and depot-level ship and boat repair will have a secondary SIC Code of 3731 and/or 3732 if the Sector (s) has structural best management practices to control discharges.

Under these parameters, Sectors are required to have an NPDES permit in accordance with Sector R facility requirements.

Stations
Primary SIC Code: 9621
Secondary SIC Code: None

The primary activity at most Stations would be the activities associated with SIC code 9621. A review of the types of personnel at Stations shows that engineering and maintenance personnel represent an average of 22% of the USCG staffing. In addition, USCG policy indicates that Stations should not be conducting boat repair and should only be conducting boat and engine maintenance. Therefore, there should be no co-located activities under another SIC code.

Aid to Navigation Teams ANTs
Primary SIC Code: 9621
Secondary SIC Code: None

The primary activity at ANTs would be the activities associated with SIC Code 9621. There should be no co-located industrial activities under another SIC code. Additionally, ANTs are typically a tenant on a larger host command.

Air Stations and Air Facilities
Primary SIC Code: 9621
Secondary SIC Code: 4581

The primary activity at Air Stations and Air Facilities would be the activities associated with SIC code 9621. Since Air Stations and Air Facilities would also operate an airfield, the secondary SIC code is 4581.

Air Stations and Air Facilities are required to have an NPDES permit in accordance with Sector S facility requirements which includes specified benchmark sampling.

Note that leased hangars should include MSGP requirements in lease agreements, outlining if 'lessor' or 'lessee' is the MSGP permit holder and what operational responsibilities apply under the permit.

Communication Stations
Primary SIC Code: 9621
Secondary SIC Code: None

The primary activity at Communication Stations would be the activities associated with SIC code 9621. There should be no co-located industrial activities under another SIC code.

Areas & Districts

Primary SIC Code:	9621
Secondary SIC Code:	None

The primary activity at Area and District facilities would be the activities associated with SIC code 9621. There should be no co-located industrial activities under another SIC code.

ATTACHMENT D

SPECIAL CONSIDERATIONS BY STATE

ATTACHMENT D – SPECIAL CONSIDERATIONS BY STATE

1. SPECIAL CONSIDERATIONS BY STATE

The following are some known permitting considerations that are either not included in the EPA's Multi-Sector General Permit or vary slightly from it. These are considered more stringent than the MSGP or pertain to items not explicitly covered under the MSGP. The stormwater permitting regulations for each state that has USCG operations were researched for these permitting special considerations. The list of state requirements below changes with time as regulations are updated. Also, interpretations of the regulations and who may or may not require permit coverage varies. Consult with storm water subject matter experts (SMEs) at your servicing Sector, CEU, or SILC EMD if you have any questions regarding special considerations.

1.1 ALABAMA

Washing Operations

Bilge and ballast water, wash water (including pressure wash water resulting from the removal of marine growth), or waters from hydrostatic and pressure testing, all which have NOT come into contact with product, waste, or waste residuals, are allowable discharges under the *State of Alabama MSGP for Boat and Ship Building and Repair,* with the following limitations:

1. Only biodegradable and phosphate-free detergents and/or surfactants are to be used in cleaning activities from which a discharge occurs.

2. Direct deck drainage to a collection system sump for settling and/or additional treatment if contaminants are present. Discharges of treated deck drainage must be included as one of the designated outfalls for which coverage has been granted under the permit prior to discharge. Otherwise, treated or contaminated deck drainage must be collected and properly disposed of offsite.
3. Pressure washing is allowed as necessary to remove marine growth from vessels. However, unless the pressure wash water is collected and treated prior to discharge, (a) pressure shall be maintained at or below a level which will not result in the removal of the existing paint coating; (b) it shall be performed only in areas of the hull; and (c) the paint coating will not be significantly disturbed by pressure washing. Pressure wash water contaminated with paint chips shall be collected and properly disposed of off-site or treated prior to discharge. Discharges of treated pressure wash water must be included as one of the designated outfalls for which coverage has been granted under the permit prior to discharge.
4. The discharge shall have no sheen, and there shall be no discharge of visible oil, floating solids, or visible foam in other than trace amounts.
5. Additives other than chlorine and detergent to the source water such as biocides and corrosion inhibitors for pressure wash water, hydrostatic or pressure test waters must be reviewed by the Alabama Department of Environmental Management prior to their use from which a discharge occurs.
6. Bilge and ballast water which have come into contact with product, waste, or waste residuals or which contain oils, solvents, or other additives (except chlorine and biodegradable and phosphate-free detergents) are not authorized for discharge and must be disposed of properly to a certified waste disposal company.

1.2 ALASKA

All CWA wastewater permit required submissions (e.g., Notices of Intent (NOIs), Notice of Termination (NOT) and Annual reports) are to be submitted electronically through Environmental Data Management System (EDMS), unless prior approval has been obtained from DEC for an alternative means. EDMS is a one-stop portal for submitting permit applications, requesting permit changes, and submitting reports to Alaska DEC Division of Water. EDMS replaces the DEC Online Application System (OASys).

NPDES permittees in Alaska and other regulated entities that are currently required to submit the following reports must begin submitting them electronically by December 21, 2025 (unless their regulatory authority directs otherwise, or they received a waiver):

- Notices of Intent to discharge (NOIs)
- Notices of Termination (NOTs)
- No Exposure Certifications (NOEs)
- Low Erosivity Waivers (LEWs)
- Municipal Separate Storm Sewer System (MS4) Program Reports
- Pretreatment Program Annual Reports
- Significant Industrial User (SIU) Bi-annual Compliance Reports
- Sewer Overflow/Bypass Event Reports
- Clean Water Act (CWA) Section 316(b) Annual Reports

Alaska NPDES/APDES permittees must submit DMR data electronically through NetDMR per this requirement. Alaska DEC has established an e-Reporting Information website that contains general information about this new reporting format. Training materials and webinars for NetDMR can be found at NetDMR website.

1.3 CALIFORNIA

Authorized Non-Stormwater Discharges

The following is a direct excerpt from California's Industrial Stormwater Permit:

Non-Storm Water Discharges (NSWDs) Discharges that do not originate from precipitation events. including but not limited to, discharges of process water, air conditioner condensate, non-contact cooling water, vehicle wash water, sanitary wastes, concrete washout water, paint wash water, irrigation water, or pipe testing water.

Section IV: AUTHORIZED NON-STORM WATER DISCHARGES (NSWDs)

A. The following NSWDs are authorized provided they meet the conditions of Section IV.B:

1) *Fire-hydrant and fire prevention or response system flushing;*
2) *Potable water sources including potable water related to the operation, maintenance, or testing of potable water systems;*
3) *Drinking fountain water and atmospheric condensate including refrigeration, air conditioning, and compressor condensate;*
4) *Irrigation drainage and landscape watering provided all pesticides, herbicides and fertilizers have been applied in accordance with the manufacturer's label;*
5) *Uncontaminated natural springs, groundwater, foundation drainage, footing drainage;*
6) *Seawater infiltration where the seawater is discharged back into the source: and,*
7) *Incidental windblown mist from cooling towers that collects on rooftops or adjacent portions of your facility, but not intentional discharges from the cooling tower (e.g., "piped" cooling tower blowdown or drains).*

Section IV B. The NSWDs identified in Section IV.A are authorized by this General Permit if the following conditions are met:

1) *The authorized NSWDs are not in violation of any Regional Water Board Water Quality Control Plans (Basin Plans) or other requirements, or statewide water quality control plans or policies requirement;*
2) *The authorized NSWDs are not in violation of any municipal agency ordinance or requirements;*
3) *BMPs are included in the SWPPP and implemented to: a. Reduce or prevent the contact of authorized NSWDs with materials or equipment that are potential sources of pollutants; b. Reduce, to the extent practicable, the flow or volume of authorized NSWDs; c. Ensure that authorized NSWDs do not contain quantities of pollutants that cause or contribute to an exceedance of a water quality standards; and, d. Reduce or prevent discharges of pollutants in authorized NSWDs in a manner that reflects best industry practice considering technological availability and economic practicability and achievability.*
4) *The Discharger conducts monthly visual observations (Section XI.A.1) of NSWDs and sources to ensure adequate BMP implementation and effectiveness; and,*
5) *The Discharger reports and describes all authorized NSWDs in the Annual Report.*

Section IV C. Firefighting related discharges are not subject to this General Permit and are not subject to the conditions of Section IV.B. These discharges, however, may be subject to Regional Water Board enforcement actions under other sections of the Water Code. Firefighting related discharges that are contained and are later discharged may be subject to municipal agency ordinances and/or Regional Water Board requirements.

1.4 CONNECTICUT

Compressor Blowdowns

Compressor blowdown wastewater may not be discharged onto the ground without a permit.

1.5 ILLINOIS

Washing Operations

Waters used to wash vehicles without the use of detergents is an allowable discharge under the *State of Illinois General NPDES Permit for Storm Water Discharges from Industrial Activities* with the following limitations:

1. Certification must be provided that the discharge has been tested or evaluated for the presence of non-storm water discharges. The certification must include a description of any tests for the presence of non-storm water discharges, the methods used, the dates of the testing, and any onsite drainage points that were observed during the testing. Any facility that is unable to provide this certification must describe the procedure of any test conducted for the presence of non-storm water discharges, the test results, potential sources of non-storm water discharges to the storm sewer, and why adequate tests for such storm sewers were not feasible.

1.6 INDIANA

Washing Operations

Vehicle wash water, where uncontaminated water without detergents or solvents is utilized, is an allowable discharge under the *State of Indiana NPDES General Permit*.

1.7 MISSISSIPPI

Washing Operations

Water used to wash vehicles where detergents are not used is an allowable discharge under the *State of Mississippi Baseline Storm Water General Permit for Industrial Activities*.

1.8 NEBRASKA

Washing Operations – Invasive Species

Hot water or drying is the recommended invasive species decontamination method in Nebraska.

1.9 OREGON

Washing Operations

Vehicle and equipment wash waters are an allowable discharge under the *State of Oregon General Permit for Wash water* (1700-A). All detergent must be phosphate-free. Boat and aircraft washing activities are not covered under this permit.

ATTACHMENT E

ADVANCED STRUCTURAL BMPS

ATTACHMENT E - ADVANCED STRUCTURAL BMPS *

*Note: The mention of trade names or commercial products does not constitute recommendation or endorsement for use by the U.S. Coast Guard.

This set of Advanced Structural BMPs consist of proprietary engineered systems that may be installed to remove pollutants from stormwater runoff or prevent pollutants from entering stormwater. These may be used when needed and resources allow. Structural BMPs typically require regular inspection and maintenance.

HYDRODYNAMIC SEPARATORS

APPLICABILITY

Hydrodynamic separators (Figure 1) are structures with a settling or separation unit that remove suspended solids, floatable trash, and petroleum products as stormwater runoff flows through the device. The energy of the flowing water allows the pollutants to efficiently separate, alleviating the need for an outside power source. Hydrodynamic separators are most effective when the pollutants are heavy particulates that can settle out or floatables that can be captured. Hydrodynamic separators vary in size and are ideal when land availability is limited, as some are small enough to fit in conventional manholes. They are also ideal for use in pollutant "hotspots" such as maintenance yards, parking lots and fueling areas/gas stations where high concentrations of pollutants are likely to occur.

Figure 1 - The Downstream Defender®

ADVANTAGES AND DISADVANTAGES

Hydrodynamic separators are designed primarily for removing floatable and gritty materials; they may have difficulty removing the less-settleable solids generally found in stormwater. The reported removal rates differ depending on the vendor. Site constraints including soil depth and soil stability may limit the applicability. Hydrodynamic separators are not maintenance-intensive; however, maintaining the system properly is very important in ensuring it is operating efficiently. Frequent inspections are required within the first year, and cleaning may generally be performed with a sump vacuum or vacuum truck when the unit has reached the recommended capacity for removal. A licensed waste management company should remove oil and sediment and dispose of it properly. Inspections may become less frequent once a typical rate of accumulation from runoff from the site has been established.

COST

The capital costs for hydrodynamic separators will depend on site-specific conditions, including the amount of runoff to be treated, land availability, and whether the unit is being used in combination with other best management practice (BMP) treatment measures. Capital costs can range from $2,300 to $40,000 for a pre-cast unit. Maintenance costs are generally low (around $1,000 per year) and will vary depending on the company contracted to clean the unit.

REFERENCES

U.S. Environmental Protection Agency (EPA), September 1999. *Storm Water Technology Fact Sheet, Hydrodynamic Separators*. EPA 832-F-99-017.

ADDITIONAL INFORMATION

https://www.stormwater360.co.nz/products/stormwater-management/gross-pollutant-traps/prod/vortechs

https://www.conteches.com/stormwater-management/treatment/stormceptor-systems

https://www.hydro-int.com/en/products/downstream-defender

https://parkusa.com/index.php/products/stormwater/stormwater-interceptors/stormtrooper

OIL/WATER SEPARATORS

APPLICABILITY

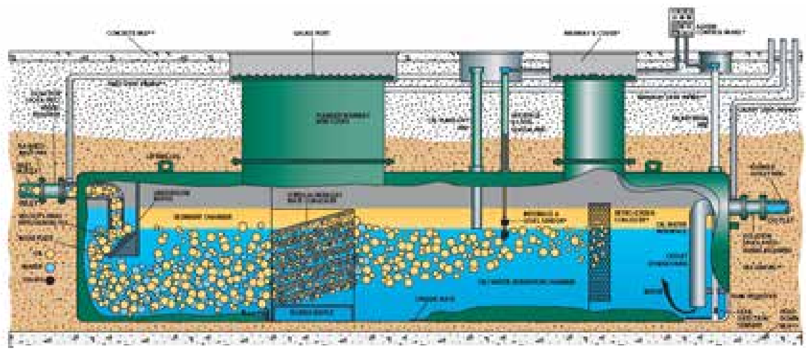


Figure 2 – Oil/Water Separator

Source: Highland Tank

Oil/water separators (OWSs) (*Figure 2*) consist of a series of chambers that are designed to remove oil, grease, light petroleum products, and oily coated solids from wastewater discharges. OWSs are ideally located in industrial areas that receive oily wastewater discharges such as fueling areas/gas stations, fuel storage areas, parking areas, loading areas, maintenance areas, and designated wash areas. The addition of a coalescing unit will greatly increase the effectiveness in oil/water separation. The effluent from OWSs is typically discharged to either the storm or sanitary sewer system. OWSs are typically used where land requirements and cost prohibit the use of larger BMP devices.

ADVANTAGES AND DISADVANTAGES

OWSs can effectively trap trash, debris, oil, and grease; however, they have demonstrated limited ability to separate dissolved or emulsified oil. OWSs are generally recommended for drainage areas of 1 acre or less. OWSs should not be implemented at sites that generate large amounts of sediment in the runoff.

Regular maintenance is essential to ensuring effective pollutant removal. Basic maintenance should consist of regularly checking and cleaning out the sediment that has accumulated in the separator, preferably after every storm event, and removal of trash and debris. OWSs can also be fitted with sensing units that indicate the need for maintenance. Lack of maintenance can cause suspension of previously settled pollutants.

COST

OWSs can be purchased as pre-manufactured units or constructed on-site. The typical cost for a pre-manufactured unit is around $5,000. Pre-manufactured units are generally less expensive than cast in place.

Higher residual hydrocarbon concentrations in trapped sediments can cause maintenance costs associated with OWSs to be higher than those of other BMPs due to hazardous waste disposal requirements. Maintenance costs will vary depending on the size of the drainage area, amount of residuals collected, and the clean-out method (e.g., confined space entry).

REFERENCES

U.S. EPA, September 1999. *Storm Water Technology Fact Sheet, Water Quality Inlets*. EPA 832-F-99-029.

Highland Tank & Manufacturing Company, Inc. Literature provided by manufacturer, 2013.

ADDITIONAL INFORMATION

www.highlandtank.com

https://parkusa.com/index.php/products/waste-water/oil-water-separators

STANDARD WASH RACK

APPLICABILITY

Vehicle washing can generate dry weather runoff contaminated with detergents, oils, grease, and heavy metals. Vehicle washing BMPs can eliminate contaminated wash water discharges to the storm sewer system. When installing a wash rack (*Figure 3*), several design features should be considered. A designated wash area should be paved and bermed or sloped to contain and direct wash water to a sump connected to the sanitary sewer or to a holding tank, process treatment system, or enclosed recycling system. Note that you must seek the permission of the sewer authority before discharging wastewater to the sanitary sewer, and that special treatment requirements may be placed on such discharges.

Figure 3 – Standard Wash Rack

ADVANTAGES AND DISADVANTAGES

The use of a designated wash rack will confine washing operations to a specific location, reducing the number of areas requiring inspection. Maintenance requirements are minimal and inexpensive; however, a wash rack plumbed to the sanitary sewer system can be expensive to build. Employees can be easily trained in washing procedures to avoid illicit discharges. A wash rack's paved surfaces and sump should be inspected and cleaned periodically to remove buildup of particulate matter and other pollutants. The area surrounding the wash rack should be visually inspected on a routine basis for leaks, overspray, and other signs of ineffective containment.

COST

The estimated construction cost for a standard wash rack ranges from $75,000 to $130,000 depending on the size of the wash rack that is required and the optional addition of a canopy structure. The maintenance costs associated with a wash rack are relatively inexpensive; however, any associated plumbing, recycling, or pretreatment systems will also require periodic inspection and additional maintenance costs.

REFERENCES

U.S. EPA. National Menu of Best Management Practices (BMPs) for Stormwater accessed September 2022.

Stormwater Best Management Practice, Municipal Vehicle and Equipment Washing (epa.gov) accessed September 2022.

ADDITIONAL INFORMATION

https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater

MOBILE WASH WATER RECOVERY SYSTEM

APPLICABILITY

For washing operations that use detergents, soaps, cleaners, hot water or steam, and other chemicals, the wash water should be collected in a manner that prevents the mixture or wash-down of pollutants with stormwater runoff. All U.S. Coast Guard (USCG) washing operations should occur in a designated wash rack when possible. In the absence of a designated wash rack, a portable vacuum recovery unit (*Figure 4*) or comparable device may be used on-site to collect wash water for proper disposal. A mobile wash water recovery unit is useful for capturing and disposing of contaminated runoff resulting from external building wash-downs and/or engine flushing areas.

Figure 4 – Mobile Water Capture System

Source: Hydro Tek Systems, Inc.

ADVANTAGES AND DISADVANTAGES

These type collection units typically have a small footprint, and some models may be used as a high-powered vacuum for dry work as well (e.g., debris collection from sanding and blasting activities). The units are limited to a certain volume of water that can be captured per minute; therefore, washing activities must be controlled and contained in a manner to allow the unit to collect all wash water runoff and overspray. Once the contaminated wash water has been collected, the water must still be disposed of properly.

COST

The cost of a mobile wash water recovery unit is relatively minimal, running between $1,300 and $1,700 per unit. However, the costs of proper wash water disposal over the life of the unit should also be considered in addition to the unit itself.

REFERENCES

Hydro Tek Systems, Inc. Literature provided by manufacturer, 2013.

ADDITIONAL INFORMATION

https://www.vikingindustrial.com/new-equipment/cleaning-equipment/water-recovery-recycling

Viking Industrial Systems - Water Recovery & Recycling

https://www.norcalpressurewash.com/

TEMPORARY CONTAINMENT SYSTEM FOR WASHING

APPLICABILITY

Figure 5 – Temporary Containment Pool

Source: Florida Department of Environmental Protection

Wastewater (including wash water) from any type of vehicle and equipment cleaning can contain significant amounts of substances such as oil and grease, petroleum products, suspended solids such as dirt and grit, heavy metals, detergents, and other pollutants. These contaminants may cause pollution of surface water or groundwater and result in violations of water quality standards if the wastewater is not properly managed. All USCG washing operations should occur in a designated wash rack when possible. In the absence of a designated wash rack or a portable vacuum recovery unit, a temporary containment pool (*Figure 5*) may be used for the collection/containment of wash water. A temporary containment pool can be used in conjunction with a portable vacuum recovery unit for maximum efficacy. Depending on the volume of water used, wash water may also be allowed to settle then evaporate from within the containment pool. Solids remaining within the pool after evaporation must be collected, characterized, and disposed of properly.

ADVANTAGES AND DISADVANTAGES

Temporary containment pools are easy to assemble, provide an immediate work area, and allow wash water to be collected in a manner that prevents pollutants from entering the storm drain system. Containment pools vary in size and material and can be used for washing a variety of equipment and vehicles. Washing activities must be conducted in a manner that minimizes overspray, and wash water or settled solids must still be disposed of properly.

COST

The typical cost for a temporary containment pool is between $1,000 and $7,000, depending on the desired size and material. For a cost savings, containment pools may also be constructed on-site by using plastic sheeting and sand bags or framing wood.

REFERENCES

Florida Department of Environmental Protection. *Recommended Best Management Practices for Mobile Vehicle and Equipment Washing*.

ADDITIONAL INFORMATION

www.newpig.com

www.polystarcontainment.com

STORM DRAIN COVERS AND FILTER INSERTS

APPLICABILITY

Storm drain inlets are entrances to the storm sewer system. They typically include a grate or curb inlet and a sump to capture sediment, debris, and pollutants. Storm drain covers (*Figure 6*), filter inserts (*Figure 7*), and filter socks (*Figure 8*) are items that can be placed on top of a storm drain grate or within a storm drain grate to either seal the drain from wastewater or to filter oil, debris, and sediment as it passes through the storm drain inlet. They come in a variety of types and can consist of a variety of materials. Storm drain filter inserts can be used on a regular basis in pollutant hotspots such as parking lots. Storm drain covers should be considered as a temporary measure when an activity with high pollutant potential (e.g., sanding, blasting, and vehicle maintenance) is occurring near a storm drain.

Figure 6 - Storm Drain Cover
Source: PIG

ADVANTAGES AND DISADVANTAGES

Figure 7 - Storm Drain Filter Insert
Source: PIG

In general, storm drain inlet protection measures are practical for areas receiving relatively clean runoff and are designed to handle drainage areas less than 1 acre. Unless frequently maintained, storm drain filter inserts can reduce the volume of water that is able to pass through the filter. This can cause stormwater ponding in the area of the storm drain as well as a resuspension of pollutants.

COST

The typical cost for a storm drain cover is $600 per unit. The typical cost for a storm drain filter insert is $100 per unit.

REFERENCES

U.S. EPA. National Menu of Best Management Practices (BMPs) for Stormwater accessed September 2022.

ADDITIONAL INFORMATION

https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater

www.newpig.com

Powered by Ferguson - ACF Environmental - Environmental Products

https://store.interstateproducts.com/products/Stormwater

www.erosionpollution.com/

Figure 8 - Storm Drain Filter Sock
Source: PIG

STANDARD SELF-CONTAINED SHED, CLAM SHELL & ROLL-TOP

APPLICABILITY

Failure to properly store hazardous materials dramatically increases the potential for those pollutants to comingle with stormwater runoff. Simple good housekeeping practices will allow for hazardous materials to be managed more effectively. Storage areas, loading and unloading areas, and new and waste materials should all be covered or enclosed. The type of storage container needed is determined by the type of material being stored and the local building requirements.

Figure 9 - Roll-Top Drum Storage
Source: PIG

ADVANTAGES AND DISADVANTAGES

Prefabricated storage containers and sheds (*Figures 9 & 10*) provide an economical means of storage and secondary containment as they eliminate the expense of designing and constructing a permanent structure. Maintenance costs associated with self-contained storage containers and sheds are generally low and consist mostly of regular inspections for leaks, signs of cracks or deterioration, rust, or any other signs of corrosion. The lifespan of the container or structure should be considered, depending on the hazardous nature of the stored materials.

It is important to note that outdoor chemical storage buildings must meet building and fire code requirements and may require local and/or state approval.

Figure 10 - Self-Contained Shed
Source: Global Equipment Company, Inc.

COST

The typical cost for hazardous materials storage containers depends on the desired storage capacity and the type of construction of the storage unit (e.g., plastic, metal). The typical cost for a single drum storage container is approximately $250 while the typical cost for a four-drum roll-top storage container is approximately $1,500. A self-contained storage shed (Figure 10) with a 16-drum capacity will cost approximately $10,000.

REFERENCES

U.S. EPA. National Menu of Best Management Practices (BMPs) for Stormwater accessed September 2022.

ADDITIONAL INFORMATION

https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater

https://www.globalindustrial.com/c/janitorial-facility-maintenance/outdoor_structures_storage/safety_storage_buildings

https://www.uschemicalstorage.com/outdoor-chemical-storage-buildings/

https://www.grainger.com/know-how/safety/emergency-response/fire-protection/kh-chemical-storage-buildings-257-qt

OIL ABSORBENT LOGS/ROLLS

APPLICABILITY

Sorbent materials are normally used in areas where storm water runoff is likely to contain oil and grease. Oil absorbent logs and rolls (Figure 11) are used around storm drains near parking lots, equipment storage areas, and vehicle and equipment maintenance areas to prevent oil or oil laden runoff from entering the storm sewer system. The sorbent materials quickly capture the targeted pollutant to prevent exposure and contamination. Sorbents can be divided into three basic categories: natural organic, natural inorganic and synthetic. A variety of sorbent types can be purchased, ranging from oil-only, to hazmat, to universal materials.

Figure 1 – Oil Absorbent Logs

ADVANTAGES AND DISADVANTAGES

Oil absorbent logs and rolls are inexpensive and are extremely easy to operate and maintain. Many sorbent materials change colors or use some type of indicator when they need to be replaced. They are also not subject to mechanical failure or breakdown.

Sorbent materials do require frequent inspection. They can also become ineffective or less effective when the storm drain is located at the bottom of a slope or if there is a large amount of waste or contaminated runoff generated. Proper handling and disposal of the used sorbent material is required.

COST

The typical cost for oil absorbent logs is approximately $75 for 40 units. The cost of proper disposal should also be considered.

REFERENCES

U.S. EPA. National Menu of Best Management Practices (BMPs) for Stormwater accessed September 2022.

Florida Department of Environmental Protection. *Recommended Best Management Practices for Mobile Vehicle and Equipment Washing*.

ADDITIONAL INFORMATION

https://www.epa.gov/npdes/national-menu-best-management-practices-bmps-stormwater

www.newpig.com

www.globalindustrial.com

www.emedco.com

ATTACHMENT F

WASTEWATER TREATMENT PLANTS

USCG-OPERATED WASTEWATER TREATMENT PLANTS

Facility Name	Design Flow	NPDES Individual Permit # / Authority	Industrial Multi-Sector General Permit Required? *(Design flow > 1.0 mgd [1])*
Sector Eastern Shore			
Station Eatons Neck	0.002 mgd 0.00075 mgd three outfalls: 0.0003 mgd each		No
Station Fire Island	three outfalls: 0.0006 mgd each	NY0180661/ NYDEC	No
Sector Fort Macon			
Station Grand Isle			
Station Montauk		NY0175188/ NYDEC	No
Base Kodiak	1.5 mgd	AK-002064-8/ AK DEC	Yes
Boston Light			
Sector Moriches	0.001 mgd two outfalls: 0.00075 mgd each	NY0180688/ NYDEC	No
COMMSTA New Orleans	Less than 5,000 gpd [2]	LAG470442/ LADEQ	No
CST Sallisaw			
TRACEN Petaluma	0.197 mgd	R1-2012-0033/ North Coast RWQCB	No
Station Shinnecock	0.003 mgd	NY0254223/ NYDEC	No
Station Venice			
Station Milford Haven			
Station Sturgeon Bay	0.0016875 mgd	490545/ Door County, WI	No
Station Alexandria Bay	0.0041 mgd (Permitted for)	0022284/ NYS	No

[1] million gallons per day (MGD)

[2] gallons per day (GPD)